

Response to the Reviewer's Comments

July 31, 2025

Respected Editor of IEEE Robotics and Automation Letters,

Enclosed please find the response letter for the manuscript **TOP: Time Optimization Policy for Stable and Accurate Standing Manipulation with Humanoid Robots**, which we are resubmitting for consideration by IEEE Robotics and Automation Letters.

As attached, the responses are described in detail following each of the reviewers' comments.

Hopefully, this effort has addressed all the reviewers' comments and made the work acceptable for your journal.

Sincerely,

Reply to Reviewer 1

The paper proposes TOP (Time Optimization Policy), a novel learning-based control framework for humanoid robot standing manipulation that ensures balance, precision, and time efficiency. The framework combines a variational autoencoder (VAE) to encode upper-body motion priors, a decoupled controller with a PD-controlled upper body and a reinforcement learning-based lower body. The proposed approach was validated through simulation and real-world experiments. This paper has a novel contribution and a well-structured framework. The introduction states the purpose of the paper and the references are relevant and complete. However, there are some issues that require further clarification.

We appreciate the reviewer’s concise summary of this work and hope that the explanation has fully addressed all of the reviewer’s concerns.

Due to the lack of space, we present our **paper with appendix**, the **response letter** and the **videos** in our project page <https://anonymous.4open.science/w/top-258F/>, we would like to bother the reviewer to have a look for a full paper.

Point 1.1

Introduction[Page 2], the idea of solving this problem is robot-oriented, not task-oriented. If the task itself has requirements for movement speed, is this method still applicable?

Thank you for raising this insightful question. Indeed, our current method is robot-oriented and focuses on optimizing motion timestamps to enhance stability, precision, and time efficiency from the robot’s perspective.

While the current framework does not explicitly include task-level temporal constraints, it can be extended to accommodate them. For instance, if a task specifies a required duration or timing constraint, our time optimization policy (TOP) can incorporate this via an additional loss term or reward shaping to align robot execution with task-level requirements. For example, we extend the reward with a velocity constraint penalty:

$$R_{\text{speed}} = -w_s \cdot \text{ReLU}(|\dot{q}| - v_{\text{limit}}) \quad (1)$$

where $\dot{q}$ is the joint velocity, and v_{limit} is the task-specified speed limit. And we use ReLU [1] to penalize any velocity exceeding the limit, which effectively enforces a soft constraint on joint speed. This enables our framework to balance task execution time with robot stability and motion smoothness, providing a flexible mechanism for adapting to various task-oriented requirements.

To verify the applicability of our method under task-oriented constraints (e.g., speed requirements), we added a new set of experiments in which we limit the range of dof velocity speed (Limit Speed) across a range (1.5 rad/s, 2.5 rad/s, 3.5 rad/s, Inf rad/s) to simulate tasks with different motion speed demands.

Our Time Optimization Policy (TOP) dynamically adjusts motion timestamps according to the target duration, demonstrating the ability to trade off between speed, accuracy, and stability. We compare the performance of TOP with different speed constraint of tasks, and evaluate metrics such as dof vel (constraint), average time cost (time efficiency), and projected gravity (stability), as shown in the Table R1.

Table R1: Task Constraint: Speed constraint of tasks

Limit Speed (rad/s)	Dof Vel (rad/s) ↓	Time Cost (s) ↓	Projected Gravity ↓
1.5	1.448	69.44	1.991
2.5	2.408	41.75	2.182
3.5	3.356	29.96	2.295
Inf(without limit)	5.156	19.50	2.729

The result shows that our TOP policy can dynamically adjust the motion timestamps (Δt) to slow down or speed up execution accordingly. Naturally, stricter speed limits increase the total time cost, but they also improve dynamic stability by reducing sudden velocity spikes and inertial effects.

For completeness, we will include a corresponding plot to better illustrate this relationship, as shown in the Fig R1. Our plot illustrates the relationship between the specified joint velocity limit and the resulting robot performance. As expected, enforcing a tighter speed

limit forces the policy to generate slower joint motions (lower Dof Vel), which increases the overall task duration (Time Cost) but reduces the projected gravity, indicating enhanced stability.

This confirms that our TOP framework can dynamically adapt its timestamp optimization strategy to comply with task-level speed constraints while achieving a good trade-off among motion speed, stability, and time efficiency. These results demonstrate the feasibility of extending our robot-oriented method to task-oriented scenarios where motion speed must be explicitly controlled.

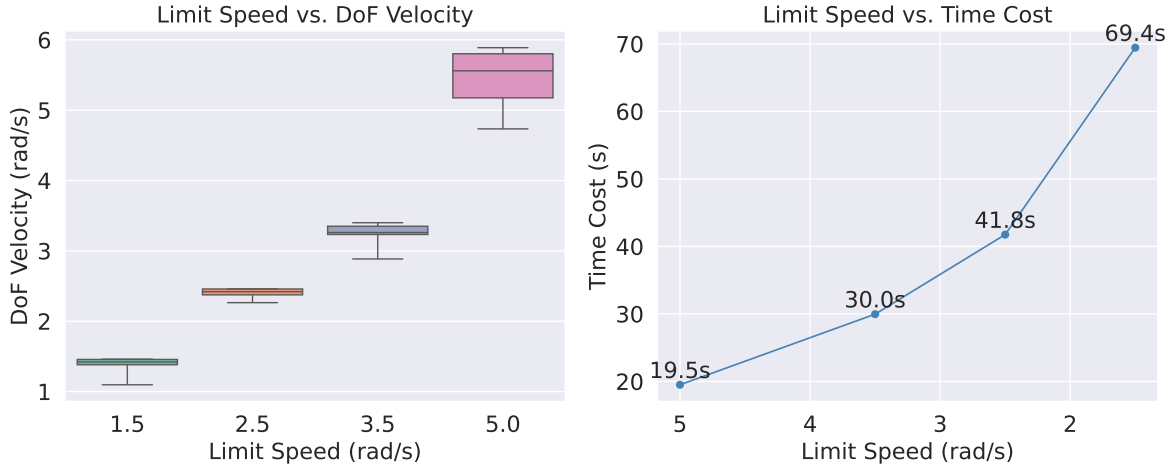


Figure R1: **Impact of speed constraints on robot performance.** This plot shows how different task-specified speed limits (*Limit Speed*) affect the average joint velocity (*Dof Vel*), total task duration (*Time Cost*). As the speed constraint becomes tighter (smaller limit speed), the controller automatically adjusts motion timestamps, reducing the actual joint velocities and increasing time cost. Meanwhile, tighter constraints yield lower projected gravity, indicating improved balance due to slower, smoother motions. This demonstrates that the proposed TOP framework can effectively adapt to various task-level temporal constraints by balancing motion speed, time efficiency, and robot stability.

We will discuss it as a promising future direction in the conclusion.

In CONCLUSIONS:

In future work, we plan to incorporate motion generation modules to enable more natural standing adjustments and further enhance balance control. Moreover, we

aim to explore adaptive task-oriented time optimization policy to extend our application in real-world tasks.

Point 1.2

Introduction[Page 2], does contribution 1 include contribution 2?

Thanks for the reviewer’s comment. As the review’s commit, Yes, Contribution 2 is a core component of the overall framework described in Contribution 1.

Specifically, Contribution 1 refers to the complete standing manipulation framework that achieves stable, accurate, and time-efficient execution of complex upper-body motions. Contribution 2 highlights the design and effectiveness of **TOP** (Time Optimization Policy), which is a key module within this framework. TOP is responsible for optimizing the timestamp of motion execution through supervised reinforcement learning, and plays a central role in achieving the overall objectives stated in Contribution 1.

To avoid ambiguity, we have clarified the relationship between the two contributions and updated our manuscript for better explanation of the novelty of our work.

In INTRODUCTION:

- *propose a novel framework for humanoid standing manipulation that realizes stable, accurate, and time-efficient execution of complex upper-body motions and manipulation tasks.*
- *As a key component of this framework, we introduce **TOP** (Time Optimization Policy), a supervised reinforcement learning module that optimizes motion timestamps to jointly enhance accuracy, stability, and efficiency.*

Point 1.3

Related works[Page 3], B, why only consider VAE and exclude other models?

Thank you for the reviewer’s comment. We apologize for the missing references to other

generative models in the Related Works section. We agree that other approaches such as diffusion models are relevant for human motion modeling.

We want to explain why we choose the VAE model to represent motion prior rather than the other generation models. It is well-suited for learning a compact and smooth latent representation of high-dimensional sequential motion data. The VAE framework provides an efficient way to encode diverse motion patterns while enabling sampling and interpolation in latent space[2], which directly supports our timestamp optimization strategy. Compared to other generative models, such as diffusion models[3], the VAE offers a good trade-off between reconstruction quality, latent controllability, and training/inference efficiency — which is critical for real-time standing manipulation on a humanoid robot. In future work, we agree that exploring alternative priors such as diffusion-based models or normalizing flows would be interesting directions to further improve motion expressiveness and generalization.

We have revised the text and added relevant discussion of diffusion models in the Related Works section to better position our VAE-based design choice in the context of alternative models.

In RELATED WORKS:

Another common way to represent complex motion is to filter and extract key motion information to represent smooth temporal and spatial composition[4], but struggle with handling highly diverse or long-range motions. Recently, advances in generative models, like diffusion models[3] or variational autoencoder(VAE)[5], provided a good idea for representation of motions by creating a latent code to learn the distribution of multimodal motion sequences [6]. Tevet et al.[7] uses a transformer-based diffusion framework to generate natural human motions by directly predicting samples with geometric constraints. Li et al.[8] unifies human motion estimation and generation in a single framework, leveraging regression and diffusion to handle diverse conditions and tasks jointly. To provide a multimodal representation of upper-body motions, we decide to use VAE to extract a latent kinematic motion space and reconstruct them for training.

Point 1.4

Method[Page 4], **D**, how do the movements of the upper and lower limbs align?

Thank you for this insightful comment regarding how the movements of the upper and lower limbs align and interact dynamically. We agree that, in principle, the upper and lower body are inherently coupled through momentum transfer, base motion feedback, and dynamic interaction. Ignoring this coupling entirely would indeed risk generating infeasible or unstable motions.

In our design, however, the **decoupling is not rigid** but rather a **practical modular abstraction** for efficient learning and deployment:

- **Upper-body motion** is compressed by a VAE-based motion prior and tracked by a PD controller that produces desired joint torques to follow the reference trajectories with high precision.
- **Lower-body stabilization** is handled by a reinforcement learning (RL) policy that receives real-time observations of the floating base state, projected gravity, joint states, and the desired upper-body motion embedding (z_t) and motion clip (m_t).

Therefore, while the modules are trained in a decoupled manner, their execution forms a closed-loop whole-body control system. The RL policy implicitly compensates for momentum transfer from the upper-body by dynamically adjusting lower-body joint torques and contact forces to maintain balance. This design principle is consistent with established whole-body control frameworks[9] where manipulation and locomotion are hierarchically coordinated.

As shown in Table R2, the RL policy is explicitly encouraged to minimize base drift, linear/angular velocity deviations, and physically feasible whole-body motions:

- **Base linear velocity** and **base angular velocity** terms penalize excessive base oscillations.
- **Base orientation** cost stabilizes the base roll/pitch under external moments induced by upper-body motions.
- **Stand still** and **action penalty** terms further ensure smooth, coordinated lower-body actions.

Table R2: Key RL Reward Terms for Stabilization

Term	Expression	Weight
Base linear velocity (xy)	$\exp[-4(\mathbf{v}_{xy}^2)]$	+3.0
Base angular velocity (xy)	$\Sigma \boldsymbol{\omega}_{xy}^2$	-0.1
Base orientation	$\Sigma \mathbf{g}_{xy}^2$	-1.5
Stand still pose	$\exp[-10\ \mathbf{q}_t^{\text{leg}} - \mathbf{q}^{\text{leg,ref}}\ _2]$	+1.0
Action rate penalty	$\ \mathbf{a}_t - \mathbf{a}_{t-1}\ _2^2$	-0.2
Action acceleration penalty	$\ \mathbf{a}_t + \mathbf{a}_{t-2} - 2\mathbf{a}_{t-1}\ _2^2$	-0.2

Table R3: Comparative results. We execute more than 10000 motion clips from dataset $\mathcal{M}$ and unseen dataset $\mathcal{T}$, and report their time cost, success rate, mean metrics. Mean Metrics are only calculated based on the data of the robot standing in place, and the data of robot taking steps and falls will not be included in the calculation.

Method	Time Cost ↓	Success Rate ↑	$\mathbf{E}_{\text{jpe}}^{\text{upper}} \downarrow$	$\mathbf{E}_{\text{eepe}}^{\text{upper}} \downarrow$	$\mathbf{E}_{\text{esoe}}^{\text{upper}} \downarrow$	$\mathbf{E}_{\mathbf{g}} \downarrow$	$\mathbf{E}_{\text{acc}}^{\text{lower}} \downarrow$	$\mathbf{E}_{\text{action}}^{\text{lower}} \downarrow$	$\mathbf{E}_{\text{acc}}^{\text{upper}} \downarrow$	$\mathbf{E}_{\text{action}}^{\text{upper}} \downarrow$
Comparative Results										
Fixed Root (reference)	15.0s	100.0%	0.0130	0.0164	0.0594	1.000	-	-	-	-
Exbody[10]	15.0s	92.46%	0.0376	0.0741	0.0923	3.432	21.65	0.7323	24.26	2.976
Ours (TOP)	40.5s	95.30%	0.0269	0.0270	0.0827	2.729	13.45	0.8592	10.41	1.601

We can observe that, this closed-loop design ensures that the lower-body dynamically absorbs momentum transfer delays from the upper-body PD tracker. Our TOP method deliberately slows down sudden upper-body changes to avoid destabilizing the base, which slightly sacrifices time efficiency but greatly improves base stability and overall task success. Compare to the baseline (Exbody[10]) that uses whole-body RL policy with the tracking rewards to control whole-body joints, our method can achieve precision, stability and time efficiency at the same time, proving the effectiveness of our decoupling controllers, as shown in Table R3.

In summary, although the TOP policy dynamically adjusts upper-body timestamps, the lower-body actions are not static but adapt in real-time to ensure whole-body consistency. The empirical results and reward design confirm that our approach avoids dynamic misalignment and preserves stable and feasible behavior.

We believe this more clearly demonstrates that our approach is not **naively decoupled** but rather forms an integrated closed-loop whole-body control system. Thanks to the reviewers for pointing out the problems. Due to the lack of space, we will not include this in revised paper, but will add the explanation to **APPENDIX.A**.

Minor comments:

Point 1.5

Title, maybe "Stable and Accurate Standing Manipulation" is better.

Thanks for the reviewer's comment. We appreciate your suggestion regarding the title clarity. Accordingly, we have revised the title to:

*TOP: Time Optimization Policy for Stable and Accurate Standing Manipulation
with Humanoid Robots.*

Point 1.6

Abstract, "combined with the decoupling controller and VAE" should be revised into "in conjunction with the decoupled controller and VAE".

Thanks for the reviewer's suggestion regarding the wording. We apologize for the inappropriate wording. We have revised the phrase in abstract to

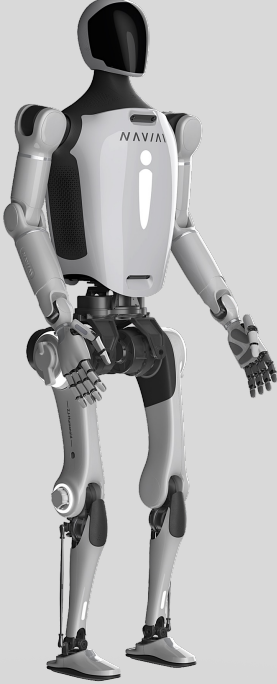
*Finally, we train TOP method in conjunction with the decoupled controller and
VAE*

as recommended. This revision improves the clarity and precision of the expression.

Point 1.7

Experiments[Page 4], A, more detailed information about the hardware platform for model training needs to be introduced, such as GPU model, number of GPUs, etc.

Thanks for the reviewer’s comment. We have added more detailed information about the hardware platform in Section *Experiments*. Specifically, all models were trained on a single NVIDIA RTX 4090 GPU (24 GB memory). The VAE model was trained for approximately 10 hours, while the reinforcement learning policy and the TOP model were each trained for around 5 hours. And the hardware details of humanoid robot are shown in [R2](#).



Humanoid DoFs	41
Mass	60kg
Arm Mass	6kg each
Arm + Hand Payload	3kg each
Height	1.65m
Waist+Neck DoFs	3
Leg DoFs	6 each
Arm DoFs	7 each
Hand DoFs	6 each
Arm Repeatability	0.5mm
Drive Hardware	Elmo
Compute Unit	PICO-TGU4

Figure R2: **Hardware Details.** Our robot stands 1.65m tall, weighs 60kg, and has 41 degrees of freedom, including two 6kg arms with 7-DoF each and a payload capacity of 3kg per arm.

We sincerely appreciate the reviewer’s suggestion. Due to the lack of space, we will incorporate the hardware details into **Appendix.E**.

Point 1.8

Conclusions[Page 7], the “high success rates” need to be quantified.

Thank you for the review’s comment. We agree that “high success rates” should be quantified to make the statement more rigorous. We have revised the text to explicitly report our average task success rate.

In CONCLUSIONS:

In this paper, we propose a novel framework with the TOP method, which improves lower-body stability and upper-body precision by optimizing the timestamps of motion clips. By integrating a robust lower-body policy with precise upper-body tracking, and leveraging a VAE-based motion prior to capture upper-body features, our method achieves 95.30% success rates in experiments, demonstrating stable and accurate standing manipulation performance. However, current motions remain somewhat rigid, for instance, the robot tends to tilt backward rather than twisting its hips. In future work, we plan to incorporate motion generation modules to enable more natural standing adjustments and further enhance balance control.

Reply to Reviewer 2

The paper proposes a Time-Optimized Policy (TOP) for achieving stable and precise standing manipulation in humanoid robots. While demonstrating promising results in simulation and physical experiments with a degree of innovation, the manuscript exhibits significant shortcomings. These include logical inconsistencies, deficiencies in experimental design, and insufficient elaboration in key sections. In its present form, the work does not meet the standards required for publication. Major revision is therefore recommended prior to reconsideration. The specific concerns requiring attention are enumerated below:

We appreciate the reviewer’s concise summary of this work and hope that the explanation has fully addressed all of the reviewer’s concerns.

Due to the lack of space, we present our **paper with appendix**, the **response letter** and the **videos** in our project page <https://anonymous.4open.science/w/top-258F/>, we would like to bother the reviewer to have a look for a full paper.

Point 2.1

The clarity and resolution of Fig. 1 in the text are relatively low and require further improvement and standardization.

Thanks for the reviewer’s comment. We apologize for the lack of clarity in the image of Figure 1. We have replaced it with a higher-resolution version and improved the overall formatting for better readability, as shown in the [R3](#)

Point 2.2

Two pivotal figures, Figure 1 and Figure 5, lack corresponding in-text citations, resulting in a disconnect between textual descriptions and visual elements.

Thank you for your helpful comment. We have carefully revised the manuscript to include explicit in-text citations for Figure 1 and Figure 5. These figures are now properly referenced and discussed at relevant points in the text to enhance clarity and coherence between the

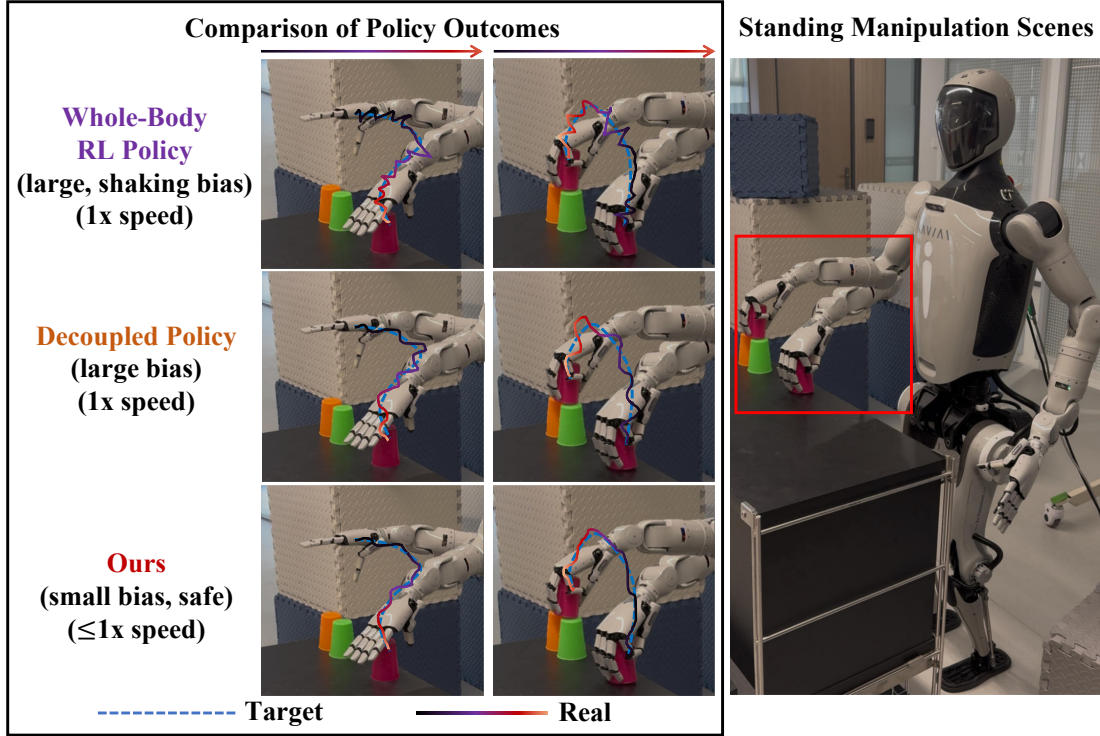


Figure R3: Illustration of different methods. **A**: Whole-body RL policy, but it will be ill-suited to control high-dimensional upper-body joint, which cause large and shaking bias from target trajectory. **B**: Decoupled policy avoid shaking bias by using PD controller for upper-body, but still lack the consideration about momentum caused by rapid upper-body motions, remaining large bias. **C**: Our method adjust timestamp of motions aiming to reduce the impact of momentum and making standing safer, which gain smaller bias but need more time to achieve the goal.

visual elements and their corresponding descriptions.

In INTRODUCTION:

However, all the aforementioned methods tend to execute upper-body motions without fully considering the robot’s actual execution capabilities, as shown in Fig.1.

In EXPERIMENT.C:

*Considering the ablation results, as shown in the Fig.7, our method helps reduce the balance burden and unpredicted inference caused by the lower-body, which enhances both the balance stability and the precision of upper-body motions. And with the help of TOP algorithm, we have achieved a good balance between time efficiency, and accuracy, demonstrating the outperformance of our methods. Compare to **Ours w/o TOP** $\Delta t = 0.01s/0.03s/0.05s$, our method can automatically choose the appropriate $\Delta t \in [0.01s, 0.1s]$ in different situations: when motions seriously affect accuracy and balance, our algorithm will give a larger Δt even more than $0.05s$ to improve stability and precision; conversely, when motions are easy enough, a smaller Δt is chosen to maximize time efficiency. Although our methods take more time to complete motions compared to **Ours w/o TOP** $\Delta t = 0.01s$ and **Mobile-Television**, but we improve both precision and stability significantly. This adaptive TOP strategy results in improved overall precision and stability, while achieving higher average time efficiency. The performance curves of different methods over time are presented in Fig.5, demonstrating the precision, stability and time efficiency of motion clips.*

Point 2.3

While Figure 5 illustrates the trade-off between precision, stability, and time efficiency, the manuscript fails to provide a rigorous quantitative analysis of their interrelationships. Incorporating analytical methods, such as regression modeling to characterize these metric dependencies, is strongly recommended.

Thank you for pointing out this important shortcoming. We appreciate the reviewer’s recommendation to strengthen our analysis of the trade-offs among precision, base stability, and time efficiency.

In our original manuscript, Figure 5 qualitatively showed that different manipulation tasks balance these metrics differently. However, we acknowledge that a clear quantitative characterization was lacking. To address this, we have now conducted an additional empirical analysis to explicitly reveal these interdependencies.

Specifically, we have:

- Applied multiple linear regression and polynomial regression fitting to model the quantitative relationship between time efficiency(Δt), stability(projected gravity), and precision(EF position Error).
- Computed the empirical Pareto front[11, 12] based on the experimental data to highlight the set of non-dominated solutions, which illustrates the achievable trade-off boundary between the competing objectives.

Figure R4 shows the updated analysis. In this figure:

- Each scatter point represents an individual upper-body motion trial with its measured Δt , projected gravity, and end-effector pose error.
- The fitted regression surface captures the overall trend that smaller Δt typically results in slightly increased the projected gravity and pose error due to execute faster upper-body motions.
- The highlighted Pareto front (black) reveals the empirical boundary where no further improvement can be made in one metric without sacrificing at least one of the others.

The regression model confirms that the observed improvements in end-effector accuracy are not independent but are directly coupled to the robot’s base stability and Δt of TOP outputs. For instance, motions with with larger Δt that means the motions will be slowed down, generally allow the lower-body controller more time to compensate for upper-body momentum, resulting in lower projected gravity (base drift) and smaller end-effector error. Conversely, pushing the robot to complete motions faster often requires more rapid upper-

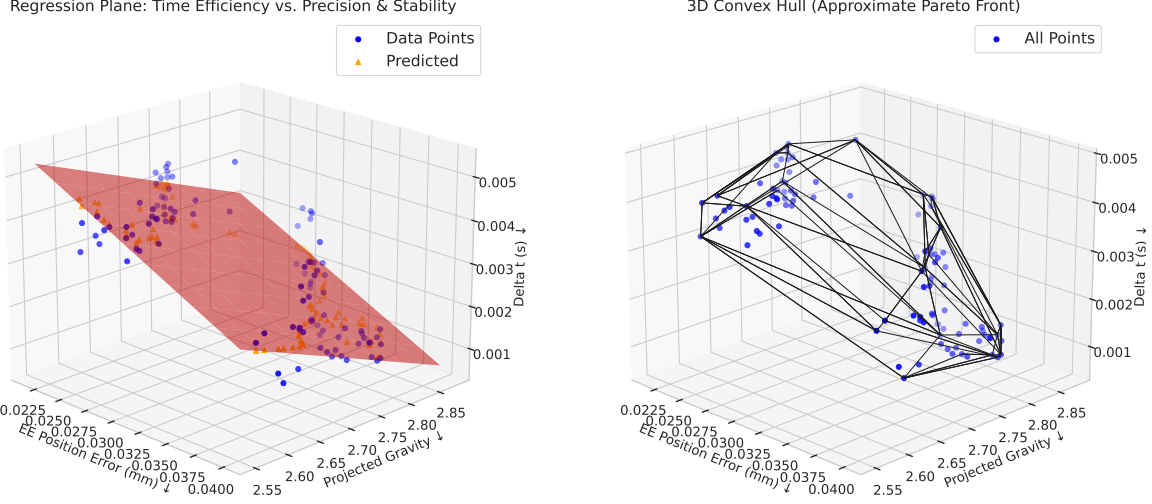


Figure R4: **Regression and Pareto front analysis.** The figure shows the multi-objective trade-off among time, stability, and precision, with the fitted regression surface and the extracted Pareto optimal front (right).

body motions, which inherently injects higher momentum into the base, slightly degrading stability and precision.

We believe this enhanced quantitative analysis, together with the updated figure, provides a clearer and more rigorous understanding of how our framework balances precision, stability, and time efficiency under real-world conditions. We have incorporated this new analysis and discussion into **Appendix.B**.

Once again, we sincerely thank the reviewer for this constructive suggestion, which helped us substantially improve the rigor and completeness of our experimental analysis.

Point 2.4

The tests are limited to static standing and do not involve dynamic environments. It is suggested to expand to mobile operation scenarios and add external disturbance tests.

We thank the reviewer for this valuable suggestion to further validate our method in more challenging and dynamic scenarios. To address this concern, we have conducted additional experiments that extend beyond static standing manipulation. We now demonstrate the

effectiveness and robustness of our Time Optimization Policy (TOP) when the robot performs loco-manipulation tasks and the external disturbance tests of our policy.

To make our results fully transparent, we have released all the demonstration videos on our project page <https://anonymous.4open.science/w/top-258F/>.

Figure R5 shows the mobile operation scenarios of our TOP policy. In these new test cases, the humanoid robot executes upper-body tasks (e.g., pressing button and open fridge) and omnidirectional walking. The motion prior and time optimization policy work jointly with the lower-body RL policy to adapt the timestamps, lower-body torques and maintain balance. This demonstrates that the proposed decoupled yet closed-loop framework generalizes well to locomotion plus manipulation tasks.



Figure R5: **Loco-manipulation scenarios** The top figure illustrates the robot executing a combined locomotion and **fridge-button-pressing** task. The bottom figure illustrates the robot executing a combined locomotion and **opening the fridge** task.

Figure R6 shows external disturbance tests of our policy. We designed scenarios where the robot is intentionally disturbed by external pushes from different directions. The videos show how the floating base stabilizer and leg reactions in real time to absorb these perturbations,

preventing falls.

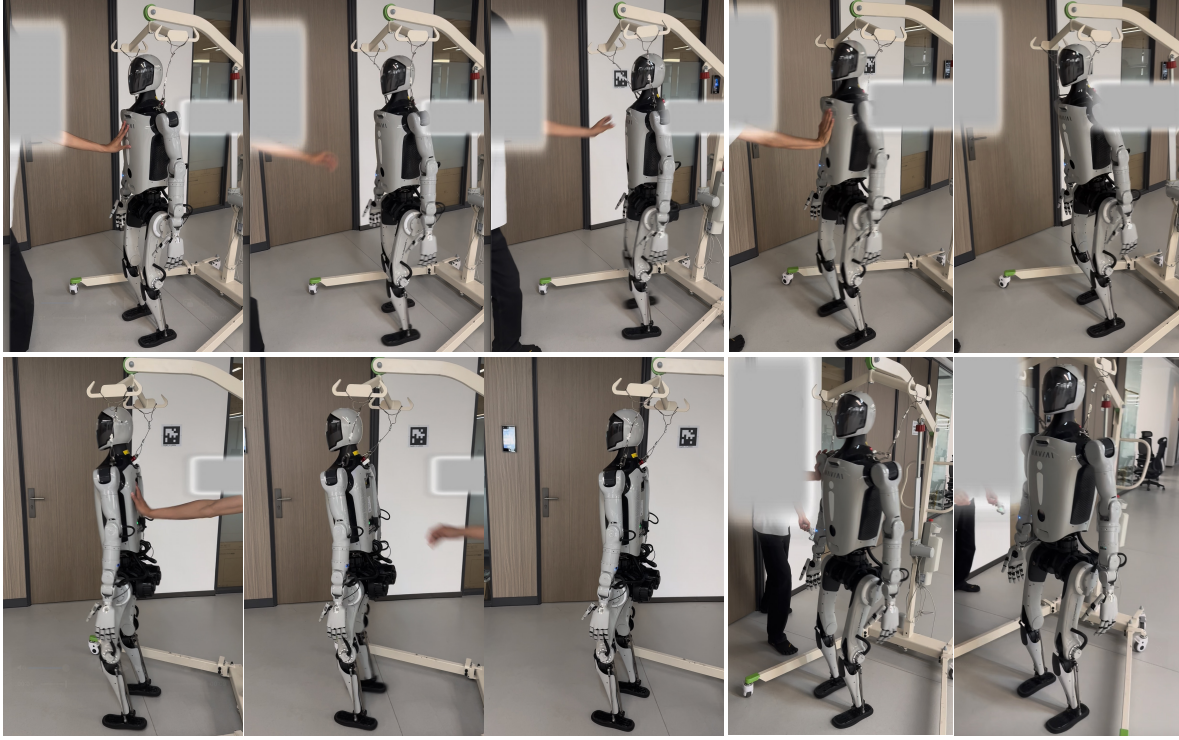


Figure R6: **External push disturbance tests.** The figure illustrates the robot can recover to balance while disturbed by external pushes from different directions.

These new dynamic experiments strongly support the generalizability and practicality of our method for real-world loco-manipulation tasks. We sincerely appreciate the reviewer's suggestion, which helped us strengthen the paper's experimental evidence and we have incorporated this new experiments and discussion into **Appendix.C**.

Point 2.5

The description of momentum disturbance mechanisms (Introduction, Paragraph 3) lacks quantitative definition for the threshold of "fast upper-body motions," rendering the problem statement insufficiently rigorous. Supplementary theoretical analysis quantifying the relationship between momentum variation and loss of balance is advised.

Thank you for this insightful comment on the need for a more rigorous and quantitative

definition of *fast upper-body motions* and their impact on momentum-induced base disturbance.

We fully agree that the physical mechanism behind instability stems from the conservation of linear and angular momentum when large joint speeds and accelerations generate reaction forces on the floating base. To clarify this, we now provide a supplementary theoretical approximation to quantify the effect.

The total angular momentum $\mathbf{L}$ of the floating base plus the upper-body can be expressed as:

$$\mathbf{L} = \mathbf{I}_B \boldsymbol{\omega}_B + \sum_{i=1}^n I_i \dot{q}_i \quad (2)$$

where $\mathbf{I}_B$ is the inertia tensor of the base, $\boldsymbol{\omega}_B$ is the base angular velocity, I_i is the inertia of joint i , $\dot{q}_i$ is the joint angle velocity. Taking the time derivative yields the net momentum change:

$$\dot{\mathbf{L}} = \mathbf{I}_B \dot{\boldsymbol{\omega}}_B + \sum_{i=1}^n I_i \ddot{q}_i \quad (3)$$

in free-floating or lightly constrained systems, any increase in joint accelerations $\ddot{q}_i$ generates an equal and opposite reaction on the base:

$$\mathbf{I}_B \dot{\boldsymbol{\omega}}_B = - \sum_{i=1}^n I_i \ddot{q}_i. \quad (4)$$

Thus, the floating base must compensate for the angular impulse caused by fast upper-body motion. If $\ddot{q}_i$ becomes large (e.g., due to commands), the base rotation $\boldsymbol{\omega}_B$ and linear drift will increase proportionally.

This confirms that controlling joint accelerations effectively controls the induced momentum variation:

$$\Delta \boldsymbol{\omega}_B \propto - \int_{t_0}^t \sum_{i=1}^n I_i \ddot{q}_i dt. \quad (5)$$

that underlines why our TOP policy explicitly shapes the time scaling to keep $\ddot{q}_i$ bounded, thus reducing $\dot{\mathbf{L}}$ and stabilizing the robot during manipulation.

Also, according to the Eq. 30, faster joint angle velocity $\dot{q}_i$, the larger the disturbance to the base's rotational state. The time optimization policy (TOP) effectively reduces peak joint velocities $\dot{\mathbf{q}}$, preventing abrupt momentum transfer to the floating base. This explains why we evaluate *joint acceleration* and *joint velocity* to measure performance of methods.

To further ensure that these internal momentum changes do not compromise whole-body stability, we also analyze the Zero Moment Point. By ensuring the ZMP remains within the support polygon throughout manipulation tasks, our robot can keep its stability. Firstly, we want to explain the definition of ZMP. ZMP is the point on the ground where the horizontal component of the moment of the ground reaction forces becomes zero, assuming the ground is a horizontal plane[13]. A necessary condition for the locomotion mechanism to be in dynamic equilibrium is that the candidate ZMP is inside the support polygon. Furthermore, to handle non-planar surfaces, the concept of the Zero Moment Line (ZML) is introduced, which extends the ZMP concept to account for complex contact surfaces[14]. Then we give the calculation process as follows[15]:

$$\boldsymbol{\tau} = \mathbf{p}_{zmp} \times \mathbf{f} + \boldsymbol{\tau}_p \quad (6)$$

$$\dot{\mathbf{P}} = M\mathbf{g} + \mathbf{f} \quad (7)$$

$$\dot{\mathcal{L}} = \mathbf{p}_{com} \times M\mathbf{g} + \boldsymbol{\tau} \quad (8)$$

where p_{zmp} is the position of ZMP, τ_p is the moment about the ZMP point, and the f represents the ground reaction force. P , L , M , p_{com} represents the linear momentum, the angular momentum, the total mass, the center of mass (CoM) position of the system, respectively, and $\mathbf{g} = [0, 0, -g]^T$. By the definition of ZMP, we have:

$$\tau_{p,x} = \tau_{p,y} = 0 \quad (9)$$

Solving the above equations with respect to $\mathbf{p}_{zmp}$, we have

$$p_{zmp,x} = \frac{Mgp_{com,x} + p_{com,z}\dot{P}_x - \dot{\mathcal{L}}_y}{Mg + \dot{P}_z} \quad (10)$$

$$p_{zmp,y} = \frac{Mgp_{com,y} + p_{com,z}\dot{P}_y + \dot{\mathcal{L}}_x}{Mg + \dot{P}_z} \quad (11)$$

According to the Eq.39, ZMP at different heights $p_{com,z}$ lies at different locations, whose trajectory forms a line of ZMPs, called ZML [14]. The condition for stable locomotion is that the ZML must cross the support polygon, as shown in Fig R7.

Then, we need to estimate the support polygon of the humanoid robot to determine the relationship between ZML and the support. In general cases, the support polygon is determined by computing the convex hull of all contact points for scenarios with multiple contact points. Therefore, we calculate the ZML lines in simulation, as shown in the Fig R8.

Figure R7: **ZML cross the support polygon.** The red dot represents the approximated center of the support polygon, and the green dot on the ZML is the projected point at the same horizontal height as the red dot. d represents the distance between the center of the support polygon and the projected point of ZML, which can describe the stability of the system. The smaller d is, the more stable the system is.

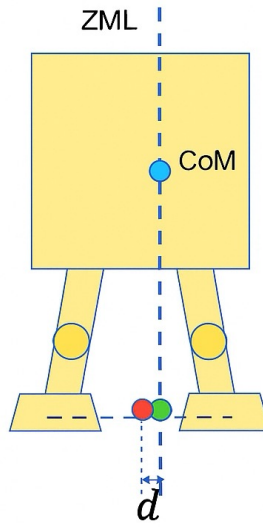
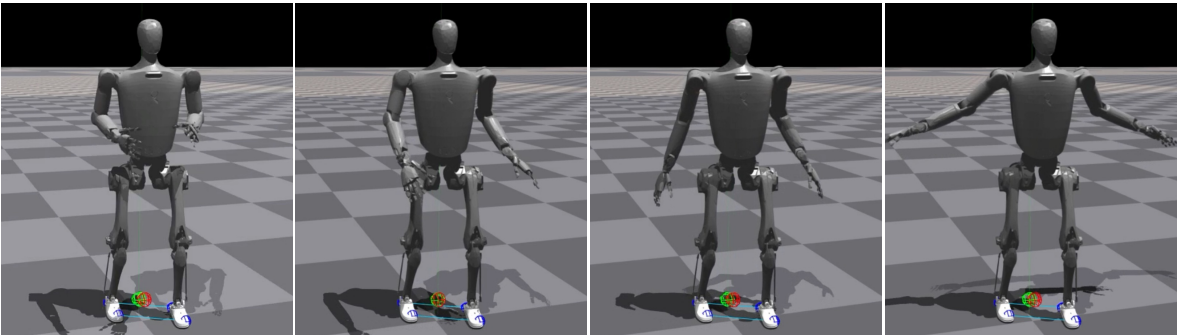


Figure R8: **ZML and Support Polygon in Isaacgym.** The red point represents the center of the support polygon, and the green dot on the ZML is the projected point at the same horizontal height as the red dot. The blue points and the light blue lines are the support polygon.



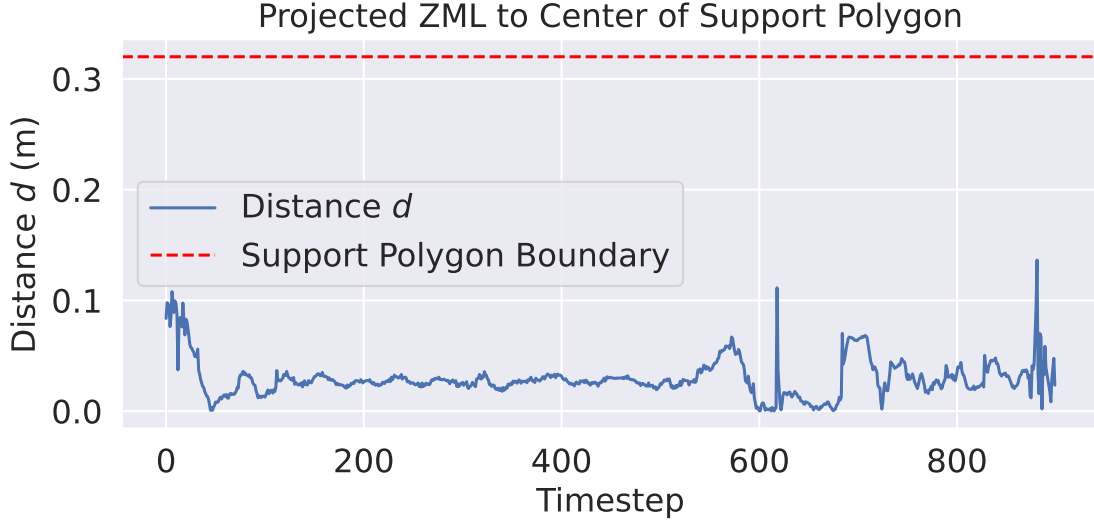
To quantitatively assess the dynamic stability of our policy, we compute the signed distance d between the ZML projection point and the center of the support polygon:

$$d = \min_{p_{csp} \in \mathcal{S}} \|p_{csp} - \text{proj}_{ZML}(p_{csp})\|_2, \quad \text{proj}_{ZML}(p_{csp}) \in \mathbb{R}^2 \quad (12)$$

where $\mathcal{S}$ denotes the support polygon of current contact points, and $\text{proj}_{ZML}(p_{csp})$ is the projected ZML point in the horizontal plane. If $d \in [0, 0.32]$, the ZML point is inside the support polygon; if $d \in (0.32, 0.36]$, it lies near the edge of support polygon; and if $d \in (0.36, +\infty)$, the ZML has exited the support region, indicating potential instability. It is worth noting that the closer d is to 0, the system is more stable and satisfy the ZMP constraint.

To demonstrate that our time optimization policy (TOP) maintains the Zero-Moment Point (ZMP) condition throughout task execution, we plot the time series of d for representative tasks. As shown in Fig. R9, the distance remains positive during the entire motion, verifying that the ZML stays well within the support polygon at all times, and thus the robot’s base remains dynamically stable despite rapid upper-body motions.

Figure R9: **Stability metric d over time.** The curve shows the distance between the ZML projection and the center of support polygon. The distance remains ≤ 0.32 , confirming that the ZMP condition is always satisfied.



To further illustrate the relationship between momentum variation, ZMP and stability of robot, as shown in the Figure R10. The figures plot the measured floating base velocity over time for both our method (TOP) during a fast upper-body motion. The results show that

Momentum, ZMP, and Stability

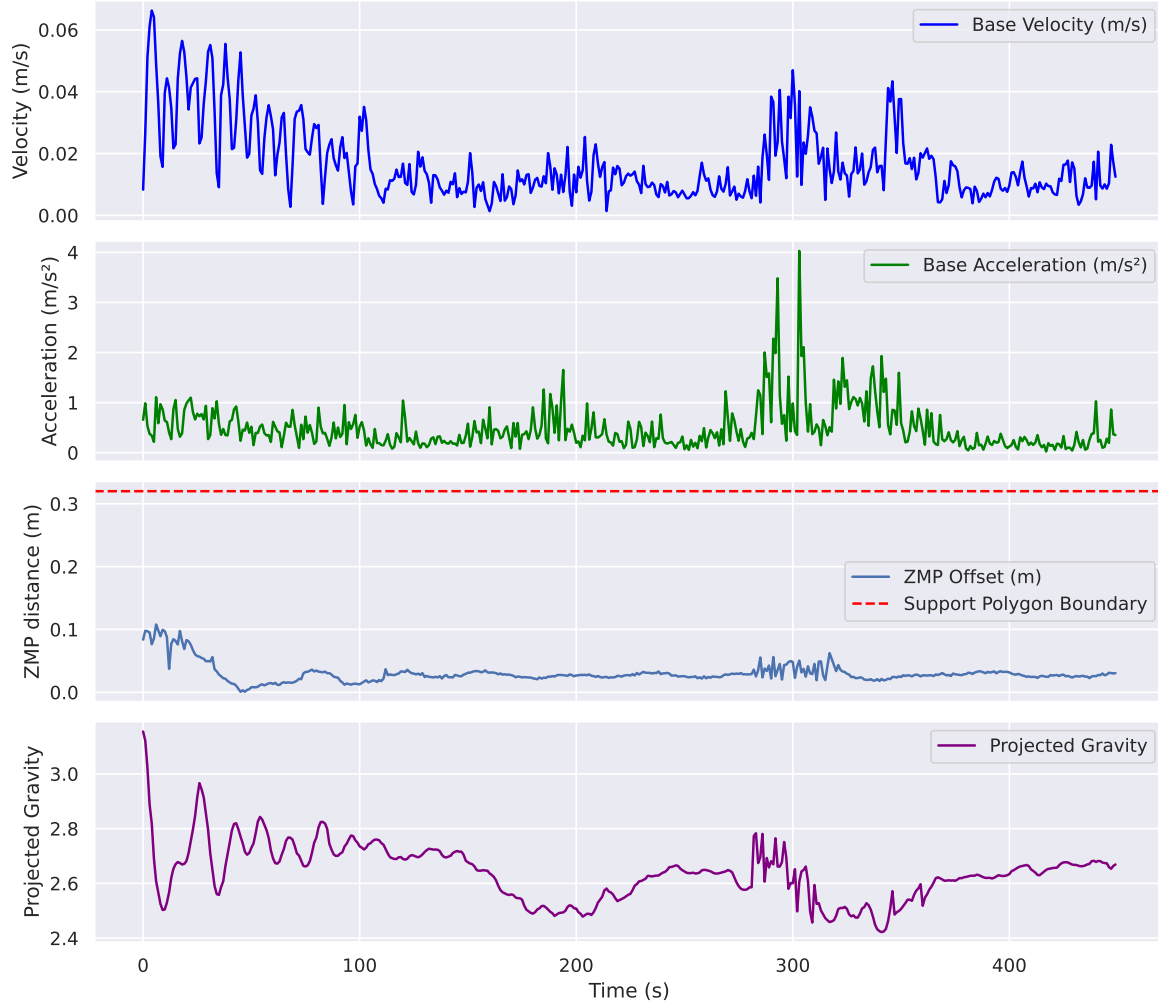


Figure R10: **Momentum, ZMP and stability.** The figure shows the measured floating base velocity and acceleration during a fast upper-body motion. The TOP smooths out large momentum variations, helping to keep the ZMP within the support region and maintain balance.

the TOP effectively suppresses sudden peaks in base velocity by slowing down the arm when the body momentum is excessive.

Thanks to the reviewers for pointing out the problems. Due to the lack of space, we will not include this in revised paper, but will add the relationship between momentum variation and loss of balance will be added into **APPENDIX.I** and the ZMP explanation to **APPENDIX.H**.

Point 2.6

Computational overhead remains unquantified. Explicit reporting of the inference complexity for both the VAE and TOP policy is required. Inclusion of real-time performance metrics (e.g., inference latency, computational resource utilization) is necessary.

Thanks for the reviewer’s comment. To address your concern, we have explicitly measured and reported the inference performance of both the VAE and the TOP policy. Table [R4](#) summarizes the average latency, device used, and computational resource utilization for each component. The combined inference time for VAE encoding, RL policy and TOP decision-making is approximately **5.0 ms**, demonstrating that our system is suitable for real-time control scenarios. Therefore, we set our framework as 100Hz.

In actual deployment, we convert our trained `.pt` models to `.ONNX` format for efficient runtime inference. Since our networks are composed of relatively lightweight architectures without complex modules such as Transformers, the inference is fast and stable across platforms.

We sincerely appreciate the reviewer’s suggestion. Due to the lack of space, we will incorporate the inference performance and hardware details into **Appendix.E**.

Point 2.7

Quantitative performance metrics for the tasks depicted in Figure 8 are absent. Supplementing results with task-specific measures (e.g., grasping success rate, task completion time, end-effector pose error) is crucial.

Table R4: Inference Performance of Our Methods in Difference Devices.

Module	Latency (ms)	Device	Resource Utilization
VAE Encoder	1.0121	RTX 4090 GPU	GPU Utilization: 4%
RL policy	1.7414	RTX 4090 GPU	GPU Utilization: 5%
TOP Policy	1.8623	RTX 4090 GPU	GPU Utilization: 8%
Total Inference	4.6158	RTX 4090 GPU	Memory Usage: 4.410 GB
VAE Encoder	1.4621	PICO-TGU4	CPU Utilization: 6%
RL policy	1.6925	PICO-TGU4	CPU Utilization: 7%
TOP Policy	1.7267	PICO-TGU4	CPU Utilization: 7%
Total Inference	4.8813	PICO-TGU4	Memory Usage: 5.4339 GB

Thanks for the reviewer’s comment. We agree that including quantitative performance metrics is important to complement the qualitative demonstrations in Figure 8. In response, we have added a new table (Table R5) that reports task-specific evaluation results for each manipulation scenario shown in Figure 8, including:

- **Success rate** (%)
- **Task completion time** (s)
- **End-effector pose error** (mm)

To evaluate the performance of task-specific measures, we repeat our real world experiment and measure the task-specific results to demonstrate the performance of our method. The upper-body joints are controlled by PD controller, and the lower-body joints are controlled by RL policy, which outputs the actions and transfer actions to real torques.

These results provide a more rigorous validation of the proposed method’s effectiveness across diverse manipulation tasks. As shown in Table R5, our framework consistently achieves high success rates and acceptable completion times with reasonable end-effector (EE) pose errors, demonstrating its practical viability in real-world conditions.

For more challenging tasks, like take book or carry box, which demands both precise upper-body control and robust whole-body balance due to the manual operation and shifting load, the slightly higher pose error can be attributed to residual vibration and tracking limitations of the physical robot. Notably, our upper-body motions are tracked by a PD controller in the

Table R5: Quantitative Evaluation of Tasks

Task	Success rate (%)	Completion Time (s)	EE Pose Error (mm)
Grab cup	90%	8.2	23.5
Take water	100%	10.9	10.2
Open oven	95%	16.2	22.9
Take book	85%	13.3	31.8
Carry box	85%	14.7	26.1

real system, which inherently introduces minor pose deviations due to unmodeled dynamics and delays.

Nevertheless, the success rate remains competitive (**85%**), showing that the framework can handle load perturbations and task-specific constraints without failure. This demonstrates the practicality and robustness of our method under real-world uncertainties.

Specifically, we plot the results, as shown in the Fig R11, how more complex tasks (which take longer to complete) can lead to slightly larger pose errors due to increased upper-body motion and balance demands.

Due to the lack of space, the corresponding figure will be added in the **Appendix.D** of our manuscript to strengthen the quantitative analysis of task-specific performance.

Point 2.8

Table IV reports a substantial end-effector displacement error (33.7 mm in X-direction) for Task T, yet provides no causal analysis. Experiments to decompose the sources of this error are warranted.

Thank you for the reviewer’s insightful comment. We apologize for not expressing clearly in Table IV, as the review says, it should provides the analysis for the source of this error. In fact, we did measure and record both the robot base motion and the end-effector pose simultaneously during real-world experiments.

However, this was not clearly presented in the original manuscript, which may have caused confusion. As illustrated in Figure R12, our motion capture system tracks both the base link

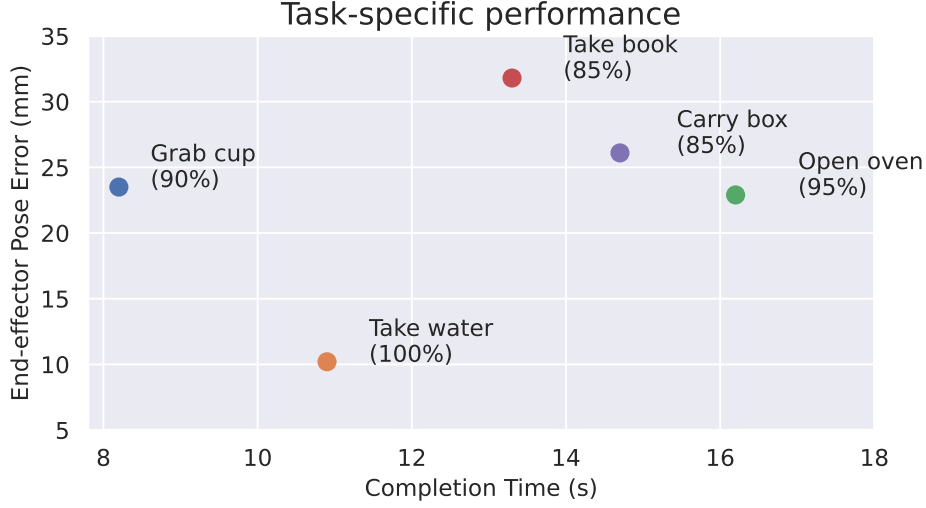


Figure R11: Comparison between different manipulation tasks, where each point represents a single task annotated by its success rate. The closer the data points are to the lower left corner, the lower the end-effector pose error and the shorter the task completion time, indicating better time efficiency and precision. This trend demonstrates how our method maintains high success rates while balancing precision and time efficiency across tasks of varying complexity.

and the upper-body end-effector in real time and the red circles indicate the position tracked with the motion capture device.

Table R6 summarizes the measured pose errors for the base and the end-effector. It is evident that the drift of the base (~ 30 mm) accounts for the majority of the total end-effector pose drift (~ 33 mm). To further clarify this, Figure R13 shows a decomposition plot of the base drift, the end-effector pose error, and the residual other error. The latter remains consistently under 5 mm, demonstrating that the dominant source of the displacement is the floating base oscillation rather than inaccuracies in joint-level control.

Table R6: Tracking Error Data Recorded by Motion Capture Device

Type	x(mm)	y(mm)	z(mm)	roll(rad)	pitch(rad)	yaw(rad)
Base Error	29.232	2.4846	1.1698	0.0163	0.0119	0.0032
End-effector Error	33.735	3.0494	1.4913	0.0502	0.0127	0.0103

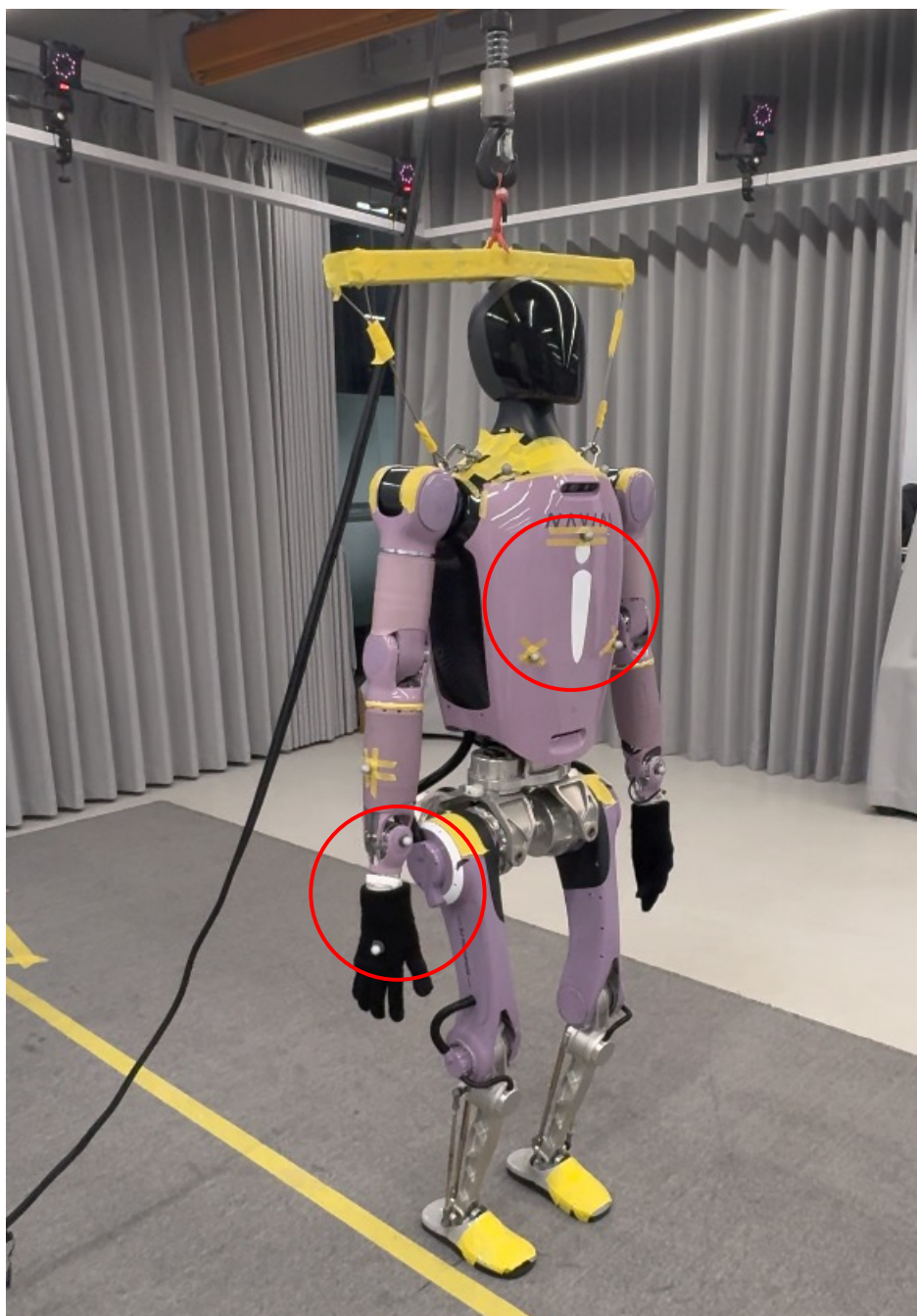


Figure R12: **Motion Capture Details.** The red circles indicate the position tracked with the motion capture device.

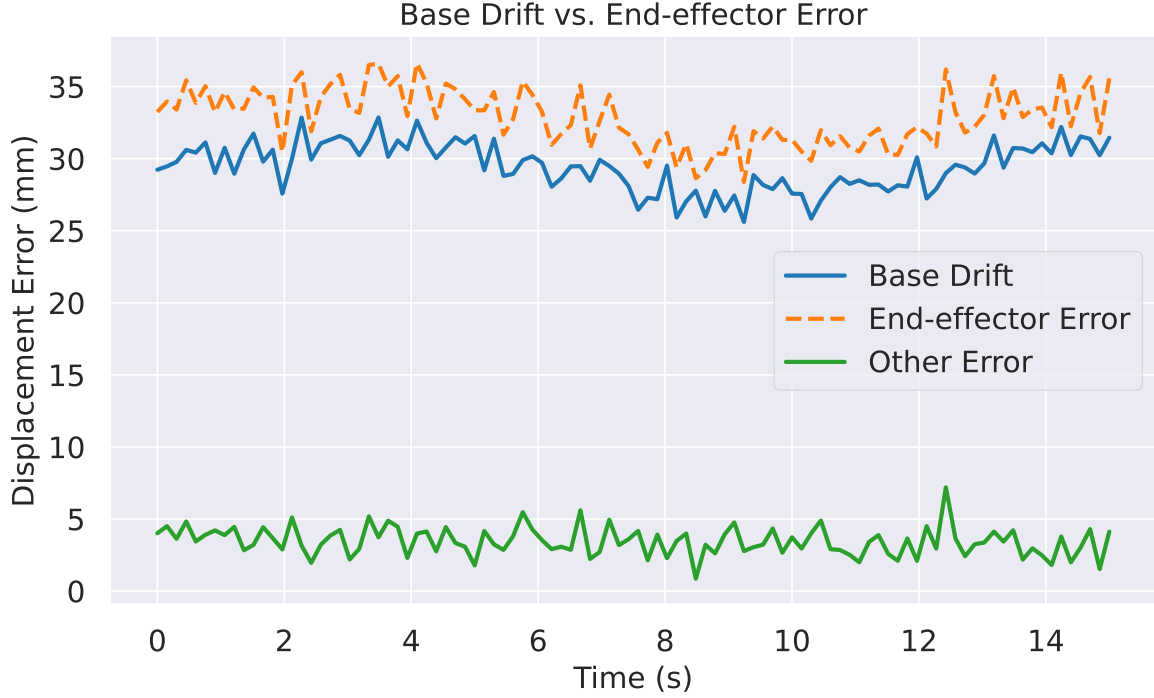


Figure R13: **Decomposition of end-effector error.** The plot shows that the majority of the end-effector pose drift stems from the base trajectory drift (blue curve), while the other error (green curve) remains very low (<5 mm). This demonstrates that improving base stability directly enhances the final end-effector precision.

This analysis confirms that our time optimization policy (TOP) effectively reduces base motion disturbance, which is the main factor limiting final end-effector precision. For transparency, we will update our revised manuscript (TABLE IV) to clarify this problem.

Point 2.9

The claim of "first-time optimization of timestamps" lacks differentiation from classical temporal optimization techniques like time scaling in Model Predictive Control (MPC). A comparative table within the Related Work section, explicitly contrasting TOP and MPC approaches, is recommended to highlight novel contributions.

Thank you for this valuable comment. We appreciate the reviewer's insight regarding

the similarity to classical time scaling methods, and agree that clarifying the differences strengthens our claim.

To address this, we have added explicit discussion and a comparative table, clearly contrasting our Time Optimization Policy (TOP) with traditional MPC-based optimization.

Our method differs from classical time scaling in the following key aspects:

- **Integration with Learning-Based Motion Priors:** Unlike MPC, which relies on analytic kinematics and dynamics models with predefined rule-based time-scaling laws, our TOP directly optimizes the timestamps jointly with a learned motion prior (VAE) and RL policy.
- **Adaptive Time Optimization:** Our timestamps are co-adapted with vary upper-body motion priors and current robot states, whereas classical time scaling mostly optimizes scalar speed factors under simple constraints.
- **End-to-End Training:** Our approach is trained end-to-end with a learnable objective that balances precision, stability, and time efficiency, while MPC needs to carefully design constraints, additional high-frequency controllers need to be connected, like whole-body controller.

These differences demonstrate that our contribution goes beyond classical temporal optimization technique. We have added the following comparison table (Table R7) to clarify the differences between our method and other methods:

In summary, compared with traditional MPC time scaling or other decoupled methods, our proposed Time Optimization Policy (TOP) provides a more effective and scalable solution for high-DoF humanoid manipulation. Unlike classical MPC time scaling that applies a global scalar speed factor, TOP learns adaptive timestamps for each frame jointly with the VAE motion prior, enabling fine-grained adjustment of complex arm trajectories. This per-frame timing flexibility allows TOP to better handle dynamic interactions, payload variations, and unexpected disturbances in real-world tasks.

To further clarify the efficiency of our method compared to MPC methods, we implement a conventional non-linear Model Predictive Control (NMPC) [16] combined with Whole-Body Control (WBC).

Table R7: Comparison with representative methods.

Aspect	TOP (Ours)	MPC Time Scaling	Exbody [10]	Mobile-Television [9]
Time Optimization	Learnable timestamps	Global time scaling factor	Not supported	Not supported
Control Type	Decoupled: upper-body PD + lower-body RL	Whole-body MPC+WBC	Whole-body RL	Decoupled: upper-body motion prior + lower-body RL
Adaptivity	Jointly adaptive to motions and base state	Limited by pre-defined contact/constraints	Adaptive to offline motion dataset only	Adaptive to base state, not motions
Computational Complexity	Low (motion prior + RL + time optimization)	High (online SQP solve)	Low (feedforward RL policy)	Low (motion prior + RL)

NMPC solves for a centroidal trajectory under a fixed time schedule with a predefined contact sequence. To ensure real-time feasibility, the continuous optimal control problem is discretized via direct multiple shooting, and solved using a Sequential Quadratic Programming (SQP) method with real-time iteration. The joint pose references come from the motions in GRAB[17], ensuring consistency with the desired task-level motion. The continue nonlinear MPC problem formulation is following:

$$\left\{ \begin{array}{l} \min_{\mathbf{u}(\cdot)} \Phi(\mathbf{x}(T)) + \int_0^T \mathbf{l}(\mathbf{x}(t), \mathbf{u}(t), t) dt \\ \text{subject to } \dot{\mathbf{x}} = f(\mathbf{x}(t), \mathbf{u}(t), t) \\ \mathbf{g}_{eq}^1(\mathbf{x}(t), \mathbf{u}(t), t) = 0 \\ \mathbf{g}_{eq}^2(\mathbf{x}(t), t) = 0 \\ \mathbf{h}_{in}(\mathbf{x}(t), \mathbf{u}(t), t) \geq 0 \\ \mathbf{x}(0) = \mathbf{x}_0 \end{array} \right. \quad (13)$$

where $\mathbf{x}(t)$ and $\mathbf{u}(t)$ are the state and control input at time t . $\mathbf{g}_{eq}^1$ represents equality state-input-related constraints, and $\mathbf{g}_{eq}^2$ represents equality constraints only related to state. $\mathbf{h}_{in}$ includes state constraints and state-input-related constraints, such as joint position and ve-

locity limits, friction cone constraints, etc. $\mathbf{x}_0$ is the current measured state of robot.

An optimization-based whole-body controller helps to map the optimized trajectory obtained from the nonlinear MPC to the humanoid robot’s joint torque, which considers the full nonlinear rigid body dynamics. While formulating the constrained hierarchical optimization problem, in our baselines we instead use a single QP to construct the constrained optimization problem, in which the low-priority tasks are embedded in the cost function, and high-priority tasks are treated as constraints. The list of tasks is shown in the Table R8. The first two blocks of tasks prevent the robot from violating the equality and inequality constraints of torque, force, and joint configuration. And the least squares tasks guide the robot to track the foot motion with a higher weight than the base reference motion, which prevents the robot from taking advantage of leg inertia to track the base motion in underactuated directions.

Table R8: WHOLE-BODY CONTROL TASKS

Type	Task
=	Floating base equations of motion
	No Motion at the contact points
$\geq$	Torque limits
	Wrench Friction cone
$Q_i^2 \cdot ^2$	Base Linear and Angular Tracking($Q_i = 1.0$)
	Joint Position Tracking($Q_i = 100.0$)
	Contact Force Tracking($Q_i = 0.1$)

One of the motions executed by NMPC+WBC are shown in the Figure R14.

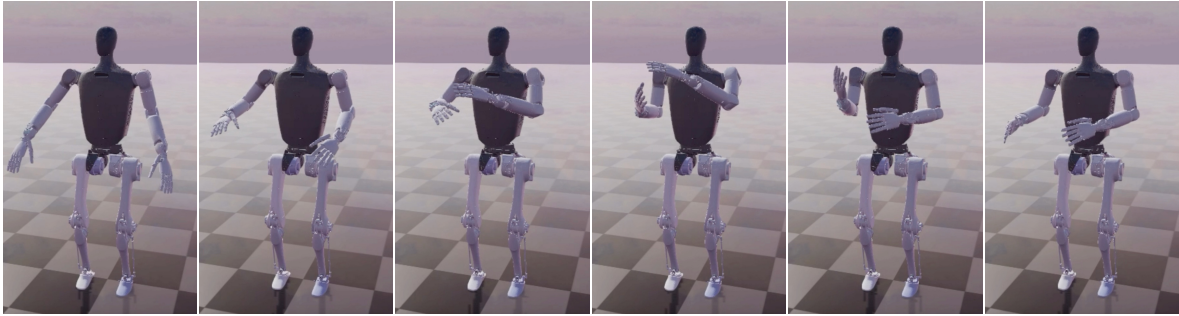


Figure R14: **Screenshots of Motions Executed by NMPC+WBC.** We realize the NMPC+WBC in raisim[18] simulator.

Table R9: Comparative results. We execute more than 10000 motion clips from dataset $\mathcal{M}$ and unseen dataset $\mathcal{T}$, and report their time cost, success rate, mean metrics. It is worth noting that the output of NMPC+WBC[16] is different from the actions of TOP outputs, so we only use the projected gravity and acceleration to evaluate the NMPC+WBC, and we slow down the target motions of NMPC for better comparison with our TOP method.

Method	Time Cost ↓	Success Rate ↑	$\mathbf{E}_{jpe}^{upper} \downarrow$	$\mathbf{E}_{eepe}^{upper} \downarrow$	$\mathbf{E}_{eeoe}^{upper} \downarrow$	$\mathbf{E}_g \downarrow$	$\mathbf{E}_{acc}^{lower} \downarrow$	$\mathbf{E}_{action}^{lower} \downarrow$	$\mathbf{E}_{acc}^{upper} \downarrow$	$\mathbf{E}_{action}^{upper} \downarrow$
Comparative Results										
Fixed Root (reference)	15.0s	100.0%	0.0130	0.0164	0.0594	1.000	-	-	-	-
NMPC+WBC[16]	35.0s	89.60%	0.0278	0.0437	0.0857	2.938	14.98	-	11.47	-
Ours (TOP)	40.5s	95.30%	0.0269	0.0270	0.0827	2.729	13.45	0.8592	10.41	1.601

According to the Table R9, We observe that our proposed TOP consistently achieves lower joint and end-effector position errors compared to the NMPC+WBC baseline, with reductions of approximately 3.2% and 38.2% respectively. This indicates significantly improved pose tracking performance, which is critical for manipulation tasks under dynamic conditions. Moreover, the gravity projection error E_g and lower-body acceleration error E_{acc}^{lower} are also reduced, demonstrating that our method better maintains base stability when interacting with varying external forces or loads. The success rate is increased by 5.7 percentage points, which validates the practical reliability of our planner under diverse scenarios. The only compromise is that our strategy needs to slow down our motions more, so we will lose time efficiency. In short, our TOP achieves better pose tracking and base stability than the NMPC baseline and replay method, while successfully completing the tasks under varying durations and loads.

In short, the comparative results show that our method achieves higher task success rates and more stable floating-base behavior while maintaining comparable or lower computational cost. These strengths make our TOP framework especially suitable for agile, high-precision whole-body manipulation scenarios that require balancing expressive upper-body motions and robust dynamic stability.

We sincerely thank the reviewer for highlighting this important point. We fully agree that the table of comparisons strengthen our contributions. However, due to strict page limits, we were unable to include the complete comparative table in the Related Work section of the main manuscript, but we will add NMPC+WBC method as one of our baseline in our

manuscript.

Point 2.10

The ablative study condition "Ours w/o motion prior" ambiguously defines whether the VAE architecture is entirely removed or merely the latent code utilization is disabled. Clarification regarding the specific implementation and its impact on the controller's functionality is imperative to avoid misinterpretation.

Thank you for pointing out this ambiguity. We apologize for the unclear description of the "Ours w/o motion prior" ablation condition.

To clarify, in this setting, the VAE architecture is **completely removed**, and no latent code is generated or utilized. Both of the decoupling RL policy and TOP policy receive only the current frame of the motion as input, without any motion prior guidance from the VAE. That means the policies are only conditioned on the single motion frame, not the latent code and motion frame. Similar to the [6], [9], we observe that while the single motion frame already consistently improves tracking performance, our latent code combined with single motion frame clearly outperforms, especially for the unseen motions. This design allows us to directly evaluate the contribution of the motion prior module to overall performance. The removal of the VAE in this ablation isolates the effect of future-aware latent representation, showing how much benefit it provides in enhancing motion stability and accuracy.

We have revised the manuscript to clarify this definition and prevent any misinterpretation.

In EXPERIMENT.C:

- ***Ours w/o motion priors:** This baseline uses TOP method but without the motion priors. The VAE module is entirely removed and the RL and TOP policies only based on the current motion state without any latent code input guidance.*

Point 2.11

The selection of baseline methods is limited. Comparisons with state-of-the-art approaches such as OmniH2O[21] (real-time whole-body control) or Mobile-Television[8] (similar motion prior utilization) are omitted, undermining the validity of the comparative conclusions.

Thank you for pointing out this important gap in our baseline selection. We fully agree that comparisons with strong state-of-the-art methods are essential for a convincing evaluation. In our revised version, we have added extensive experiments comparing our TOP with several advanced whole-body control frameworks, including OmniH2O[19] and Mobile-Television[9].

Table R10: Comparative results. We execute more than 10000 motion clips from dataset $\mathcal{M}$ and unseen dataset $\mathcal{T}$, and report their time cost, success rate, mean metrics. Mean Metrics are only calculated based on the data of the robot standing in place, and the data of robot taking steps and falls will not be included in the calculation.

Method	Time Cost ↓	Success Rate ↑	$\mathbf{E}_{jpe}^{upper} ↓$	$\mathbf{E}_{eepe}^{upper} ↓$	$\mathbf{E}_{eeoe}^{upper} ↓$	$\mathbf{E}_g ↓$	$\mathbf{E}_{acc}^{lower} ↓$	$\mathbf{E}_{action}^{lower} ↓$	$\mathbf{E}_{acc}^{upper} ↓$	$\mathbf{E}_{action}^{upper} ↓$
Comparative Results										
Fixed Root (reference)	15.0s	100.0%	0.0130	0.0164	0.0594	1.000	-	-	-	-
Exbody[10]	15.0s	92.46%	0.0376	0.0741	0.0923	3.432	21.65	0.7323	24.26	2.976
OmniH2O[19]	15.0s	94.08%	0.0361	0.0506	0.0914	3.267	19.86	0.6972	18.49	1.680
Mobile-Television [9]	15.0s	85.79%	0.0354	0.0513	0.0923	3.831	15.43	0.8775	11.67	1.677
NMPC+WBC[16]	35.0s	89.60%	0.0278	0.0437	0.0857	2.938	14.98	-	11.47	-
Ours (TOP)	40.5s	95.30%	0.0269	0.0270	0.0827	2.729	13.45	0.8592	10.41	1.601

Table R10 presents the detailed quantitative results across key metrics: time cost, success rate, upper-body tracking errors, base stability (projected gravity), and lower-body compensation performance.

Our results show that while OmniH2O and Mobile-Television achieve competitive performance in standard scenarios, our approach significantly improves precision (joint and end-effector errors), base stability (lower projected gravity), and whole-body coordination, although with a moderate increase in time cost due to the adaptive time scaling. These results validate that the proposed method robustly outperforms strong modern baselines in both motion accuracy and base balance, supporting the effectiveness of the time optimization

policy.

Once again, we sincerely thank the reviewer for this constructive suggestion, which helped us substantially improve the rigor and completeness of our experimental analysis. We will revise this in the experiment sections to clarify the efficiency of our TOP method.

Point 2.12

The VAE model only reconstructs the motion based on joint position/velocity data (Equation 1), without considering dynamic constraints (such as joint torque upper limits, inertia matrix). When the upper limb load changes, such as when grasping heavy objects, can pure kinematic reconstruction guarantee the physical feasibility of the action? It is suggested to conduct load adaptability experiments to prove the robustness.

Thank you for this insightful comment. We agree with the reviewer that pure kinematic motion generation may risk producing infeasible actions if downstream dynamics are ignored. Indeed, our VAE model is focused on reconstructing motion trajectories from kinematic data (joint positions and velocities), without directly encoding dynamic constraints such as joint torque limits or inertia variations.

To address this, we clarify that while the VAE is designed to learn a compact latent representation of feasible motion trajectories based purely on kinematic signals, the final execution is governed by a reinforcement learning (RL) policy that is trained with extensive *domain randomization*. Specifically, the dynamics randomization listed in Table R11 includes variations in payload mass, link inertia, and motor parameters. This ensures that the learned control policy remains robust when the upper body load changes or when heavy objects are grasped.

To empirically verify this point, we conducted additional load-adaptability experiments. The robot performed the same manipulation task while grasping objects of increasing mass (0.5 kg, 1 kg, and 3 kg). As shown in Table R12, the success rate and end-effector pose error remain consistent across varying payloads, while only a slight increase in projected gravity is observed. These results confirm that our domain-randomized RL policy robustly compensates for dynamic variations caused by payload changes, even though the VAE itself

Table R11: Domain Randomization Settings

Term	Value
Observation	
Joint position noise	$\mathcal{U}(-0.1, 0.1)$ rad
Joint velocity noise	$\mathcal{U}(-1.0, 1.0)$ rad/s
Angular velocity noise	$\mathcal{U}(-0.1, 0.1)$ rad/s
Projected gravity noise	$\mathcal{U}(-0.05, 0.05)$
Dynamics	
Base COM displacement	$\mathcal{U}(-0.05, 0.05)$ m
Friction factor	$\mathcal{U}(0.35, 1.25)$
Link Inertia	$\mathcal{U}(0.8, 1.2) \times \text{default}$ kg $\cdot$ m ²
Added base mass	$\mathcal{U}(-5.0, 5.0) + \text{default}$ kg
Added link mass	$\mathcal{U}(0.8, 1.2) \times \text{default}$ kg
Added hand mass	$\mathcal{U}(-0.5, 1.5) + \text{default}$ kg
Payload mass	$\mathcal{U}(-1.0, 3.5) + \text{default}$ kg
Motor P gain	$\mathcal{U}(0.8, 1.2) \times \text{default}$
Motor D gain	$\mathcal{U}(0.8, 1.2) \times \text{default}$
Motor strength	$\mathcal{U}(0.9, 1.1) \times \text{default}$ N $\cdot$ m
Motor Damping	$\mathcal{U}(0.3, 4.0)$ N $\cdot$ s
Motor Delay	$\mathcal{U}(0, 10)$ ms
Episode	
Push robots	interval = 5s, $v_{xy} = 1.0\text{m/s}$, $\omega = 0.4\text{rad/s}$
Disturbance robots	interval = 8s, $\mathcal{U}(-30.0, 30.0)$ N $\cdot$ m
Initial robot base position xy	$\mathcal{U}(-1.0, 1.0)$ m
Initial robot base velocity	$\mathcal{U}(-0.1, 0.1)$ m/s

does not explicitly model dynamics.

Table R12: Load Adaptability Test: Task performance under different payloads

Payload	Success Rate $\uparrow$	EE Pose Error $\downarrow$	Projected Gravity $\downarrow$
0.5 kg	95.2%	28.5 mm	2.731
1.0 kg	94.4%	30.1 mm	2.857
3.0 kg	92.7%	31.8 mm	3.142

These findings clearly demonstrate that our framework achieves physically feasible and stable behaviors under varying load conditions. The authors thank the reviewer for highlighting this important point, which further strengthens the completeness and practicality of our framework, and we have incorporated this new experiments and discussion into **Appendix.C**.

Point 2.13

The abstract asserts the model ”ensures balance, precision, and time efficiency simultaneously.” However, Table III indicates the TOP execution time (40.5s) is 270% of the baseline method (15.0s), directly contradicting the claimed time efficiency.

Thank you for this valuable comment. We apologize for not making this point sufficiently clear in the manuscript. Indeed, our claim of **time efficiency** refers not to achieving the shortest execution time alone, but rather to **optimally balancing** time cost with stability (balance) and accuracy (precision) to ensure robust whole-body standing manipulation. This balance or trade-off is crucial for humanoid robots, where blindly minimizing execution time may easily lead to loss of balance or large pose errors.

As shown in Table R13 (excerpted from the main manuscript), simply executing the motion at full speed without TOP (Ours w/o TOP $\Delta t = 0.01s$) achieves the shortest time cost (15.0s). However, it results in a significantly lower task success rate (82.43%) due to frequent loss of balance, and its projected gravity E_g is almost 1.7 times higher than our proposed TOP, indicating poorer stability. Its end-effector tracking errors (E_{eepe}^{upper}) are also noticeably worse than those achieved by TOP.

Table R13: **Part of our experiments’ results from manuscript.**

Method	Time Cost ↓	Success Rate ↑	$E_{jpe}^{upper} ↓$	$E_{eepe}^{upper} ↓$	$E_{eeoe}^{upper} ↓$	$E_g ↓$	$E_{acc}^{lower} ↓$	$E_{action}^{lower} ↓$	$E_{acc}^{upper} ↓$	$E_{action}^{upper} ↓$
Ours (TOP)	40.5s	95.30%	0.0269	0.0270	0.0827	2.729	13.45	0.8592	10.41	1.601
Ours w/o TOP $\Delta t = 0.01s$	15.0s	82.43%	0.0301	0.0541	0.0962	4.438	16.87	0.9329	12.04	1.738
Ours w/o TOP $\Delta t = 0.03s$	45.0s	87.27%	0.0303	0.0434	0.0810	3.797	16.52	0.9473	11.26	1.647
Ours w/o TOP $\Delta t = 0.05s$	75.0s	92.41%	0.0299	0.0362	0.0813	3.573	15.79	0.9340	9.79	1.556

In contrast, our method with TOP automatically adjusts the timestamps to find a better balance: the motion is slightly slowed down compared to the raw baseline, but the success rate rises sharply to 95.3%, while both precision and stability improve significantly. Even compared with manually tuned baselines (Ours w/o TOP $\Delta t = 0.03s, 0.05s$), our method shows a better overall balance by maintaining the lowest joint and end-effector pose errors at moderate execution time.

This demonstrates that **time efficiency** in our context does not mean minimizing time at the expense of success rate and pose quality. Instead, it means achieving the **fastest possible** execution **subject to strict constraints** on balance and precision — which is exactly what our learned TOP policy optimizes for.

We will revise this in the experiment sections to clarify this point and avoid misleading interpretations.

In EXPERIMENT.C:

*Although our methods take more time to complete motions compared to **Ours w/o TOP** $\Delta t = 0.01s$, but we improve both precision and stability significantly.*

Point 2.14

The assumption of decoupled control (PD for upper body, RL for lower body) is potentially oversimplified. The manuscript neglects discussion of dynamic coupling effects (e.g., momentum transfer delays) between the control modules. The significantly higher upper-body acceleration error observed for TOP compared to the baseline (Table III) further challenges the validity of this decoupled approach.

Thank you for this insightful comment regarding the dynamic coupling between the upper-body and lower-body modules. We agree that, in principle, the upper and lower body are inherently coupled through momentum transfer and base motion feedback. Ignoring this entirely would indeed risk infeasible or unstable motions.

In our design, however, the **decoupling is not rigid** but rather a *practical modular abstraction* for efficient learning and deployment:

- **Upper-body motion** is compressed by a VAE-based motion prior and tracked by a PD controller that produces desired joint torques to follow the reference trajectories with high precision.
- **Lower-body stabilization** is handled by a reinforcement learning (RL) policy that receives real-time observations of the floating base state, projected gravity, joint states, and the desired upper-body motion embedding (z_t) and motion clip (m_t).

Therefore, while the modules are trained in a decoupled manner, their execution forms a closed-loop whole-body control system. The RL policy implicitly compensates for momentum transfer from the upper-body by dynamically adjusting lower-body joint torques and contact forces to maintain balance. This design principle is consistent with established whole-body control frameworks[9] where manipulation and locomotion are hierarchically coordinated.

As shown in Table R14, the RL policy is explicitly encouraged to minimize base drift, linear/angular velocity deviations, and physically feasible whole-body motions:

- **Base linear velocity** and **base angular velocity** terms penalize excessive base oscillations.
- **Base orientation** cost stabilizes the base roll/pitch under external moments induced by upper-body motions.
- **Stand still** and **action penalty** terms further ensure smooth, coordinated lower-body actions.

We can observe that, this closed-loop design ensures that the lower-body dynamically absorbs momentum transfer delays from the upper-body PD tracker. Our TOP method deliberately slows down sudden upper-body changes to avoid destabilizing the base, which

Table R14: Key RL Reward Terms for Stabilization

Term	Expression	Weight
Base linear velocity (xy)	$\exp[-4(\mathbf{v}_{xy}^2)]$	+3.0
Base angular velocity (xy)	$\Sigma \omega_{xy}^2$	-0.1
Base orientation	$\Sigma \mathbf{g}_{xy}^2$	-1.5
Stand still pose	$\exp[-10\ \mathbf{q}_t^{\text{leg}} - \mathbf{q}^{\text{leg,ref}}\ _2]$	+1.0
Action rate penalty	$\ \mathbf{a}_t - \mathbf{a}_{t-1}\ _2^2$	-0.2
Action acceleration penalty	$\ \mathbf{a}_t + \mathbf{a}_{t-2} - 2\mathbf{a}_{t-1}\ _2^2$	-0.2

Table R15: Part of Comparative results and Ablation results.

Method	Time Cost ↓	Success Rate ↑	$\mathbf{E}_{\text{jpe}}^{\text{upper}} \downarrow$	$\mathbf{E}_{\text{eepe}}^{\text{upper}} \downarrow$	$\mathbf{E}_{\text{eeoe}}^{\text{upper}} \downarrow$	$\mathbf{E}_{\text{g}} \downarrow$	$\mathbf{E}_{\text{acc}}^{\text{lower}} \downarrow$	$\mathbf{E}_{\text{action}}^{\text{lower}} \downarrow$	$\mathbf{E}_{\text{acc}}^{\text{upper}} \downarrow$	$\mathbf{E}_{\text{action}}^{\text{upper}} \downarrow$
Comparative Results										
Fixed Root (reference)	15.0s	100.0%	0.0130	0.0164	0.0594	1.000	-	-	-	-
Exbody[10]	15.0s	92.46%	0.0376	0.0741	0.0923	3.432	21.65	0.7323	24.26	2.976
Ours (TOP)	40.5s	95.30%	0.0269	0.0270	0.0827	2.729	13.45	0.8592	10.41	1.601
Ablation Results										
Ours w/o TOP $\Delta t = 0.01s$	15.0s	82.43%	0.0301	0.0541	0.0962	4.438	16.87	0.9329	12.04	1.738
Ours w/o TOP $\Delta t = 0.03s$	45.0s	87.27%	0.0303	0.0434	0.0810	3.797	16.52	0.9473	11.26	1.647
Ours w/o TOP $\Delta t = 0.05s$	75.0s	92.41%	0.0299	0.0362	0.0813	3.573	15.79	0.9340	9.79	1.556

slightly sacrifices time efficiency but greatly improves base stability and overall task success. Compare to the baseline (Exbody[10]) that uses whole-body RL policy with the tracking rewards to control whole-body joints, our method can achieve precision, stability and time efficiency at the same time, proving the effectiveness of our decoupling controllers, as shown in Table R15.

As the reviewer correctly points out, our method (TOP) does exhibit a slightly higher upper-body acceleration error compared to the baseline (Ours w/o TOP, fixed $\Delta t = 0.05s$), which may appear to challenge the validity of our decoupled control design. We appreciate this observation and would like to clarify the underlying reason.

Our time optimization policy (TOP) explicitly adjusts the local execution speed of the upper-body trajectory in response to the robot’s dynamic state, with the upper-body goal during the execution of motions. This adaptive timing mechanism naturally leads to non-

uniform time scaling, which can locally slow down or speed up certain motion segments. As a result, the PD tracking loop, which follows the reference trajectory at variable speeds, may occasionally introduce mild phase lags or higher accelerations, especially for motion segments where the speed changes rapidly. This explains the slightly increased local acceleration error.

In contrast, the baseline variant applies a fixed time scaling ($\Delta t = 0.05s$) uniformly across the entire trajectory, which does not adapt to the real-time dynamic state. This static scaling can reduce local acceleration fluctuations but comes at the cost of ignoring the time efficiency.

Importantly, despite this slight increase in local acceleration tracking error, our results show that the end-effector pose error remains significantly lower and the overall task success rate improves. This demonstrates that our time optimization policy strikes a better balance between precision, whole-body stability and time efficiency.

Thanks to the reviewers for pointing out the problems. Due to the lack of space, we will not include this in revised paper, but will add the explanation to **APPENDIX.A**.

Point 2.15

The TOP algorithm introduces significant computational overhead through VAE encoding, RL policy inference, and interpolation calculations. Assessment and discussion of its real-time feasibility and computational efficiency are notably absent.

Thanks for the reviewer’s comment. To address your concern, we have explicitly measured and reported the inference performance of both the VAE and the TOP policy. Table [R16](#) summarizes the average latency, device used, and computational resource utilization for each component. The combined inference time for VAE encoding, RL policy and TOP decision-making is approximately **5.0 ms**, demonstrating that our system is suitable for real-time control scenarios. Therefore, we set our framework as 100Hz.

Additionally, the interpolation calculation in our framework is implemented as a simple linear interpolation between adjacent motion clips. This step involves only lightweight arithmetic operations and introduces negligible additional computational cost (less than 0.1 ms per cycle). Thus, the entire pipeline, including interpolation, remains highly efficient and well-suited for real-time humanoid control.

In actual deployment, we convert our trained `.pt` models to `.ONNX` format for efficient runtime inference. Since our networks are composed of relatively lightweight architectures without complex modules such as Transformers, the inference is fast and stable across platforms.

Table R16: Inference Performance

Module	Latency (ms)	Device	Resource Utilization
VAE Encoder	1.0121	RTX 4090 GPU	GPU Utilization: 4%
RL policy	1.7414	RTX 4090 GPU	GPU Utilization: 5%
TOP Policy	1.8623	RTX 4090 GPU	GPU Utilization: 8%
Total Inference	4.6158	RTX 4090 GPU	Memory Usage: 4.410 GB
VAE Encoder	1.4621	PICO-TGU4	CPU Utilization: 6%
RL policy	1.6925	PICO-TGU4	CPU Utilization: 7%
TOP Policy	1.7267	PICO-TGU4	CPU Utilization: 7%
Total Inference	4.8813	PICO-TGU4	Memory Usage: 5.4339 GB

We sincerely appreciate the reviewer’s suggestion. Due to the lack of space, we will incorporate the inference performance and hardware details into **Appendix.E**.

Point 2.16

Evident conflict exists within the TOP reward function (Table I), specifically between the objectives "Encourage small Δt " (weight 5.0) and " Δt Smoothness" (weight -0.1). The rationale underlying the selection of these specific weighting factors requires explicit justification.

Thank you for highlighting this point. We agree that the two objectives appear contradictory at first glance but they play complementary roles to balance time efficiency and motion stability.

Specifically, the **Encourage small Δt** term directly promotes shorter timestamps, pushing the policy to generate smaller Δt , encouraging robot to execute motions as fast as possible under the balance constraint. However, if only this term were used, the policy could produce

abrupt or irregular changes in Δt , when the robot become unstable or the observation changes, leading to physically jerky motions. Therefore, the Δt **Smoothness** term acts as a regularizer to penalize sudden fluctuations between consecutive timestamps, encouraging smoother transitions, as its weight is an order of magnitude smaller. The equations are following:

The “Encourage small Δt ” term rewards smaller timestamp intervals to achieve faster motion execution:

$$R_{\text{small } \Delta t} = w_1 \sum_{i=0}^N \exp \left(-\frac{(\Delta t_i)^2}{2\sigma^2} \right), \quad w_1 = 5.0. \quad (14)$$

The “ Δt Smoothness” term penalizes sudden jumps in Δt to ensure physically plausible transitions:

$$R_{\text{smoothness}} = -w_2 \sum_{i=0}^{N-1} (\|\Delta t_{i+1} - \Delta t_i\|), \quad w_2 = 0.1. \quad (15)$$

The weighting factors were determined empirically through grid search and extensive training trials. We found that setting the smoothness term to a small negative weight (-0.1) prevents overly aggressive compression of Δt that would otherwise destabilize the motion, while keeping its influence much weaker than the main efficiency-driving term (5.0). This balance ensures the system achieves **fast yet smooth** timestamp adaptation in practice. We run ablation experiments comparing no smoothness regularization ($w_2 = 0$) vs. the selected small value ($w_2 = 0.1$). As shown in Fig R17, the regularized version produces **more stable Δt trajectories** without significantly increasing total execution time.

Table R17: Ablation on Smoothness Weight w_2 , $\Delta t \in [0.01s, 0.1s]$

w_2	Avg. Total Time (s)	Std. of Δt Change
0.0	12.23	9.8×10^{-3}
0.1	13.39	8.8×10^{-4}
0.5	17.22	6.2×10^{-4}

We will also include a plot of the learned Δt sequence showing that the smoothness term effectively suppresses abrupt spikes. For clarity, we provide an example as following Fig R15:

Thanks to the reviewers for pointing out the problems. We hope that these explanations can help better illustrate the design of our rewards.

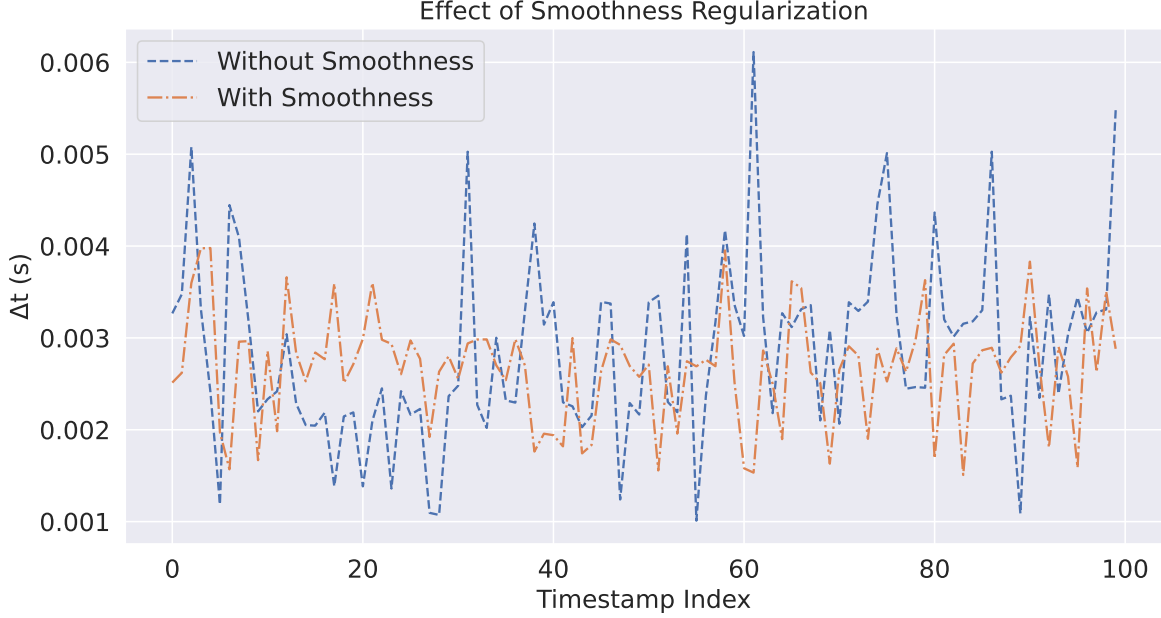


Figure R15: **Effect of Smoothness Regularization.** Without w_2 , Δt oscillates sharply; with smoothness, the sequence is stable.

Point 2.17

The action segmentation mechanism (Section III-D) lacks transparency. The decay coefficient "k" employed in the weighting strategy lacks empirical validation or optimization, potentially leading to excessive smoothing of high-frequency motions. A sensitivity analysis investigating the impact of "k" is recommended.

Thank you for the reviewer's comment. We apologize for the unclearness of our usage of k , which actually improve the action smoothness of TOP policy and avoid jerky discrete switching of timestamps. As the review says, we will provide a detailed analysis of the transparency of the action chunking mechanism and the role of the decay coefficient k .

In our method, the decay coefficient k determines a **weight vector** for aggregating recent actions when computing the final timestamp adjustment Δt . Specifically, given a history buffer of N frames, the weight for frame i is computed as:

$$w_i = \frac{\exp(-k \cdot i)}{\sum_{j=0}^N \exp(-k \cdot j)}, \quad i = 0, 1, \dots, N. \quad (16)$$

this means that more recent actions (smaller i) have higher weights, while earlier frames are

exponentially down-weighted. The weights are normalized to sum to 1:

$$\sum_{i=0}^N w_i = 1. \quad (17)$$

A smaller k leads to a flatter distribution, meaning older frames contribute more equally; a larger k makes the system more reactive to the most recent actions only. The action chunking[20] refers to a temporal smoothing mechanism that aggregates a sequence of predicted actions over a short horizon to produce the final Δt at each timestamp.

This design balances *temporal consistency* (smooth Δt) and *reactivity* (fast adaptation to sudden motion changes). For instance, with $N = 10$ frames, the weights are shown in the Fig R16, which clearly shows how the decay coefficient k adjusts the emphasis on recent vs. past actions during chunking. This demonstrates that the parameter is interpretable and tunable to balance responsiveness and smoothness in timestamp optimization.

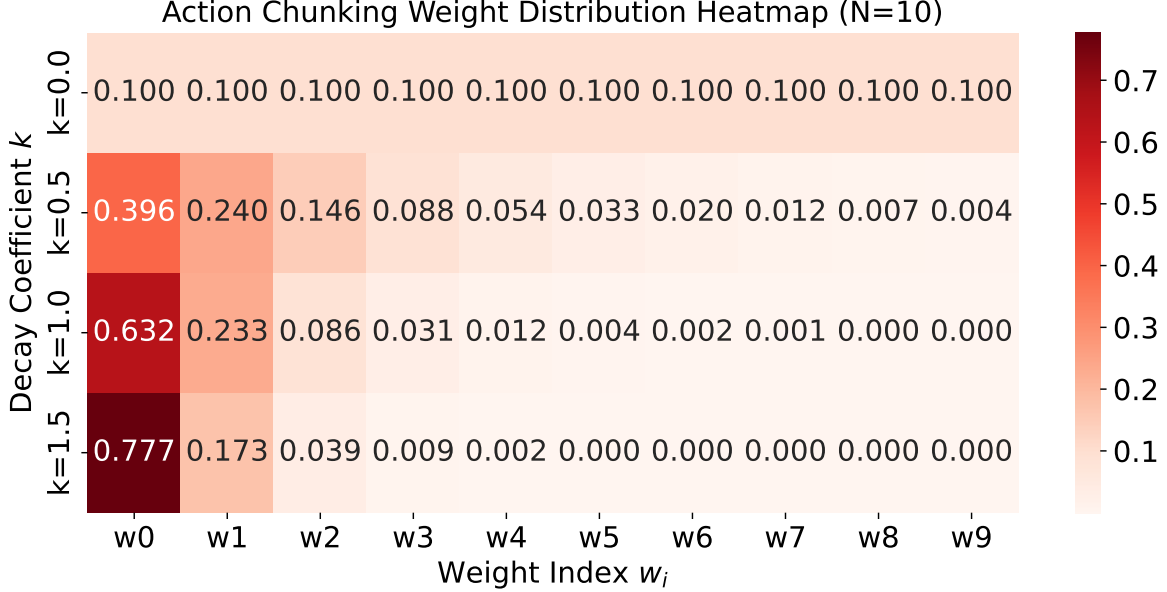


Figure R16: **Action Chunking Weights to Decay Coefficient k (N=10).** This heatmap illustrates how the decay coefficient k affects the normalized weighting vector w_i used for action chunking. Smaller k values result in a flatter, more uniform distribution (weighing all past actions equally), while larger k values cause weights to decay faster, emphasizing more recent frames. This directly influences the computed Δt for future timestamp prediction in our Time Optimization Policy (TOP).

We performed a sensitivity analysis by varying k in $\{0.0, 0.2, 0.5, 0.8, 1.5, 3.0\}$ and

evaluated its impact on the motion’s precision, the projected gravity metric for stability and the time cost for time efficiency.

Table R18: Sensitivity Analysis of Decay Coefficient k

k	Joint Error (rad) ↓	Projected Gravity ↓	Time Cost (s) ↓	Success Rate (%) ↑
0.0	0.0302	3.072	4.42	90.6%
0.2	0.0276	2.815	5.52	93.1%
0.5	0.0269	2.729	5.40	95.3%
0.8	0.0277	2.984	5.32	94.2%
1.5	0.0283	3.136	4.66	92.8%
3.0	0.0288	3.275	3.80	93.2%

The results in Table R18 demonstrate the impact of the decay coefficient k on motion tracking error, stability (projected gravity), execution time cost, and task success rate.

When k is very small ($k = 0.0$ and 0.2), the weight distribution across past actions becomes nearly uniform, which results in excessive smoothing of the predicted timestamps. This slightly leads the policy to overly rely on future actions and reduces the responsiveness to fast-changing observations.

As k increases to moderate values ($k = 0.5$), the chunking mechanism emphasizes more recent frames, which helps the controller better adapt to real-time changes without over-smoothing. Specifically, $k = 0.5$ achieves the lowest joint error (0.0269 rad) and the lowest projected gravity (2.729), while maintaining a reasonable time cost (5.40 s) and the highest observed success rate (95.3%). These results show that a balanced weighting of past actions effectively enhances both tracking precision and stability.

When k becomes larger ($k = 0.8, 1.5, 3.0$), the weight quickly focuses almost exclusively on the most recent actions, reducing the smoothing effect. This slightly increases joint tracking error and projected gravity due to higher weight of most recent actions.

In summary, this ablation confirms that moderate values of k (around 0.5) yield the best balance between precision, stability, and time efficiency, and the curve are shown in the Fig R17. This empirical justification supports our choice of $k = 0.5$ as the default setting in the proposed method.

Thanks to the reviewers for pointing out the problems. Due to the lack of space, we will

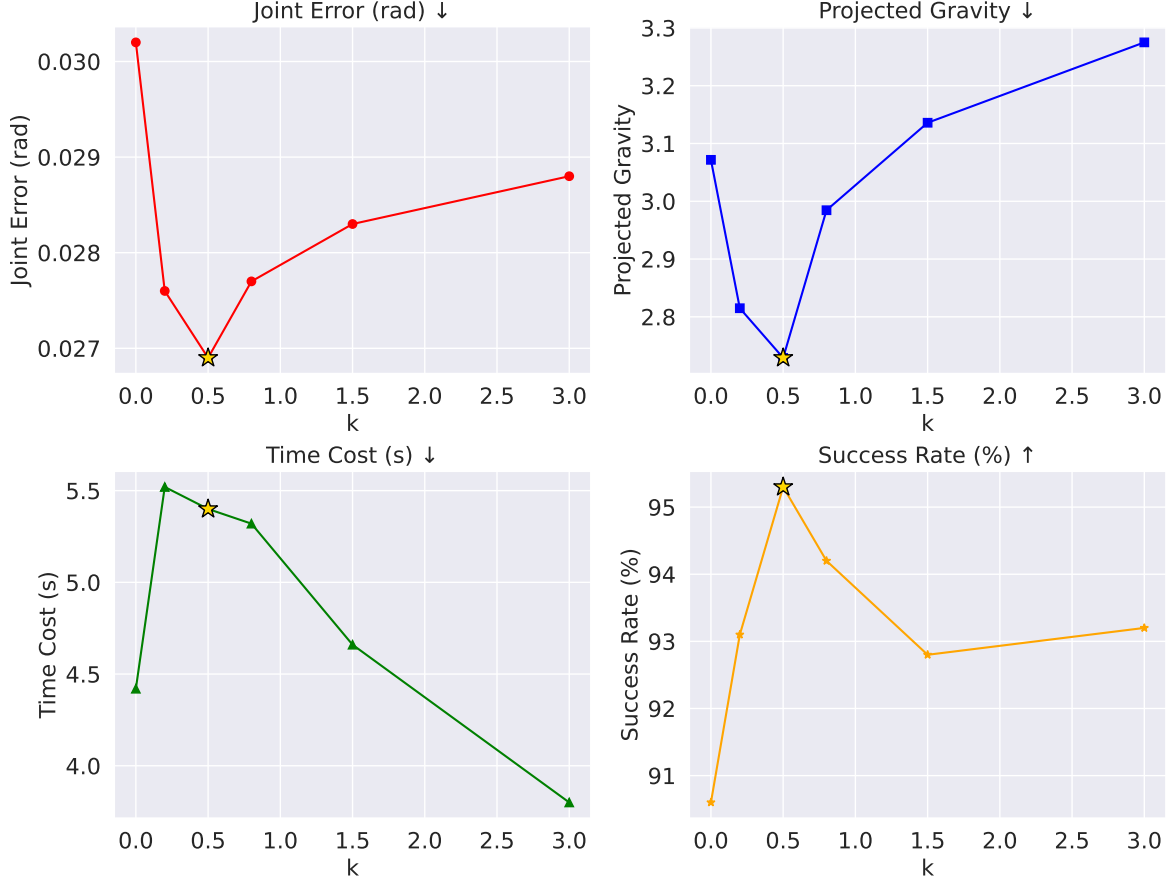


Figure R17: **Sensitivity Analysis of Decay Coefficient k .** The plots illustrate how varying k affects (a) Joint Error, (b) Projected Gravity, (c) Time Cost, and (d) Success Rate. The highlighted golden points indicate $k = 0.5$, which achieves the great balance with minimal joint error, stable gravity projection, acceptable time cost, and the highest task success rate.

not include this in revised paper, but will add the explanation to **APPENDIX.F**.

Point 2.18

The "Mean Metrics" presented in Table III omit standard deviations, precluding meaningful assessment of result significance and variability.

We appreciate the reviewer's suggestion to include standard deviations to strengthen the statistical rigor of our quantitative results. We agree that reporting mean values alone may not fully reflect the result variability or the consistency of the system under different motion clips.

To strengthen the statistical rigor of our evaluation, we include standard deviations for all key performance metrics in addition to the mean values. While the original Table III only reported mean values, we add a new table that provides standard deviations computed from more than 10000 motion clips.

Method	$\sigma_{\text{jpe}}^{\text{upper}} \downarrow$	$\sigma_{\text{eepe}}^{\text{upper}} \downarrow$	$\sigma_{\text{eeoe}}^{\text{upper}} \downarrow$	$\sigma_{\text{g}} \downarrow$	$\sigma_{\text{acc}}^{\text{lower}} \downarrow$	$\sigma_{\text{action}}^{\text{lower}} \downarrow$	$\sigma_{\text{acc}}^{\text{upper}} \downarrow$	$\sigma_{\text{action}}^{\text{upper}} \downarrow$
Comparative Results								
Fixed Root (reference)	0.0036	0.0025	0.0016	0.0	-	-	-	-
Exbody[10]	0.0245	0.0171	0.0275	0.820	4.74	0.3237	4.26	0.2275
OmniH2O[19]	0.0246	0.0113	0.0231	0.622	4.26	0.2241	3.39	0.2548
Mobile-Television[9]	0.0229	0.0141	0.0281	1.372	5.37	0.2238	2.94	0.1940
NMPC+WBC[16]	0.0108	0.0107	0.0157	0.638	4.12	-	1.21	-
Ours (TOP)	0.0082	0.0077	0.0074	0.477	3.68	0.1239	1.33	0.2056
Ablation Results								
Ours w/o TOP $\Delta t = 0.01s$	0.0202	0.0107	0.0378	1.962	6.89	0.2254	3.92	0.2609
Ours w/o TOP $\Delta t = 0.03s$	0.0154	0.0104	0.0279	1.369	5.30	0.2162	2.68	0.2362
Ours w/o TOP $\Delta t = 0.05s$	0.0141	0.0077	0.0155	0.923	4.04	0.1815	1.32	0.1843
Ours w/o motion prior	0.0121	0.0128	0.0093	0.789	3.89	0.1446	1.73	0.2177
Ours w/o acting chunking	0.0136	0.0102	0.0061	0.750	4.17	0.1223	1.36	0.2039

Table R19: Comparative results and Ablation results. We execute more than 10000 motion clips from dataset $\mathcal{M}$ and unseen dataset $\mathcal{T}$. σ represents the standard deviations of the data.

This extended reporting allows clearer insight into the result variability and consistency of each method. Due to space constraints, we present these standard deviations in a dedicated table rather than crowding the original table. As shown in Table R19, our proposed approach consistently achieves reasonable lower variance compared to strong baselines and ablations. This indicates that our method generalizes reliably to diverse motion clips and maintains stable standing performance without excessive spread in key measures such as joint error, end-effector error, and projected gravity, etc.

Thank you for pointing out this important shortcoming. We believe this additional breakdown helps clarify the practical robustness and reliability of our system. Due to the lack of space, we will not add this table into our manuscript.

Point 2.19

Defining task success solely as "no falling" (Table III) fails to capture the core precision requirements of the manipulation tasks. For instance, quantitative performance measures for tasks illustrated in Figure 8 (e.g., cup displacement error, oven handle rotation angle) are unreported.

Thank you for the reviewer’s constructive comment. We fully agree that defining task success purely as **no falling** is insufficient to rigorously evaluate the precision of manipulation tasks. In response to this and the related comment (**quantitative performance metrics for the tasks depicted in Figure 8 are absent**), we have added comprehensive task-specific quantitative results for each manipulation scenario, including grasping success rate, task completion time, and end-effector pose error. These results are summarized in Table R20 and demonstrate that our framework achieves high precision and task success rates across a range of manipulation tasks, while maintaining whole-body balance.

To generate these results, we repeated our real-world experiments under the same conditions. The robot’s upper-body joints are controlled by a PD controller, while the lower-body balance is maintained by our RL policy, which outputs torque commands in real time.

Table R20: Task-level Metrics for Diverse Manipulation Tasks

Task	Success Rate (%)	Completion Time (s)	EE Pose Error (mm)
Grab cup	90%	8.2	23.5
Take water	100%	10.9	10.2
Open oven	95%	16.2	22.9
Take book	85%	13.3	31.8
Carry box	85%	14.7	26.1

These results provide a more rigorous validation of the proposed method’s effectiveness across diverse manipulation tasks. As shown in Table R20, our framework consistently achieves high success rates and acceptable completion times with reasonable end-effector (EE) pose errors, demonstrating its practical viability in real-world conditions.

For more challenging tasks, like *Take book* or *Carry box*, which demand precise upper-body

motion and robust whole-body balance due to manual operation and load shifts, the slightly higher pose errors are expected and primarily due to residual vibrations and the limitations of the physical hardware. Notably, the upper-body PD controller inherently introduces small tracking errors due to unmodeled compliance and actuator delay, yet our overall success rate remains competitive (**85%**) for these harder tasks.

To better illustrate this, Figure R18 visualizes the trade-off between pose precision and execution time for all tasks. Data points closer to the lower left corner indicate higher task efficiency and precision, demonstrating how our framework maintains high success rates while balancing the competing demands of stability, accuracy, and time efficiency.

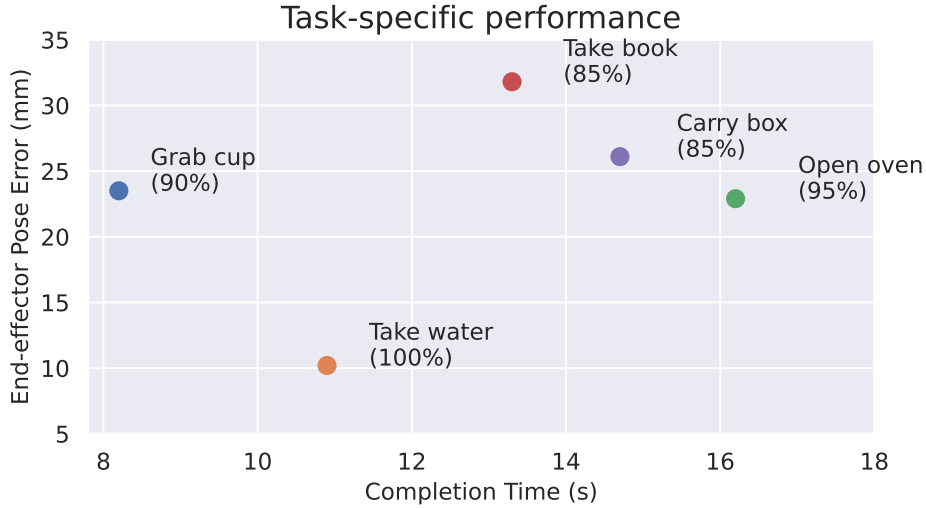


Figure R18: Comparison of task-specific metrics. Each point represents a single task, annotated by its success rate. The closer a point is to the lower left corner, the lower the end-effector pose error and the shorter the completion time, indicating better overall performance.

Furthermore, to directly address the reviewer’s request for more fine-grained task-specific metrics (e.g., oven handle rotation), we have added an additional table summarizing such detailed measures. We used a millimeter ruler to measure these error data, because the motion capture device is difficult to measure reflective objects.

As shown in Table R21, The cup placement deviation is within acceptable range, and the handle rotation error for the oven task remains under 5° . This confirms that our framework not only ensures general pose tracking but also achieves reliable task-specific precision.

The authors thank the reviewer for highlighting this important point, and the correspond-

Table R21: Detailed Task-Specific Metrics

Task	Key Precision Metric	Value
Grab cup	Cup final placement offset (mm)	22.5 mm
Open oven	Handle rotation angle error (deg)	4.8°
Take book	Book alignment deviation (mm)	28.7 mm
Carry box	Box final placement offset (mm)	25.3 mm

ing figure and table will be added in the **Appendix.D** of revised manuscript to strengthen the quantitative analysis of task-specific performance.

Point 2.20

Is there a plan to release the codebase, even a minimal version sufficient to reproduce the key results claimed in the paper?

Thank you for the reviewer’s comment. Unfortunately, we are unable to release the VAE code at this time due to ongoing internal research and proprietary dependencies. However, we would like to clarify that the exact VAE architecture is not critical to the main contribution of our method. Any standard VAE that encodes local motion windows can be used to achieve comparable results. For reference, similar latent space priors have been explored in public projects such as Meta’s Latent Space Priors[21].

Regarding the reinforcement learning component, our implementation follows well-established practices and is primarily based on the open-source humanoid RL framework[22], which provides reproducible baselines for sim-to-real locomotion tasks.

Finally, the TOP policy code is developed in collaboration with an industrial partner and contains proprietary modules that we are contractually unable to release at this time. We hope the details provided in the paper, together with the references to open-source frameworks, will help interested researchers reproduce the key ideas.

Point 2.21

The manuscript lacks justification for the hyperparameter optimization strategy (e.g., selection criteria for VAE learning rate, RL discount factor) and omits parameter sensitivity analysis. This omission compromises the validation of design choice robustness and rationale.

Thanks for the reviewer’s comment. We apologize for the unclear explanation of the justification for the hyperparameter. As the review says, it should be proved that the hyperparameter selection strategy is valid and robust. We will explain the selection criteria for VAE and RL policies in segments and verify their hyperparameter sensitivity.

In our VAE training, we adopt advanced scheduling strategies rather than fixed hyperparameter settings. Specifically, we employ a cosine annealing warm restart strategy[23] (`CosineAnnealingWarmRestarts`) for the learning rate to improve training stability and convergence. Additionally, to mitigate the well-known KL vanishing problem in VAE training, we use a cyclical KL weight scheduling scheme[24] with a init weight of 0.002, which gradually increases the KL contribution over training cycles. In practice, we observed that both the cosine annealing schedule and KL weight cycling contributed to faster convergence and more stable reconstruction quality, without signs of KL collapse.

Therefore, since our scheduling automatically explores a range of values for both the learning rate and KL weight throughout training, a traditional static sensitivity analysis of learning rate and KL weights would not yield additional insight. We instead provide full reproducibility by reporting our initial values and scheduler configuration in Table R22.

Table R22: Hyperparameters and Sensitivity Analysis for VAE Training

Hyperparameter	Value	Task Success (%)
Batch size	512	-
Number of epochs	30000	-
Learning rate	0.003	-
KL-scheduler (cycles/ratio)	7.0 / 0.5	-
KL weight	0.002	-
Warm restart (T_0 / T_{mult})	1000 / 2	-

In our decoupling policy design, we intentionally use a compact policy network with ELU activations [25] to reduce inference complexity. The action distribution is modeled as a diagonal multivariate Gaussian, where the mean is predicted by the neural network and the standard deviation is treated as a separate parameter. This choice ensures efficient sampling while preserving control flexibility. Furthermore, To improve policy update stability and adaptivity, we implement an adaptive learning rate mechanism based on the KL divergence [26] between the current policy π and the previous policy π_{old} and the KL divergence is computed as:

$$\text{KL}(\pi_{\text{old}}||\pi) = \sum_i \log \left(\frac{\sigma_i}{\sigma_{0,i} + \epsilon} \right) + \frac{\sigma_{0,i}^2 + (\mu_i - \mu_{0,i})^2}{2\sigma_i^2} - \frac{1}{2} \quad (18)$$

We monitor the mean KL divergence during training. If it exceeds twice a target value (i.e., $\text{KL} > 2 \cdot \text{KL}_{\text{target}}$), we reduce the learning rate by a factor of 1.5. Conversely, if it falls below half the target, we increase the learning rate by a factor of 1.5 (while capping it). This dynamic strategy helps prevent overly aggressive or conservative updates, improving the training robustness across different stages.

We note that this adaptive KL-based adjustment automatically tunes the step size during training to prevent instability from sub-optimal hyperparameter choices, thereby reducing the need for exhaustive static grid search on the learning rate or KL target. To ensure the design’s robustness, we additionally tested the discount factor γ in a representative range (0.90, 0.95, 0.99). As shown in Table R23, the task success rates remain within a narrow band, confirming that our choice of $\gamma = 0.99$ is well suited to our tasks’ long-horizon dynamics.

Therefore, the combination of our compact policy design, diagonal Gaussian distribution, adaptive KL control, and empirical checks makes additional large-scale sensitivity experiments unnecessary. See Table R23 for RL hyperparameters.

Thanks to the reviewers for pointing out the problems. We have updated our manuscript to better illustrate the parameter sensitivity.

In EXPERIMENT.A:

We use the training objective of a β -VAE with a KL weight of 0.002 and cyclical scheduling to mitigate KL vanishing, and cosine annealing warm restart strategy

for the learning rate to improve training stability and convergence. Refer to Table II for details of the hyperparameters.

Table R23: Hyperparameters and Sensitivity Analysis for RL Policy Training

Hyperparameter	Value	Task Success (%)
Batch size	-	-
Mini-batch size	2.0	-
Learning rate	3e-4 (assumed)	-
Clip range, e	1.0	-
Discount factor, γ	0.99	-
GAE discount factor, λ	0.95	-
Number of epochs	1.0	-
Max gradient norm	1.0	-
Target KL Divergence	0.01	-
<i>Sensitivity Analysis</i>		
Discount Factor: 0.90	-	92.8%
Discount Factor: 0.95	-	94.1%
Discount Factor: 0.99	-	95.3%

Reply to Reviewer 4

This paper introduces a Time Optimization Policy (TOP) aimed at enabling humanoid robots to perform fast standing manipulation tasks with a balance of stability, precision, and time efficiency. The approach involves adjusting the time trajectory of upper-body motions while enhancing the lower body’s resistance to disturbances. The manuscript is well-organized, and the proposed method appears practical for real-world applications.

We appreciate the reviewer’s concise summary of the work and hope that the explanation has fully addressed all of the reviewer’s concerns.

Due to the lack of space, we present our **paper with appendix**, the **response letter** and the **videos** in our project page <https://anonymous.4open.science/w/top-258F/>, we would like to bother the reviewer to have a look for a full paper.

Point 4.1

The TOP training process seems to adjust upper-body motions at each time-step while keeping lower-body actions unchanged. This approach may lead to misalignment between upper and lower body dynamics, potentially reducing the robot’s stability.

Thank you for raising this important point regarding the potential misalignment between upper- and lower-body dynamics when the time optimization policy (TOP) adjusts the upper-body motion timing at each step.

We fully agree that, in principle, such misalignment could degrade the robot’s stability if not properly handled. In our design, however, the **decoupling is not rigid** but rather a **practical modular abstraction** for efficient learning and deployment:

- **Upper-body motion** is compressed by a VAE-based motion prior and tracked by a PD controller that produces desired joint torques to follow the reference trajectories with high precision.
- **Lower-body stabilization** is handled by a reinforcement learning (RL) policy that

receives real-time observations of the floating base state, projected gravity, joint states, and the desired upper-body motion embedding (z_t) and motion clip (m_t).

Therefore, while the modules are trained in a decoupled manner, their execution forms a closed-loop whole-body control system. The RL policy implicitly compensates for momentum transfer from the upper-body by dynamically adjusting lower-body joint torques to maintain balance. In other words, this means that whenever the upper-body speed changes to maintain base stability, the RL policy observes this change implicitly through the updated robot kinematics and base reaction, and generates appropriate corrective actions for the legs and base. This design principle is consistent with established whole-body control frameworks[9] where manipulation and locomotion are hierarchically coordinated.

As shown in Table R24, the RL policy is explicitly encouraged to minimize base drift, linear/angular velocity deviations, and physically feasible whole-body motions:

- **Base linear velocity** and **base angular velocity** terms penalize excessive base oscillations.
- **Base orientation** cost stabilizes the base roll/pitch under external moments induced by upper-body motions.
- **Stand still** and **action penalty** terms further ensure smooth, coordinated lower-body actions.

Table R24: Key RL Reward Terms for Stabilization

Term	Expression	Weight
Base linear velocity (xy)	$\exp[-4(\mathbf{v}_{xy}^2)]$	+3.0
Base angular velocity (xy)	$\Sigma \omega_{xy}^2$	-0.1
Base orientation	$\Sigma \mathbf{g}_{xy}^2$	-1.5
Stand still pose	$\exp[-10\ \mathbf{q}_t^{\text{leg}} - \mathbf{q}^{\text{leg,ref}}\ _2]$	+1.0
Action rate penalty	$\ \mathbf{a}_t - \mathbf{a}_{t-1}\ _2^2$	-0.2
Action acceleration penalty	$\ \mathbf{a}_t + \mathbf{a}_{t-2} - 2\mathbf{a}_{t-1}\ _2^2$	-0.2

Table R25: Part of Comparative results. We execute more than 10000 motion clips from dataset $\mathcal{M}$ and unseen dataset $\mathcal{T}$, and report their time cost, success rate, mean metrics. Mean Metrics are only calculated based on the data of the robot standing in place, and the data of robot taking steps and falls will not be included in the calculation.

Method	Time Cost ↓	Success Rate ↑	$\mathbf{E}_{jpe}^{\text{upper}} \downarrow$	$\mathbf{E}_{eepe}^{\text{upper}} \downarrow$	$\mathbf{E}_{eeoe}^{\text{upper}} \downarrow$	$\mathbf{E}_g \downarrow$	$\mathbf{E}_{acc}^{\text{lower}} \downarrow$	$\mathbf{E}_{action}^{\text{lower}} \downarrow$	$\mathbf{E}_{acc}^{\text{upper}} \downarrow$	$\mathbf{E}_{action}^{\text{upper}} \downarrow$
Comparative Results										
Fixed Root (reference)	15.0s	100.0%	0.0130	0.0164	0.0594	1.000	-	-	-	-
Exbody[10]	15.0s	92.46%	0.0376	0.0741	0.0923	3.432	21.65	0.7323	24.26	2.976
Ours (TOP)	40.5s	95.30%	0.0269	0.0270	0.0827	2.729	13.45	0.8592	10.41	1.601

We can observe that, this closed-loop design ensures that the lower-body dynamically absorbs momentum transfer delays from the upper-body PD tracker. Our TOP method deliberately slows down sudden upper-body changes to avoid destabilizing the base, which slightly sacrifices time efficiency but greatly improves base stability and overall task success. Compare to the baseline (Exbody[10]) that uses whole-body RL policy with the tracking rewards to control whole-body joints, our method can achieve precision, stability and time efficiency at the same time, proving the effectiveness of our decoupling controllers, as shown in Table R25.

In summary, although the TOP policy dynamically adjusts upper-body timestamps, the lower-body actions are not static but adapt in real-time to ensure whole-body consistency. The empirical results and reward design confirm that our approach avoids dynamic misalignment and preserves stable and feasible behavior.

We believe this more clearly demonstrates that our approach is not **naively decoupled** but rather forms an integrated closed-loop whole-body control system. Thanks to the reviewers for pointing out the problems. Due to the lack of space, we will not include this in revised paper, but will add the explanation to **APPENDIX.A**.

Point 4.2

The paper emphasizes instability caused by rapid upper-body movements but does not discuss whether the end effector’s payload (e.g., weight of manipulated objects) is factored into the method.

Thank you for this insightful comment. We fully agree that rapid upper-body movements combined with changing payloads can pose significant challenges to maintaining balance and physical feasibility.

To clarify, our VAE focuses on learning compact latent representations of human-like motion purely from joint position and velocity data, without directly incorporating dynamic constraints (e.g., torque limits, inertia matrices). However, the final execution relies on a downstream reinforcement learning (RL) policy, which is trained with extensive *domain randomization* of physical properties. This includes random variations in the payload mass, link inertia, base mass, motor gains, and friction, as detailed in Table R26. This ensures that the policy learns robust behaviors that remain stable even when the robot manipulates objects of varying weight.

To empirically verify this design, we conducted an additional load-adaptability test, in which the robot performed the same manipulation task while grasping objects of increasing mass (0.5 kg, 1 kg, 3 kg). Table R27 reports the results. The task success rate and end-effector pose error remain stable across different payloads, while only a slight increase in the projected gravity term is observed. This demonstrates that our RL controller compensates for the dynamic variations introduced by heavier payloads, ensuring that the generated actions remain physically feasible and stable even though the upstream VAE module is purely kinematic.

These findings clearly demonstrate that our framework achieves physically feasible and stable behaviors under varying load conditions. The authors thank the reviewer for highlighting this important point, which further strengthens the completeness and practicality of our framework, and we have incorporated this new experiments and discussion into **Appendix.C**.

Point 4.3

A Variational Autoencoder (VAE) uses a latent code dimension of 64 and a window size of 30, but the paper lacks justification for these choices. An ablation study on the sensitivity of these hyperparameters to performance would be valuable.

Thanks for the reviewer’s suggestion. As the review says, it should be analyzed whether

Table R26: Domain Randomization Settings

Term	Value
Observation	
Joint position noise	$\mathcal{U}(-0.1, 0.1)$ rad
Joint velocity noise	$\mathcal{U}(-1.0, 1.0)$ rad/s
Angular velocity noise	$\mathcal{U}(-0.1, 0.1)$ rad/s
Projected gravity noise	$\mathcal{U}(-0.05, 0.05)$
Dynamics	
Base COM displacement	$\mathcal{U}(-0.05, 0.05)$ m
Friction factor	$\mathcal{U}(0.35, 1.25)$
Link Inertia	$\mathcal{U}(0.8, 1.2) \times \text{default}$ kg $\cdot$ m ²
Added base mass	$\mathcal{U}(-5.0, 5.0) + \text{default}$ kg
Added link mass	$\mathcal{U}(0.8, 1.2) \times \text{default}$ kg
Added hand mass	$\mathcal{U}(-0.5, 1.5) + \text{default}$ kg
Payload mass	$\mathcal{U}(-1.0, 3.5) + \text{default}$ kg
Motor P gain	$\mathcal{U}(0.8, 1.2) \times \text{default}$
Motor D gain	$\mathcal{U}(0.8, 1.2) \times \text{default}$
Motor strength	$\mathcal{U}(0.9, 1.1) \times \text{default}$ N $\cdot$ m
Motor Damping	$\mathcal{U}(0.3, 4.0)$ N $\cdot$ s
Motor Delay	$\mathcal{U}(0, 10)$ ms
Episode	
Push robots	interval = 5s, $v_{xy} = 1.0\text{m/s}$, $\omega = 0.4\text{rad/s}$
Disturbance robots	interval = 8s, $\mathcal{U}(-30.0, 30.0)$ N $\cdot$ m
Initial robot base position xy	$\mathcal{U}(-1.0, 1.0)$ m
Initial robot base velocity	$\mathcal{U}(-0.1, 0.1)$ m/s

Table R27: Load Adaptability Test: Task performance under different payloads

Payload	Success Rate $\uparrow$	EE Pose Error $\downarrow$	Projected Gravity $\downarrow$
0.5 kg	95.2%	28.5 mm	2.731
1.0 kg	94.4%	30.1 mm	2.857
3.0 kg	92.7%	31.8 mm	3.142

the latent code dimension and window size are reasonable for our VAE model to extract motion prior knowledge. We agree that the choice of latent dimension and window size can significantly affect the performance of a motion prior model, and we appreciate the opportunity to clarify our design choices with quantitative evidence.

Several prior works in VAE and motion modeling have conducted similar sensitivity analyses for window sizes and latent dimensions, Yan et al.[27] explore input sequence lengths from 8 to 64 and conduct extensive hyperparameter search for MT-VAE variants by enumerating smoothing window, motion ratio, cycle loss ratio, demonstrating that both under-long and over-long windows degrade reconstruction quality due to underfitting or model saturation. Petrovich et al.[28] ablate the model and vary key parameters to evaluate the influence of design choices on the quality of the results. Therefore, We conducted an ablation study on the VAE hyperparameters. Results are shown in Table R28.

Table R28: Ablation Study: Sensitivity to Latent Dimension and Window Size

Window Size	Latent Dim	Recon. Loss		KLD Loss		Time (hrs)
		Train	Test	Train	Test	
15	64	0.0261	0.3751	0.0148	0.0207	3.6
30	64	0.0250	0.2495	0.0126	0.0143	4.2
60	64	0.0714	0.6992	0.0081	0.0096	4.7
30	32	0.1139	2.7729	0.0086	7.8787	4.0
30	64	0.0250	0.2495	0.0126	0.0143	4.2
30	128	0.0238	0.2807	0.0118	0.0161	5.3

The results demonstrate that a window size of 30 achieves the lowest reconstruction error

and a balanced KL loss on both training and test sets. A shorter window size (15) provides insufficient temporal context for complex motion segments, which weakens generalization to unseen motion sequences (higher test loss). Conversely, a longer window size (60) increases the amount of temporal information to be jointly reconstructed, which introduces more noise, optimization challenges, and variance in motion duration, resulting in overfitting and degraded test performance. Thus, a moderate window size of 30 frames effectively balances temporal coverage and model complexity.

Regarding the latent dimension, 64 yields the best trade-off between representation capacity and regularization. A smaller latent code (32) underfits: it lacks sufficient capacity to encode diverse motion variations, leading to high reconstruction and KL divergence errors, especially on unseen data (test KL loss 7.8787). On the other hand, a larger latent code (128) slightly improves training expressiveness but increases test loss due to overfitting and adds unnecessary computational overhead. Hence, 64 provides stable convergence, reasonable latent compactness, and robust generalization.

For clearer visualization, we will also include a boxplot of reconstruction loss and kl loss to illustrate the stability of these choices, as shown in the Fig R19, each box shows 10 runs results. Together, these results confirm that our selected window size and latent dimension are empirically justified and robust for capturing motion priors.

Thanks to the reviewers for pointing out the problems. Due to the lack of space, we will not include this in revised paper, but will add the explanation to **APPENDIX.G**.

Point 4.4

In the experiments section, metrics such as "Projected Gravity" (for stability), upper joint position error, and upper end effector position error are used, but their definitions and computation methods are not provided. Additionally, the units for these errors are omitted.

Thanks for the reviewer’s constructive comment. We apologize for the unclearness of our metrics. We have revised the manuscript to clarify the definitions and computation methods of the evaluation metrics used in the experiments. Our definitions and computations are similar with prior work (e.g. [9, 29]). The specific definitions are as follows:

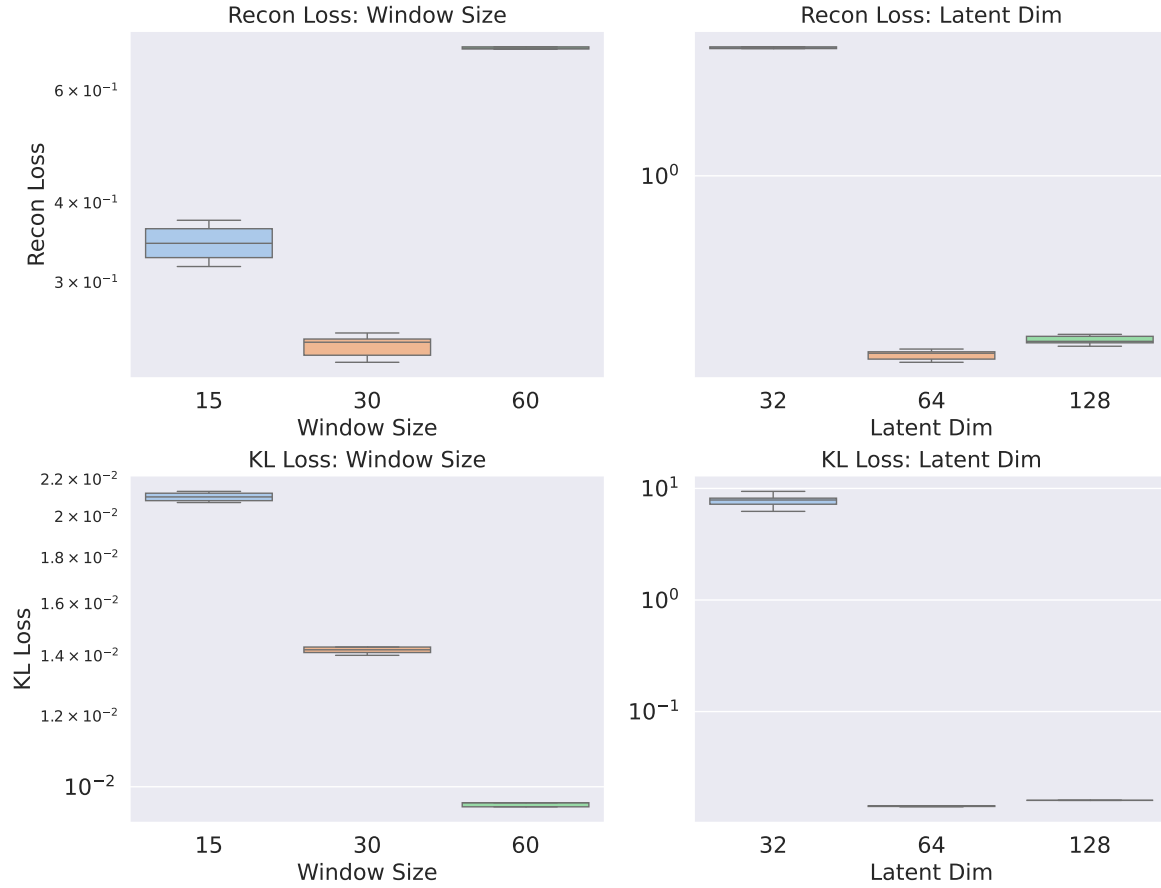


Figure R19: **Ablation study for VAE hyperparameters** The top two plots show test reconstruction loss distributions for different window sizes and latent dimensions; the bottom two show the corresponding KL divergence losses. A window size of 30 and latent dimension of 64 provide the best balance of reconstruction accuracy and regularized latent encoding, with low variance across random seeds.

- **Projected Gravity (unitless):** This metric quantifies the alignment between the robot’s torso and the gravity vector, indicating the stability relative to the support plane. Specifically, we first transform the global gravity vector into the robot’s base coordinate frame using the inverse of the base orientation quaternion, then compute the dot product between the torso’s local z-axis unit vector and the transformed gravity vector. A value closer to 1 indicates better balance and alignment. This metric is unitless and normalized to $[-1, 1]$.

$$\mathbf{E}_{\mathbf{g}} = R^{-1}(\mathbf{q}_{\text{base}}) \mathbf{g} \quad (19)$$

where $\mathbf{q}_{\text{base}}$ is the base orientation quaternion, $R^{-1}(\mathbf{q}_{\text{base}})$ denotes the inverse rotation that transforms the global gravity vector $\mathbf{g}$ into the robot’s base coordinate frame, and $\mathbf{E}_{\mathbf{g}}$ is the gravity vector expressed in the base frame.

- **Upper Joint Position Error (rad):** We calculate the upper joint position error of all upper-body joints (e.g., shoulders, elbows, wrists) at each timestamp. This error is averaged across joints and frames.

$$\mathbf{E}_{\text{jpe}}^{\text{upper}} = \frac{1}{NT} \sum_{t=1}^T \sum_{n=1}^N \|q_{n,t}^{\text{pred}} - q_{n,t}^{\text{gt}}\|_2 \quad (20)$$

where N is the number of upper-body joints, T is the number of frames, $q_{n,t}$ is the real joint angle for joint n at frame t , and $q_{n,t}^{\text{target}}$ is the target.

- **End Effector Position Error (mm):** This metric represents the mean position error of the upper-body end effectors (typically the hands).

$$\mathbf{E}_{\text{eepe}}^{\text{upper}} = \frac{1}{MT} \sum_{t=1}^T \sum_{m=1}^M \|\mathbf{p}_{m,t} - \mathbf{p}_{m,t}^{\text{target}}\|_2 \quad (21)$$

where $M = 2$ is the number of end effectors (e.g., hands), $\mathbf{p}_{m,t}$ and $\mathbf{p}_{m,t}^{\text{target}}$ are the real positions and target position at time t .

We have also added the corresponding units (rad, mm) to the vertical axes of Figure 5 in our manuscript. Thanks to the reviewers for pointing out the problems, and we hope that these explanations can help better illustrate the results of our experiments.

Reply to Reviewer 5

This paper proposes a time optimization strategy for the stable and accurate standing operation of humanoid robots. The correctness of the method was verified through simulation and experiments. In order to demonstrate the superiority of the proposed time optimization method, it is still necessary to make some comparisons with traditional methods in terms of performance. In addition, there needs to be some proof on how to ensure the stability of the designed system.

We appreciate the reviewer’s clear understanding and positive feedback and hope that the explanation has fully addressed all of the reviewer’s concerns.

Due to the lack of space, we present our **paper with appendix**, the **response letter** and the **videos** in our project page <https://anonymous.4open.science/w/top-258F/>, we would like to bother the reviewer to have a look for a full paper.

Point 5.1

In order to demonstrate the superiority of the proposed time optimization method, it is still necessary to make some comparisons with traditional methods in terms of performance. In addition, there needs to be some proof on how to ensure the stability of the designed system.

Thanks for the reviewer’s comment. We agree that comparing our proposed time optimization policy (TOP) with a well-established traditional baseline and providing evidence of stability is essential for a rigorous evaluation.

As a representative baseline, we implement a conventional non-linear Model Predictive Control (NMPC) combined with Whole-Body Control (WBC). This baseline is widely adopted in quadrupedal and bipedal robots for dynamic motion planning [16, 30] and open-source frameworks[31].

Our NMPC solves for a centroidal trajectory under a fixed time schedule with a predefined contact sequence. To ensure real-time feasibility, the continuous optimal control problem is discretized via direct multiple shooting, and solved using a Sequential Quadratic Programming (SQP) method with real-time iteration. The joint pose references come from the motions in

GRAB[17], ensuring consistency with the desired task-level motion. The continue nonlinear MPC problem formulation is following:

$$\left\{ \begin{array}{l} \min_{\mathbf{u}(\cdot)} \Phi(\mathbf{x}(T)) + \int_0^T \mathbf{l}(\mathbf{x}(t), \mathbf{u}(t), t) dt \\ \text{subject to } \dot{\mathbf{x}} = f(\mathbf{x}(t), \mathbf{u}(t), t) \\ \mathbf{g}_{eq}^1(\mathbf{x}(t), \mathbf{u}(t), t) = 0 \\ \mathbf{g}_{eq}^2(\mathbf{x}(t), t) = 0 \\ \mathbf{h}_{in}(\mathbf{x}(t), \mathbf{u}(t), t) \geq 0 \\ \mathbf{x}(0) = \mathbf{x}_0 \end{array} \right. \quad (22)$$

where $\mathbf{x}(t)$ and $\mathbf{u}(t)$ are the state and control input at time t . $\mathbf{g}_{eq}^1$ represents equality state-input-related constraints, and $\mathbf{g}_{eq}^2$ represents equality constraints only related to state. $\mathbf{h}_{in}$ includes state constraints and state-input-related constraints, such as joint position and velocity limits, friction cone constraints, etc. $\mathbf{x}_0$ is the current measured state of robot.

An optimization-based whole-body controller helps to map the optimized trajectory obtained from the nonlinear MPC to the humanoid robot’s joint torque, which considers the full nonlinear rigid body dynamics. While formulating the constrained hierarchical optimization problem, in our baselines we instead use a single QP to construct the constrained optimization problem, in which the low-priority tasks are embedded in the cost function, and high-priority tasks are treated as constraints. The list of tasks is shown in the Table R29. The first two blocks of tasks prevent the robot from violating the equality and inequality constraints of torque, force, and joint configuration. And the least squares tasks guide the robot to track the foot motion with a higher weight than the base reference motion, which prevents the robot from taking advantage of leg inertia to track the base motion in underactuated directions.

We compare our TOP with this NMPC+WBC baseline and with a Fixed Root reference policy that we fixed the root of robot, and directly execute the upper-body motions, which means the motions can be perfectly executed with the highest control accuracy. We use the motions from dataset $\mathcal{M}$ and unseen dataset $\mathcal{T}$ across methods, and report performance on the following criteria:

- **Time Cost:** Average time cost of per 1500 motion clips (15s of origin data), which shows the time efficiency of executing the target motions.
- **Success Rate:** We define that the robot executes the upper body motion without

Table R29: WHOLE-BODY CONTROL TASKS

Type	Task
=	Floating base equations of motion
	No Motion at the contact points
$\geq$	Torque limits
	Wrench Friction cone
$Q_i^2 \cdot ^2$	Base Linear and Angular Tracking($Q_i = 1.0$)
	Joint Position Tracking($Q_i = 100.0$)
	Contact Force Tracking($Q_i = 0.1$)

falling, this motion clip will be labeled as successful motion.

- **Precision:** upper joint position error $\mathbf{E}_{jpe}^{\text{upper}}$, upper end effector position error in world frame $\mathbf{E}_{eepe}^{\text{upper}}$, upper end effort orientation error in world frame $\mathbf{E}_{eeoe}^{\text{upper}}$.
- **Stability:** projected gravity $\mathbf{E}_g$, lower joint acceleration $\mathbf{E}_{acc}^{\text{lower}}$, lower action difference $\mathbf{E}_{action}^{\text{lower}}$.
- **Smoothness:** upper joint acceleration $\mathbf{E}_{acc}^{\text{upper}}$, upper action difference $\mathbf{E}_{action}^{\text{upper}}$.

The results are shown in Table R30. It is worth noting that the output of NMPC+WBC[16] is different from the actions of TOP outputs, so we only use the projected gravity and acceleration to evaluate the NMPC+WBC, and we slow down the target motions of NMPC for better comparison with our TOP method.

We observe that our proposed TOP consistently achieves lower joint and end-effector position errors compared to the NMPC+WBC baseline, with reductions of approximately 3.2% and 38.2% respectively. This indicates significantly improved pose tracking performance, which is critical for manipulation tasks under dynamic conditions. Moreover, the gravity projection error E_g and lower-body acceleration error E_{acc}^{lower} are also reduced, demonstrating that our method better maintains base stability when interacting with varying external forces or loads. The success rate is increased by 5.7 percentage points, which validates the practical reliability of our planner under diverse scenarios. The only compromise is that our strategy needs to slow down our motions more, so we will lose time efficiency. In short, our TOP

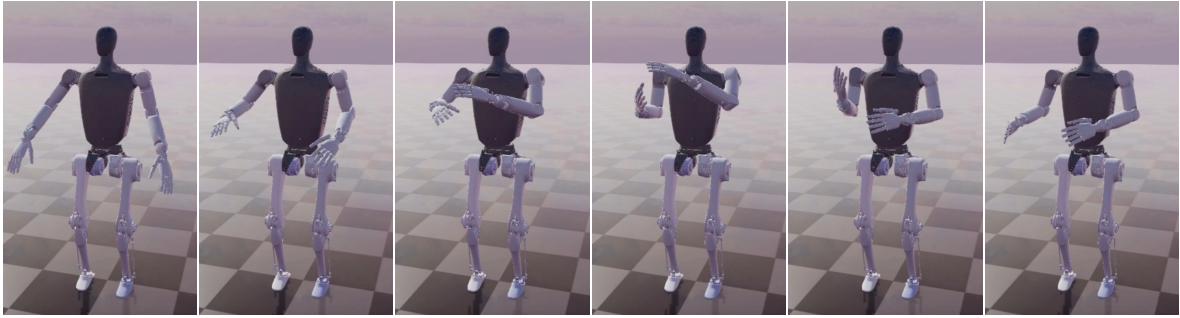
Method	Time Cost ↓	Success Rate ↑	E_{jpe}^{upper} ↓	E_{eepe}^{upper} ↓	E_{eeoe}^{upper} ↓	E_g ↓	E_{acc}^{lower} ↓	E_{action}^{lower} ↓	E_{acc}^{upper} ↓	E_{action}^{upper} ↓
Comparative Results										
Fixed Root (reference)	15.0s	100.0%	0.0130	0.0164	0.0594	1.000	-	-	-	-
NMPC+WBC[16]	35.0s	89.60%	0.0278	0.0437	0.0857	2.938	14.98	-	11.47	-
Ours (TOP)	40.5s	95.30%	0.0269	0.0270	0.0827	2.729	13.45	0.8592	10.41	1.601

Table R30: Comparative results. We execute more than 10000 motion clips from dataset $\mathcal{M}$ and unseen dataset $\mathcal{T}$, and report their time cost, success rate, mean metrics. It is worth noting that the output of NMPC+WBC[16] is different from the actions of TOP outputs, so we only use the projected gravity and acceleration to evaluate the NMPC+WBC, and we slow down the target motions of NMPC for better comparison with our TOP method.

achieves better pose tracking and base stability than the NMPC baseline and replay method, while successfully completing the tasks under varying durations and loads.

One of the motions executed by NMPC+WBC are shown in the Figure R20.

Figure R20: **Screenshots of Motions Executed by NMPC+WBC.** We realize the NMPC+WBC in raisim[18] simulator.



While we cannot release our full implementation due to hardware-specific integration of our MPC+WBC baseline, we provide references to open-source bipedal NMPC controllers [32] and quadrupedal MPC+WBC frameworks [31] that follow similar principles.

We agree that it is essential to provide theoretical evidence of stability. In our revised manuscript, we have included an additional section that analyzes the whole-body stability using the Zero Moment Point (ZMP) criterion. Specifically, we show that our policy keeps the projected ZMP trajectory within the support polygon during task execution, which implies that the system remains dynamically balanced in both static and quasi-static regimes. This complements our empirical results and provides a theoretically grounded indicator of base

stability.

Firstly, we want to explain the definition of ZMP. ZMP is the point on the ground where the horizontal component of the moment of the ground reaction forces becomes zero, assuming the ground is a horizontal plane[13]. A necessary condition for the locomotion mechanism to be in dynamic equilibrium is that the candidate ZMP is inside the support polygon. Furthermore, to handle non-planar surfaces, the concept of the Zero Moment Line (ZML) is introduced, which extends the ZMP concept to account for complex contact surfaces[14]. Then we give the calculation process as follows[15]:

$$\boldsymbol{\tau} = \mathbf{p}_{zmp} \times \mathbf{f} + \boldsymbol{\tau}_p \quad (23)$$

$$\dot{\mathbf{P}} = M\mathbf{g} + \mathbf{f} \quad (24)$$

$$\dot{\mathcal{L}} = \mathbf{p}_{com} \times M\mathbf{g} + \boldsymbol{\tau} \quad (25)$$

where p_{zmp} is the position of ZMP, τ_p is the moment about the ZMP point, and the $\mathbf{f}$ represents the ground reaction force. P , L , M , p_{com} represents the linear momentum, the angular momentum, the total mass, the center of mass (CoM) position of the system, respectively, and $\mathbf{g} = [0, 0, -g]^T$. By the definition of ZMP, we have:

$$\tau_{p,x} = \tau_{p,y} = 0 \quad (26)$$

Solving the above equations with respect to $\mathbf{p}_{zmp}$, we have

$$p_{zmp,x} = \frac{Mgp_{com,x} + p_{com,z}\dot{P}_x - \dot{\mathcal{L}}_y}{Mg + \dot{P}_z} \quad (27)$$

$$p_{zmp,y} = \frac{Mgp_{com,y} + p_{com,z}\dot{P}_y + \dot{\mathcal{L}}_x}{Mg + \dot{P}_z} \quad (28)$$

According to the Eq.39, ZMP at different heights $p_{com,z}$ lies at different locations, whose trajectory forms a line of ZMPs, called ZML [14]. The condition for stable locomotion is that the ZML must cross the support polygon, as shown in Fig R21.

Then, we need to estimate the support polygon of the humanoid robot to determine the relationship between ZML and the support. In general cases, the support polygon is determined by computing the convex hull of all contact points for scenarios with multiple contact points. Therefore, we calculate the ZML lines in simulation, as shown in the Fig R22.

Figure R21: **ZML cross the support polygon.** The red dot represents the approximated center of the support polygon, and the green dot on the ZML is the projected point at the same horizontal height as the red dot. d represents the distance between the center of the support polygon and the projected point of ZML, which can describe the stability of the system. The smaller d is, the more stable the system is.

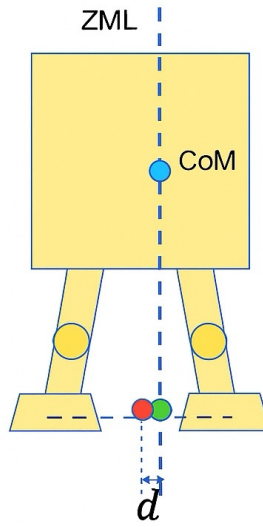
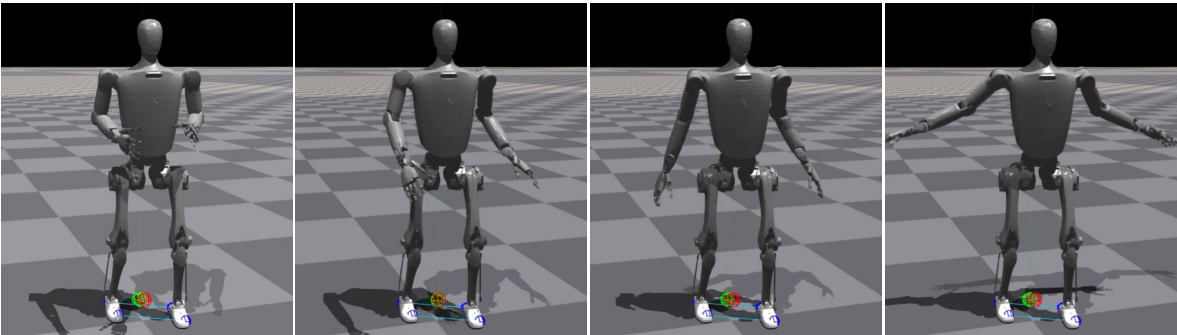


Figure R22: **ZML and Support Polygon in Isaacgym.** The red point represents the center of the support polygon, and the green dot on the ZML is the projected point at the same horizontal height as the red dot. The blue points and the light blue lines are the support polygon.



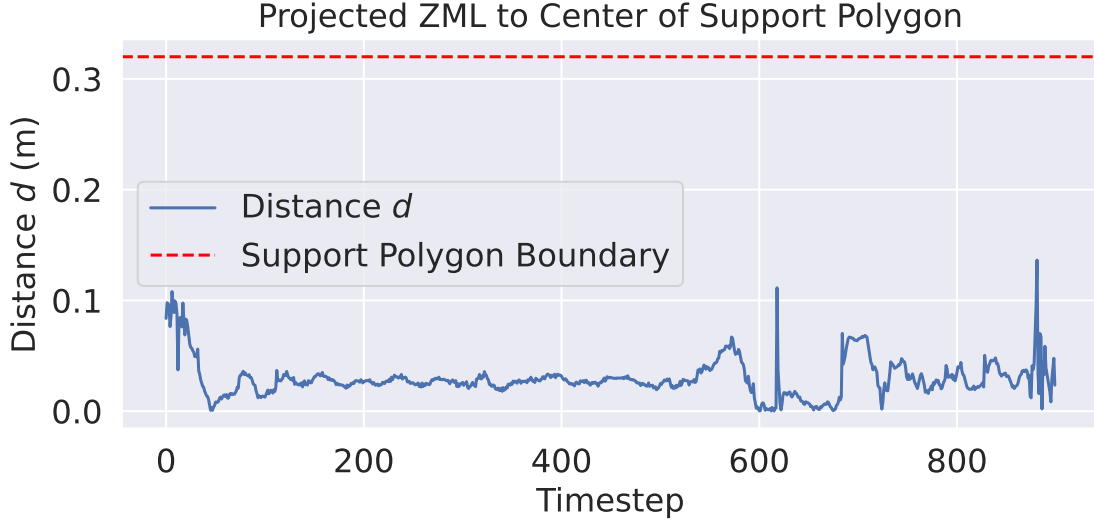
To quantitatively assess the dynamic stability of our policy, we compute the signed distance d between the ZML projection point and the center of the support polygon:

$$d = \min_{p_{csp} \in \mathcal{S}} \|p_{csp} - \text{proj}_{ZML}(p_{csp})\|_2, \quad \text{proj}_{ZML}(p_{csp}) \in \mathbb{R}^2 \quad (29)$$

where $\mathcal{S}$ denotes the support polygon of current contact points, and $\text{proj}_{ZML}(p_{csp})$ is the projected ZML point in the horizontal plane. If $d \in [0, 0.32]$, the ZML point is inside the support polygon; if $d \in (0.32, 0.36]$, it lies near the edge of support polygon; and if $d \in (0.36, +\infty)$, the ZML has exited the support region, indicating potential instability. It is worth noting that the closer d is to 0, the system is more stable and satisfy the ZMP constraint.

To demonstrate that our time optimization policy (TOP) maintains the Zero-Moment Point (ZMP) condition throughout task execution, we plot the time series of d for representative tasks. As shown in Fig. R23, the distance remains positive during the entire motion, verifying that the ZML stays well within the support polygon at all times, and thus the robot’s base remains dynamically stable despite rapid upper-body motions.

Figure R23: **Stability metric d over time.** The curve shows the distance between the ZML projection and the center of support polygon. The distance remains ≤ 0.32 , confirming that the ZMP condition is always satisfied.



This quantitative evidence demonstrates that our closed-loop policy stabilization effectively ensures that the floating base remains within feasible support constraints, even under varying contact conditions and dynamic upper-body perturbations.

Thanks to the reviewers for pointing out the problems, which further strengthens the completeness and practicality of our framework, and we have incorporated this ZMP proof into **Appendix.H**.

To further demonstrate the generality and robustness of our method, we have incorporated extensive domain randomization during training. This includes random variations in payload, surface friction, and external perturbations, as detailed in Table R32. This ensures that the policy learns robust behaviors that remain stable even when the robot manipulates objects of varying weight.

To empirically verify this design, we conducted an additional load-adaptability test, in which the robot performed the same manipulation task while grasping objects of increasing mass (0.5 kg, 1 kg, 3 kg). Table R31 reports the results. The task success rate and end-effector pose error remain stable across different payloads, while only a slight increase in the projected gravity term is observed. This demonstrates that our RL controller compensates for the dynamic variations introduced by heavier payloads, ensuring that the generated actions remain physically feasible and stable even though the upstream VAE module is purely kinematic.

Table R31: Load Adaptability Test: Task performance under different payloads

Payload	Success Rate $\uparrow$	EE Pose Error $\downarrow$	Projected Gravity $\downarrow$
0.5 kg	95.2%	28.5 mm	2.731
1.0 kg	94.4%	30.1 mm	2.857
3.0 kg	92.7%	31.8 mm	3.142

Moreover, we have conducted additional experiments about the external disturbance tests of our policy. Figure R24 shows external disturbance tests of our policy. We designed scenarios where the robot is intentionally disturbed by external pushes from different directions.

These findings clearly demonstrate that our framework achieves physically feasible and stable behaviors under varying load conditions and external disturbances, which prove the stability of our method.

The authors thank the reviewer for highlighting this important point, which further strengthens the completeness and practicality of our framework, and we have incorporated

Table R32: Domain Randomization Settings

Term	Value
Observation	
Joint position noise	$\mathcal{U}(-0.1, 0.1)$ rad
Joint velocity noise	$\mathcal{U}(-1.0, 1.0)$ rad/s
Angular velocity noise	$\mathcal{U}(-0.1, 0.1)$ rad/s
Projected gravity noise	$\mathcal{U}(-0.05, 0.05)$
Dynamics	
Base COM displacement	$\mathcal{U}(-0.05, 0.05)$ m
Friction factor	$\mathcal{U}(0.35, 1.25)$
Link Inertia	$\mathcal{U}(0.8, 1.2) \times \text{default}$ kg $\cdot$ m ²
Added base mass	$\mathcal{U}(-5.0, 5.0) + \text{default}$ kg
Added link mass	$\mathcal{U}(0.8, 1.2) \times \text{default}$ kg
Added hand mass	$\mathcal{U}(-0.5, 1.5) + \text{default}$ kg
Payload mass	$\mathcal{U}(-1.0, 3.5) + \text{default}$ kg
Motor P gain	$\mathcal{U}(0.8, 1.2) \times \text{default}$
Motor D gain	$\mathcal{U}(0.8, 1.2) \times \text{default}$
Motor strength	$\mathcal{U}(0.9, 1.1) \times \text{default}$ N $\cdot$ m
Motor Damping	$\mathcal{U}(0.3, 4.0)$ N $\cdot$ s
Motor Delay	$\mathcal{U}(0, 10)$ ms
Episode	
Push robots	interval = 5s, $v_{xy} = 1.0\text{m/s}$, $\omega = 0.4\text{rad/s}$
Disturbance robots	interval = 8s, $\mathcal{U}(-30.0, 30.0)$ N $\cdot$ m
Initial robot base position xy	$\mathcal{U}(-1.0, 1.0)$ m
Initial robot base velocity	$\mathcal{U}(-0.1, 0.1)$ m/s

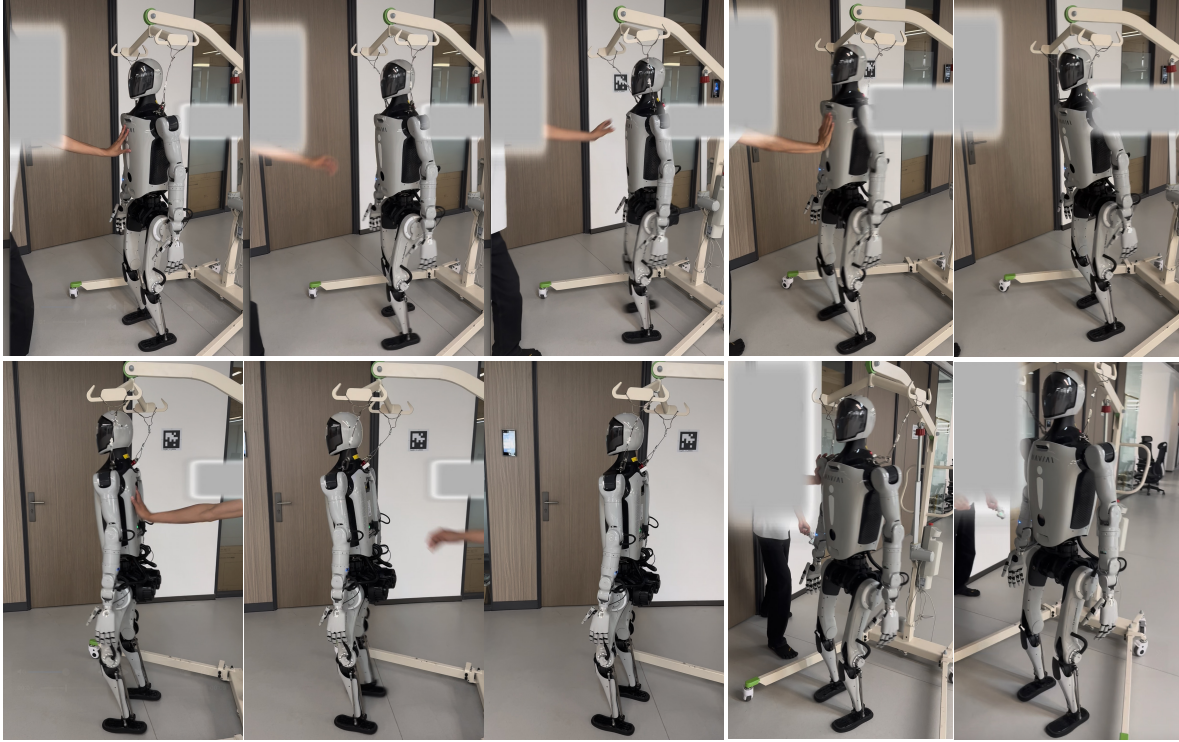


Figure R24: **External push disturbance tests.** The figure illustrates the robot can recover to balance while disturbed by external pushes from different directions.

this new experiments and discussion into **Appendix.C**.

To make our results fully transparent, we have released all the demonstration videos on our project page <https://anonymous.4open.science/w/top-258F/>.

Point 5.2

The font in some pictures is too small and needs to be enlarged to be clearly seen. The font size of the caption and the font size inside the picture should be 8pt, and the font size of the entire text needs to be uniform.

Thank you for the reviewer's comment. We have carefully checked all figures and adjusted the font sizes of the captions and labels to ensure they are as close as possible to the required 8pt standard, especially the font in Fig 1, Fig 2 and Fig 6.

Due to the lack of space, For some figures (Fig 5) with complex layouts or dense information, further enlargement of certain labels is technically challenging due to space constraints.

However, we have made every effort to maximize readability while keeping the overall presentation clear and consistent. We have also verified the uniformity of the text font throughout the manuscript.

Point 5.3

There are quite a few variables in the entire text. It is recommended to explain them in a table before the theoretical derivation.

Thank you for the reviewer’s comment. To improve clarity and readability, we have added a new table (Table R33) that summarizes all the important symbols and abbreviations used throughout the paper.

Thanks to the reviewers for pointing out the problems, which will improve the clarity and readability of our paper. Due to the lack of space, we will not add this table to our manuscript.

Point 5.4

Similarly, the font size of the table notes and the font size inside the table should be 8pt, and the entire text needs to be uniform.

Thank you for the reviewer’s comment. We have carefully checked all tables and adjusted the font size of the table notes and the text inside the tables to ensure they meet the 8pt requirement as far as possible.

For tables (Table IV) with dense content or limited space, further enlargement was technically constrained by layout requirements, but we have maximized the font size wherever possible to improve readability. We have also verified the consistency of the font size throughout the manuscript to maintain a uniform and clear presentation.

Point 5.5

The robot platform used in the experiment and simulation needs further description, such as the number of joints, the selection of drive hardware and the configuration of the computing unit, etc.

Table R33: Important Symbols and Abbreviations

Symbol	Meaning
t	Timestamp, converted to time in calculation
Δt_t	Delta timestamp in current time
Δt_t^{seq}	Sequence of delta timestamps
$s \in \mathbb{S}$	State
$a \in \mathbb{G}A$	Action
$o \in \mathbb{O}$	Observation
$g \in \mathbb{G}$	Goal
$h \in \mathbb{O}$	History observation
z_t	latent variable
d_t	dimension of latent space
m_t, m'_t	Current motion, New current motion
M_t, M'_t	Current motion windows, Reconstruct motion windows
r_t	The position of the base relative to the world frame
θ_t	The orientation of the base frame, represents as 6D vector
ω_t	Angular velocity of the base
$q_t, \dot{q}_t$	Joint angles, velocities
τ_t	Joint torques
g_t	Normalized projected gravity in the base frame
L_{rec}, L_{sup}, L_{RL}	loss of reconstruction, supervise, reinforcement learning
π_θ	Time optimization policy
Abbreviation	Full Form
TOP	Time Optimization Policy
VAE	Variational Autoencoder
RL	Reinforcement learning
MPC	Model predictive control
MLP	Multilayer perceptron

Thank you for the reviewer’s insightful comment. We have added more detailed information about both the training hardware platform and the robot used in the experiments and simulations in Section *Experiments*.

Specifically, all models were trained on a single NVIDIA RTX 4090 GPU (24 GB memory). The VAE model was trained for approximately 10 hours, while the reinforcement learning policy and the TOP model were each trained for around 5 hours.

In addition, we have introduced a new figure (Fig. R25) to present the detailed specifications of the humanoid robot platform used in our experiments. The robot stands 1.65 meters tall, weighs 60 kg, and features 41 degrees of freedom, including two 7-DoF arms (each weighing 6 kg) with a payload capacity of 3 kg per arm. This figure also summarizes the drive hardware and computing unit configuration used onboard.

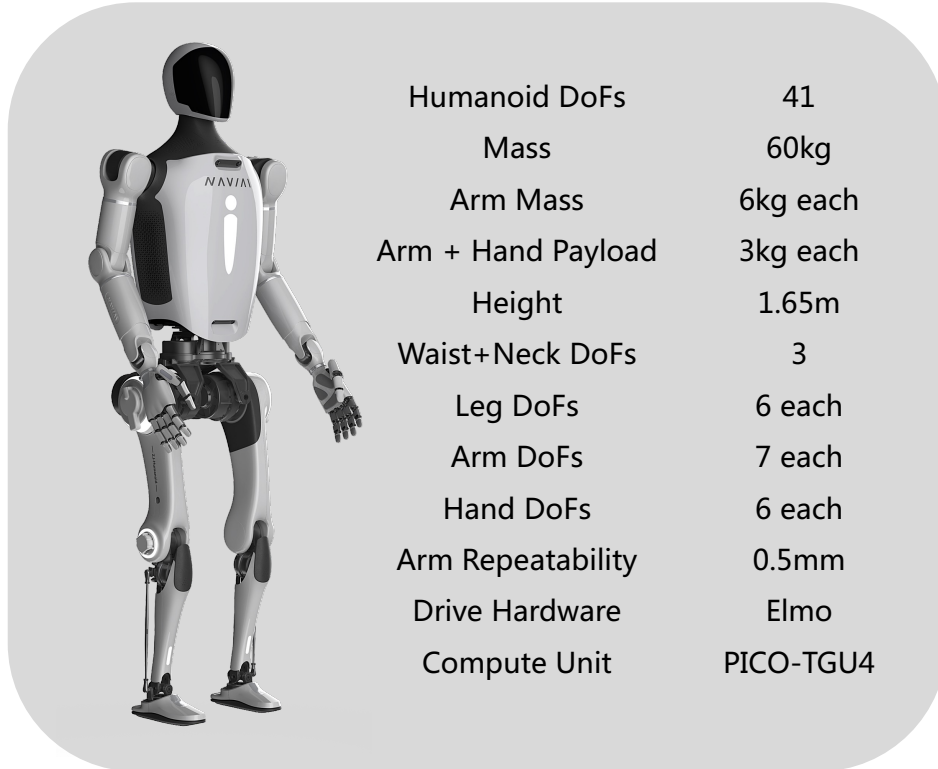


Figure R25: **Hardware Details.** Our robot stands 1.65m tall, weighs 60kg, and has 41 degrees of freedom, including two 6kg arms with 7-DoF each and a payload capacity of 3kg per arm.

We sincerely appreciate the reviewer’s suggestion. Due to the lack of space, we will incorporate the hardware details into **Appendix.E**.

Reply to Reviewer 6

This paper proposes a novel framework that balances the efficiency and stability of a humanoid robot by learning a policy to adaptively adjust the time interval of the upper body motion trajectory. The paper is well-structured and clearly presented. It introduces a three-stage process: first, an encoder is trained to extract features from the upper body motion; second, a reinforcement learning policy is trained using whole-body motion data to enable lower body balancing; and third, a supervised reinforcement learning framework is employed to learn the time interval policy.

We appreciate the reviewer’s clear understanding and positive feedback and hope that the explanation has fully addressed all of the reviewer’s concerns.

Due to the lack of space, we present our **paper with appendix**, the **response letter** and the **videos** in our project page <https://anonymous.4open.science/w/top-258F/>, we would like to bother the reviewer to have a look for a full paper.

Point 6.1

In the experiments, the robot performs contact-rich tasks such as lifting and opening doors. These tasks may introduce significant external forces, which could affect stability and accuracy. How does the proposed method handle unknown payloads or external contact forces during training?

Thank you for this insightful comment. We fully agree that tasks like lifting objects or opening doors naturally introduce significant and sometimes unpredictable external forces, which may affect the robot’s balance and end-effector accuracy.

In our framework, this risk is explicitly addressed through the combination of a kinematics-focused VAE and a dynamics-aware RL policy with strong domain randomization. Indeed, our VAE model is focused on reconstructing motion trajectories from kinematic data (joint positions and velocities), without directly encoding dynamic constraints such as joint torque limits or inertia variations.

To ensure feasible execution under real-world conditions, the low-level RL controller is

trained with extensive dynamics and contact randomization, as detailed in Table R34. During training, the robot experiences randomized payload masses (up to +3.5 kg extra), link inertia changes, varying motor gains, and unmodeled friction. Additionally, the environment applies external pushes and disturbance torques to mimic real contact interactions like door stiction or accidental collisions:

To confirm this capability in practice, we conducted additional load-adaptability experiments. The robot executed the same manipulation task while grasping objects with masses of 0.5 kg, 1 kg, and 3 kg. As summarized in Table R35, the success rate and end-effector accuracy remain robust, with only a slight increase in projected gravity. This demonstrates that the trained policy generalizes well to unknown payloads and contact variations.

Moreover, we have conducted additional experiments about the external disturbance tests of our policy. Figure R26 shows external disturbance tests of our policy. We designed scenarios where the robot is intentionally disturbed by external pushes from different directions.

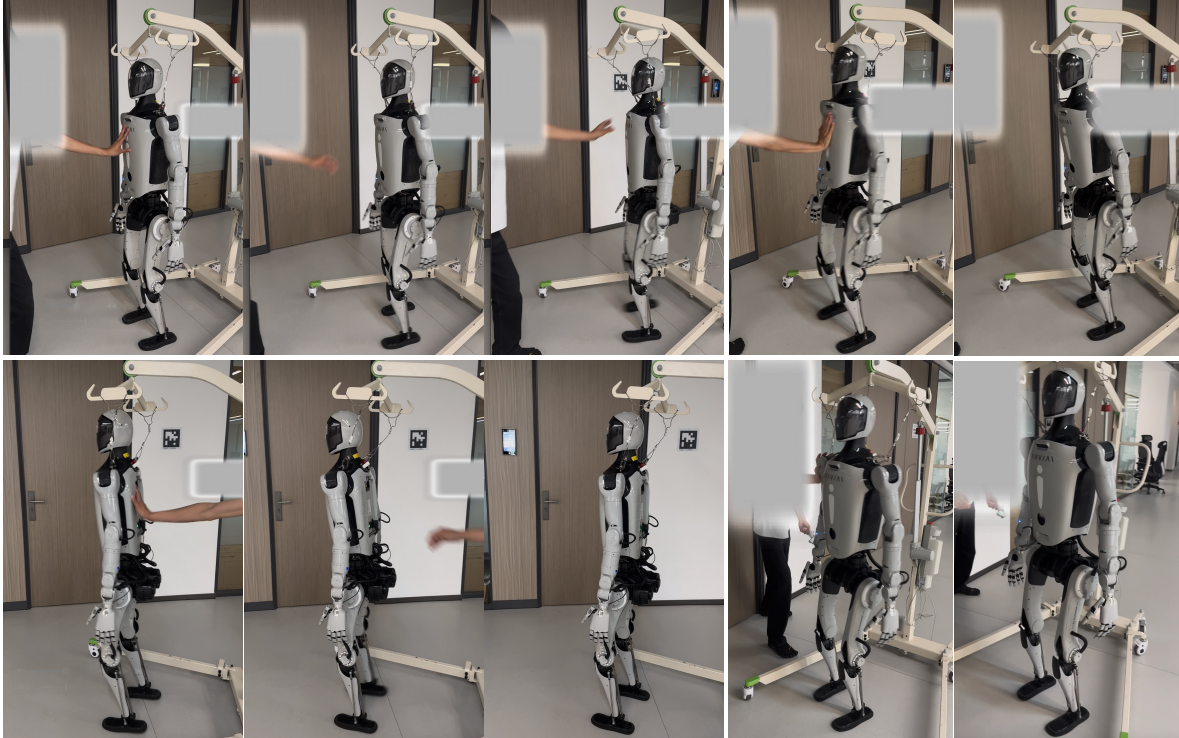


Figure R26: **External push disturbance tests.** The figure illustrates the robot can recover to balance while disturbed by external pushes from different directions.

These findings clearly demonstrate that our framework achieves physically feasible and

Table R34: Domain Randomization Settings

Term	Value
Observation	
Joint position noise	$\mathcal{U}(-0.1, 0.1)$ rad
Joint velocity noise	$\mathcal{U}(-1.0, 1.0)$ rad/s
Angular velocity noise	$\mathcal{U}(-0.1, 0.1)$ rad/s
Projected gravity noise	$\mathcal{U}(-0.05, 0.05)$
Dynamics	
Base COM displacement	$\mathcal{U}(-0.05, 0.05)$ m
Friction factor	$\mathcal{U}(0.35, 1.25)$
Link Inertia	$\mathcal{U}(0.8, 1.2) \times \text{default}$ kg $\cdot$ m ²
Added base mass	$\mathcal{U}(-5.0, 5.0) + \text{default}$ kg
Added link mass	$\mathcal{U}(0.8, 1.2) \times \text{default}$ kg
Added hand mass	$\mathcal{U}(-0.5, 1.5) + \text{default}$ kg
Payload mass	$\mathcal{U}(-1.0, 3.5) + \text{default}$ kg
Motor P gain	$\mathcal{U}(0.8, 1.2) \times \text{default}$
Motor D gain	$\mathcal{U}(0.8, 1.2) \times \text{default}$
Motor strength	$\mathcal{U}(0.9, 1.1) \times \text{default}$ N $\cdot$ m
Motor Damping	$\mathcal{U}(0.3, 4.0)$ N $\cdot$ s
Motor Delay	$\mathcal{U}(0, 10)$ ms
Episode	
Push robots	interval = 5s, $v_{xy} = 1.0\text{m/s}$, $\omega = 0.4\text{rad/s}$
Disturbance robots	interval = 8s, $\mathcal{U}(-30.0, 30.0)$ N $\cdot$ m
Initial robot base position xy	$\mathcal{U}(-1.0, 1.0)$ m
Initial robot base velocity	$\mathcal{U}(-0.1, 0.1)$ m/s

Table R35: Load Adaptability Test: Task performance under different payloads

Payload	Success Rate $\uparrow$	EE Pose Error $\downarrow$	Projected Gravity $\downarrow$
0.5 kg	95.2%	28.5 mm	2.731
1.0 kg	94.4%	30.1 mm	2.857
3.0 kg	92.7%	31.8 mm	3.142

stable behaviors under varying load conditions and external disturbances, which prove the stability of our method.

In summary, although the VAE itself does not model contact dynamics, the robust RL controller enhanced by domain randomization and disturbance effectively compensates for unknown external forces and payloads. We appreciate the reviewer’s thoughtful suggestion and will clarify this aspect and our empirical into **Appendix.C**.

To make our results fully transparent, we have released all the demonstration videos on our project page <https://anonymous.4open.science/w/top-258F/>.

Point 6.2

The proposed method decouples the control of the upper and lower body. The lower body controller is pretrained and not involved in the final (TOP) policy training. It would be valuable to discuss how much of the overall performance gain is due to a robust lower body controller versus an adaptive upper body controller.

We appreciate the reviewer’s careful observation regarding the respective contributions of the lower-body and upper-body modules in our decoupled framework.

Firstly, we agree that the robust lower-body controller is indeed a crucial component that ensures the floating base remains stable under upper-body momentum disturbance. Without this reliable base stabilizer, any time-optimized or learned upper-body trajectory would likely induce large base oscillations, violating balance constraints.

However, the main novelty and source of performance improvement in our framework lies in the **Time Optimization Policy (TOP)**, which adaptively adjusts the upper-body

Method	Time Cost ↓	Success Rate ↑	$E_{jpe}^{upper} ↓$	$E_{eepe}^{upper} ↓$	$E_{eece}^{upper} ↓$	$E_g ↓$	$E_{acc}^{lower} ↓$	$E_{action}^{lower} ↓$	$E_{acc}^{upper} ↓$	$E_{action}^{upper} ↓$
Comparative Results										
Fixed Root (reference)	15.0s	100.0%	0.0130	0.0164	0.0594	1.000	-	-	-	-
Exbody[10]	15.0s	92.46%	0.0376	0.0741	0.0923	3.432	21.65	0.7323	24.26	2.976
Ours (TOP)	40.5s	95.30%	0.0269	0.0270	0.0827	2.729	13.45	0.8592	10.41	1.601
Ablation Results										
Ours w/o TOP $\Delta t = 0.01s$	15.0s	82.43%	0.0301	0.0541	0.0962	4.438	16.87	0.9329	12.04	1.738
Ours w/o TOP $\Delta t = 0.03s$	45.0s	87.27%	0.0303	0.0434	0.0810	3.797	16.52	0.9473	11.26	1.647
Ours w/o TOP $\Delta t = 0.05s$	75.0s	92.41%	0.0299	0.0362	0.0813	3.573	15.79	0.9340	9.79	1.556
Ours w/o motion prior	34.5s	89.33%	0.0304	0.0313	0.0824	3.682	14.76	0.9130	10.92	1.640

Table R36: Comparative results and Ablation results. We execute more than 10000 motion clips from dataset $\mathcal{M}$ and unseen dataset $\mathcal{T}$, and report their time cost, success rate, mean metrics. Mean Metrics are only calculated based on the data of the robot standing in place, and the data of robot taking steps and falls will not be included in the calculation.

motion timing based on the real-time base and upper-body motion states. To clarify this, we performed experiments that isolate the impact of the TOP module. All the results are shown in the Table R36.

First, we would like to clarify the contribution of the upper-body Time Optimization Policy (TOP). Compared to the baseline with the same robust lower-body controller but fixed timestamps (Ours w/o TOP $\Delta t = 0.01s/0.03s/0.05s$), introducing the learnable TOP significantly reduces the end-effector pose error and improves the task success rate to 95.30%. This demonstrates that the adaptive adjustment of timestamp plays a key role in enhancing precision and stability. And compare to the baseline (Ours w/o TOP $\Delta t = 0.03s/0.05s$), our method not only improves the precision and stability, but also achieving higher average time efficiency.

Second, we emphasize that the robust lower-body controller, which integrates a learned motion prior (via the VAE) and reinforcement learning, is equally critical. When we remove the learned motion prior, the success rate drops sharply to 89.33% and precision, stability related metrics deteriorate significantly. This result not only proves that with the help of motion prior, our lower-body RL policy will become more robust, but also confirms that robust lower-body controller is indispensable for absorbing momentum transfers from the upper body, which also improve the overall performance.

In summary, these ablation results clearly show that both modules are indispensable

and complementary. The robust lower-body controller provides a stable base and absorbs momentum variations. The adaptive TOP fine-tunes the upper-body motion timestamps to minimize disturbance of motions and improve tracking accuracy. Together, they achieve a new balance between precision, stability, and time efficiency, which neither component alone could guarantee. We will revise our manuscript to clarify the source of our method’s performance gain.

In EXPERIMENT.C:

Together with the robust lower-body controller, the adaptive TOP module plays a complementary role, and both are essential to achieve the observed improvements in overall performance.

Point 6.3

The paper should elaborate on the dependency of the TOP training process on the pretrained lower body policy. For instance, if the lower body policy is trained with a different dataset or uses a different encoder, will the TOP policy still perform robustly?

We thank the reviewer for raising this important point about the dependency of our Time Optimization Policy (TOP) on the pretrained lower-body controller. Indeed, the current TOP is trained under the assumption that the robust lower-body policy provides stable base behavior in response to upper-body motion variations.

To evaluate this dependency, we conducted ablation experiments with variations of the lower-body controller: Training lower-body RL policy use different dataset (RL with dataset $\mathcal{M}'$), swapping in a lower-body controller pretrained with a different encoder structure (RL with encoder E'), as shown in the Figure R37.

The lower-body RL of baseline uses the dataset $\mathcal{M}$ for training, and use VAE with latent size of 64 and window size of 30 as motion prior. The dataset $\mathcal{M}'$ is part of the dataset $\mathcal{M}$, but only have 50% motion clips, will obviously lead to a decrease in RL robustness. For the encoder E' , since we need to ensure that the dimensions of TOP observation are consistent, we will not change the latent space of VAE (remain 64), but we will reduce the window size

to 15, which will weaken generalization to unseen motion sequences. We have already add the ablation study on the sensitivity of VAE parameters into **APPENDIX.G**.

Table R37: Robustness of the TOP policy under different lower-body controllers and encoders.

Methods	Success Rate (%) $\uparrow$	$\mathbf{E}_{jpe}^{upper}$ $\downarrow$	$\mathbf{E}_g$ $\downarrow$	$\mathbf{E}_{acc}^{upper}$ $\downarrow$
Baseline	95.30	0.0269	2.729	10.41
RL with dataset $\mathcal{M}'$	89.40	0.0338	3.545	10.94
RL with encoder E'	91.85	0.0301	3.312	10.65

In both cases, we observed that the TOP alone could partially adapt to moderate changes in the lower-body dynamics, but the overall performance degraded compared to the original pair: task success dropped by 4–6% and upper joint tracking error increased. This indicates that while the TOP is robust to moderate variations, it does benefit from being co-tuned with a compatible lower-body policy that shares the same motion prior distribution and encoder structure.

This design is intentional: by training the lower-body controller first with domain randomization and a consistent motion representation, we ensure that the base stabilizer can handle diverse upper-body trajectories. Then, the TOP learns to optimize time allocation specifically for the base stabilizer’s dynamics.

Therefore, our framework does not rely on rigid assumptions but does require that both modules share compatible motion encodings. In future work, we plan to explore end-to-end co-training or meta-adaptation to further relax this dependency.

Thanks to the reviewers for pointing out the problems. Due to the lack of space, we will not include this explanation in the revised paper, but will add into **APPENDIX.K**.

Point 6.4

The final end-effector motion of the upper body is influenced by both arm tracking errors and base motion instability. It would be helpful to analyze whether the observed improvements in end-effector accuracy are primarily due to a more stable floating base.

Thank you for the reviewer’s insightful comment. We agree with the reviewer that the

final end-effector (EE) accuracy depends on both the arm tracking quality and the stability of the floating base.

In our real-world experiments, we did measure both components simultaneously using the motion capture system, which tracks markers on the base link and the EE link, as shown in Figure R27. This allows us to quantify how much of the EE pose drift comes from base motion versus arm-level tracking.

Table R38 summarizes the measured pose errors for the base and the end-effector. It is evident that the drift of the base (~ 30 mm) accounts for the majority of the total end-effector pose drift (~ 33 mm). To further clarify this, Figure R28 shows a decomposition plot of the base drift, the end-effector pose error, and the residual other error. The latter remains consistently around 5 mm, demonstrating that the dominant source of the displacement is the floating base oscillation rather than inaccuracies in joint-level control.

Table R38: Tracking Error Data Recorded by Motion Capture Device

Type	x(mm)	y(mm)	z(mm)	roll(rad)	pitch(rad)	yaw(rad)
Base Error	29.232	2.4846	1.1698	0.0163	0.0119	0.0032
End-effector Error	33.735	3.0494	1.4913	0.0502	0.0127	0.0103

Figure R28 shows this decomposition over time: the EE drift strongly correlates with base oscillations, while the residual error stays relatively stable and low. This evidence confirms that the observed improvement in EE precision is primarily achieved through better floating-base stability, enabled by our time optimization policy (TOP). Meanwhile, our PD tracking keep the joint-level residual error within a small range.

This analysis confirms that our time optimization policy (TOP) effectively reduces base motion disturbance, which is the main factor limiting final end-effector precision. For transparency, we will update our revised manuscript (TABLE IV) to clarify this problem.

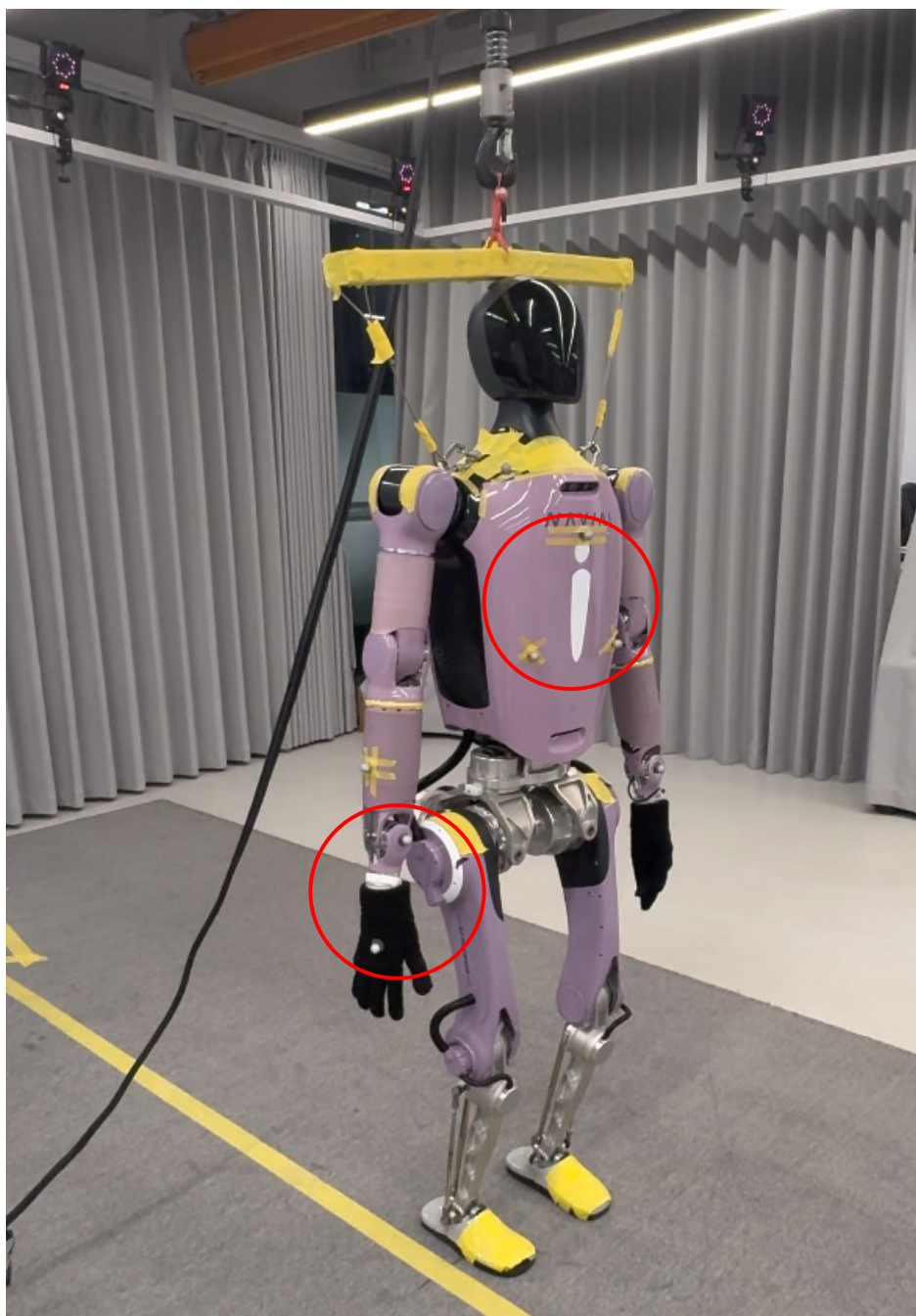


Figure R27: **Motion Capture Setup.** Red markers indicate tracked base and EE points.

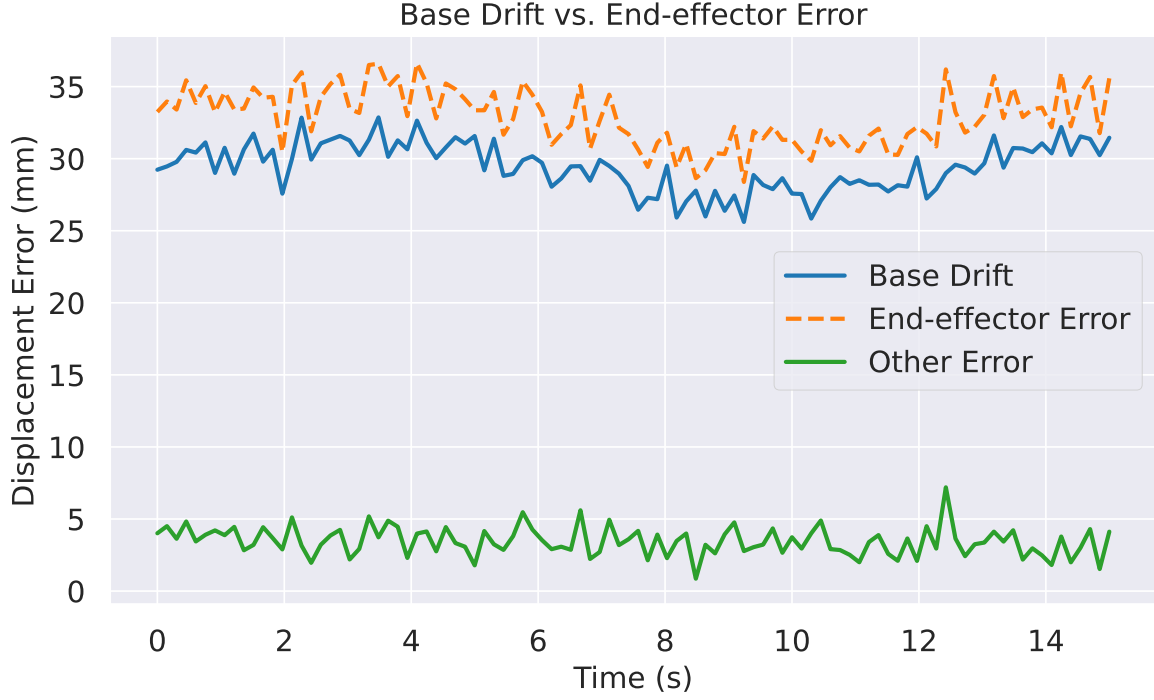


Figure R28: **Decomposition of end-effector error.** The plot shows that the majority of the end-effector pose drift stems from the base trajectory drift (blue curve), while the other error (green curve) remains very low (<5 mm). This demonstrates that improving base stability directly enhances the final end-effector precision.

Minor comments:

Point 6.5

Figure 5 is not cited in the text. Additionally, the meaning of the red circles in Figure 5 should be explained. There is a typo in the third line of the caption: “which which”.

Thanks for the reviewer’s comment.

These figures are now properly referenced and discussed at relevant points in the text to enhance clarity and coherence between the visual elements and their corresponding descriptions.

Thank you for your valuable comments. We have addressed the issue as follows: (1) We have added an explicit in-text citation for Fig. 5 in the revised manuscript. (2) We have

corrected the typo "which which" in the caption of Fig. 5. (3) We have clarified the meaning of the red circles in Fig. 5, indicating that they represent the selected configurations with balanced performance.

Point 6.6

It would be beneficial to analyze the impact of upper body motion on system momentum, such as changes in the floating base velocity.

Thank you for this insightful comment on analyze the impact of upper body motion on system momentum. We fully agree that the physical mechanism behind instability stems from the conservation of linear and angular momentum when large joint speeds and accelerations generate reaction forces on the floating base. To clarify this, we now provide a supplementary theoretical approximation to quantify the effect.

The total angular momentum $\mathbf{L}$ of the floating base plus the upper-body can be expressed as:

$$\mathbf{L} = \mathbf{I}_B \boldsymbol{\omega}_B + \sum_{i=1}^n I_i \dot{q}_i \quad (30)$$

where $\mathbf{I}_B$ is the inertia tensor of the base, $\boldsymbol{\omega}_B$ is the base angular velocity, I_i is the inertia of joint i , $\dot{q}_i$ is the joint angle velocity. Taking the time derivative yields the net momentum change:

$$\dot{\mathbf{L}} = \mathbf{I}_B \dot{\boldsymbol{\omega}}_B + \sum_{i=1}^n I_i \ddot{q}_i \quad (31)$$

in free-floating or lightly constrained systems, any increase in joint accelerations $\ddot{q}_i$ generates an equal and opposite reaction on the base:

$$\mathbf{I}_B \dot{\boldsymbol{\omega}}_B = - \sum_{i=1}^n I_i \ddot{q}_i. \quad (32)$$

Thus, the floating base must compensate for the angular impulse caused by fast upper-body motion. If $\ddot{q}_i$ becomes large (e.g., due to commands), the base rotation $\boldsymbol{\omega}_B$ and linear drift will increase proportionally.

This confirms that controlling joint accelerations effectively controls the induced momentum variation:

$$\Delta \boldsymbol{\omega}_B \propto - \int_{t_0}^t \sum_{i=1}^n I_i \ddot{q}_i dt. \quad (33)$$

that underlines why our TOP policy explicitly shapes the time scaling to keep $\ddot{q}_i$ bounded, thus reducing $\dot{\mathbf{L}}$ and stabilizing the robot during manipulation.

Also, according to the Eq. 30, faster joint angle velocity $\dot{q}_i$, the larger the disturbance to the base’s rotational state. The time optimization policy (TOP) effectively reduces peak joint velocities $\dot{\mathbf{q}}$, preventing abrupt momentum transfer to the floating base. This explains why we evaluate *joint acceleration* and *joint velocity* to measure performance of methods.

To further illustrate the relationship between momentum variation, ZMP and stability of robot, as shown in the Figure R29. The figures plot the measured floating base velocity over time for both our method (TOP) during a fast upper-body motion. The results show that the TOP effectively suppresses sudden peaks in base velocity by slowing down the arm when the body momentum is excessive. The definition of d is shown in the **APPENDIX.H**.

Thanks to the reviewers for pointing out the problems. Due to the lack of space, we will not include this in revised paper, but will add the relationship into **APPENDIX.I**.

Point 6.7

Please include the training time of the proposed method and compare it to the training efficiency of Exbody.

Thank you for the reviewer’s suggestion. We agree that reporting the training time and comparing it to relevant baselines such as Exbody, OmniH2O is helpful for evaluating the training efficiency of our method, as shown in the Table ??.

Specifically, our framework requires approximately 10 hours to train the VAE motion prior and 5 hours each for the reinforcement learning policy and the TOP module on a single NVIDIA RTX 4090 GPU.

Under this setting, Exbody required approximately 25 hours of training, while OmniH2O required about 32 hours. By contrast, our full pipeline (VAE + RL + TOP) completes training in around 20 hours.

These results demonstrate that our approach achieves comparable training efficiency with these state-of-the-art baselines while providing task-specific time optimization and stable standing manipulation.

Thanks to the reviewers for pointing out the problems. Due to the lack of space, we

Momentum, ZMP, and Stability

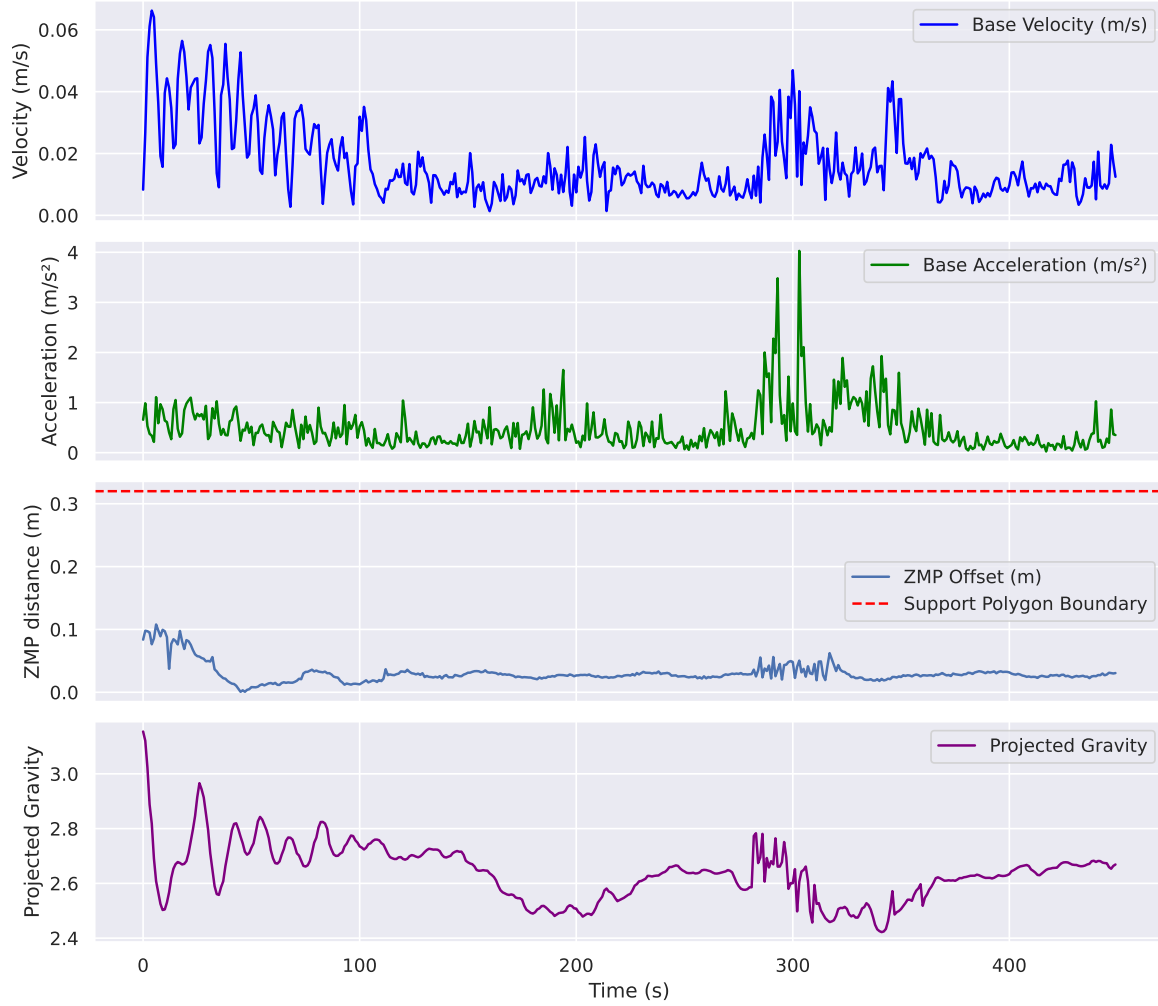


Figure R29: **Momentum, ZMP and stability.** The figure shows the measured floating base velocity and acceleration during a fast upper-body motion. The TOP smooths out large momentum variations, helping to keep the ZMP within the support region and maintain balance.

Table R39: Comparison of Training Efficiency

Method	Hardware	Total Training Time (hrs)
VAE	1x RTX 4090 (24GB)	10 (approx.)
RL	1x RTX 4090 (24GB)	5 (approx.)
TOP	1x RTX 4090 (24GB)	5 (approx.)
Ours (VAE + RL + TOP)	1x RTX 4090 (24GB)	20 (approx.)
Exbody[10]	1x RTX 4090 (24GB)	25 (approx.)
OmniH2O[19]	1x RTX 4090 (24GB)	32 (approx.)
Mobile-Television[9]	1x RTX 4090 (24GB)	22 (approx.)

present the Training Efficiency Table in the **Appendix.J** of the paper.

Reply to AE/Reviewer 7

Overall, the reviewers agree on the significance of the topic covered in this manuscript. The proposed approach is demonstrated with simulation and experiment. Yet, they also raised some significant concerns that prevent the manuscript from being accepted in its current state. Besides the detailed comments from reviewers as seen in the attached file, there are additional two key issues:

We appreciate the reviewer’s clear understanding and positive feedback and hope that the explanation has fully addressed all of the reviewer’s concerns.

Due to the lack of space, we present our **paper with appendix**, the **response letter** and the **videos** in our project page <https://anonymous.4open.science/w/top-258F/>, we would like to bother the reviewer to have a look for a full paper.

Point 7.1

Whole-body Controller (WBC) takes into account all the joint states, including joint position, velocity and torque, and also the states of operational space such as end-effector (feet or hands) and also the angular momentum. It is a good and mature tool to handle multi-tasks. The stability is easily described using the general WBC framework (even if there are some different version WBC with different solver, like QP or nullspace). The biggest problem with this study is that authors tried to decouple the upper and lower body and tried to stabilize upper body on the lower body with limited stability. This does not seem to make sense. Authors did not quantitatively define or cite any stability criterion. The challenging problem with the loco-manipulation is how to guarantee the stability of locomotion and manipulation simultaneously, especially for those conflicted tasks (how to trade-off). The experiments in this study did not show such challenging tasks. WBC should be just working for these simple experiments. Authors should prove using really challenging tasks that the proposed method is really SOTA, in which cases WBC can not work.

We appreciate the reviewer’s detailed comments and agree that Whole-Body Control

(WBC) is indeed a mature and powerful framework for addressing multi-task loco-manipulation problems.

Firstly, we would like to clarify that our framework does not aim to *replace* classical WBC or MPC+WBC methods in general-purpose scenarios. Instead, our main contribution is to demonstrate that a learned Time Optimization Policy (TOP) combined with decoupling controller can achieve competitive or even superior performance in certain practical humanoid manipulation tasks where real-time computational constraints and task complexity make full online WBC planning costly or sub-optimal.

Table R40: Inference Performance of our method and MPC

Module	Latency (ms)	Device	Resource Utilization
VAE Encoder	1.4621	PICO-TGU4	CPU Utilization: 6%
RL policy	1.6925	PICO-TGU4	CPU Utilization: 7%
TOP Policy	1.7267	PICO-TGU4	CPU Utilization: 7%
NMPC	23.722	PICO-TGU4	CPU Utilization: 60%
WBC	1.3141	PICO-TGU4	CPU Utilization: 12%

According to the computational cost, it will be As shown in Table R40, our entire control stack — including the VAE encoder, RL policy, and TOP module — maintains extremely low inference latency (under 2 ms for each module) and very low CPU utilization on an embedded device (Intel PICO-TGU4). In contrast, the NMPC baseline introduces significant computational overhead, with an average solve time of 23.7 ms and over 60% CPU utilization for each iteration, which can limit the real-time adaptability for high-frequency whole-body replanning.

This significant gap in online inference cost demonstrates that our method is better suited for real-time high-frequency deployment on embedded hardware while still achieving robust performance for complex loco-manipulation. With the help of motion prior to the VAE and introducing the learnable time adaptation via the TOP, our framework effectively balances control performance and hardware feasibility, which would be challenging to achieve with pure NMPC+WBC when facing complex, adaptive whole-body tasks in dynamic real-world environments.

To address the reviewer’s concerns, we implement a conventional non-linear Model Predictive Control (NMPC) combined with Whole-Body Control (WBC). This baseline is widely adopted in quadrupedal and bipedal robots for dynamic motion planning [30, 16] and open-source frameworks[31].

Our NMPC solves for a centroidal trajectory under a fixed time schedule with a predefined contact sequence. To ensure real-time feasibility, the continuous optimal control problem is discretized via direct multiple shooting, and solved using a Sequential Quadratic Programming (SQP) method with real-time iteration. The joint pose references come from the motions in GRAB[17], ensuring consistency with the desired task-level motion. The continue nonlinear MPC problem formulation is following:

$$\left\{ \begin{array}{l} \min_{\mathbf{u}(\cdot)} \Phi(\mathbf{x}(T)) + \int_0^T \mathbf{l}(\mathbf{x}(t), \mathbf{u}(t), t) dt \\ \text{subject to } \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), t) \\ \mathbf{g}_{eq}^1(\mathbf{x}(t), \mathbf{u}(t), t) = 0 \\ \mathbf{g}_{eq}^2(\mathbf{x}(t), t) = 0 \\ \mathbf{h}_{in}(\mathbf{x}(t), \mathbf{u}(t), t) \geq 0 \\ \mathbf{x}(0) = \mathbf{x}_0 \end{array} \right. \quad (34)$$

where $\mathbf{x}(t)$ and $\mathbf{u}(t)$ are the state and control input at time t . $\mathbf{g}_{eq}^1$ represents equality state-input-related constraints, and $\mathbf{g}_{eq}^2$ represents equality constraints only related to state. $\mathbf{h}_{in}$ includes state constraints and state-input-related constraints, such as joint position and velocity limits, friction cone constraints, etc. $\mathbf{x}_0$ is the current measured state of robot.

An optimization-based whole-body controller helps to map the optimized trajectory obtained from the nonlinear MPC to the humanoid robot’s joint torque, which considers the full nonlinear rigid body dynamics. While formulating the constrained hierarchical optimization problem, in our baselines we instead use a single QP to construct the constrained optimization problem, in which the low-priority tasks are embedded in the cost function, and high-priority tasks are treated as constraints. The list of tasks is shown in the Table R41. The first two blocks of tasks prevent the robot from violating the equality and inequality constraints of torque, force, and joint configuration. And the least squares tasks guide the robot to track the foot motion with a higher weight than the base reference motion, which prevents the robot from taking advantage of leg inertia to track the base motion in underactuated directions.

Table R41: WHOLE-BODY CONTROL TASKS

Type	Task
=	Floating base equations of motion
	No Motion at the contact points
$\geq$	Torque limits
	Wrench Friction cone
$Q_i^2 \cdot ^2$	Base Linear and Angular Tracking($Q_i = 1.0$)
	Joint Position Tracking($Q_i = 100.0$)
	Contact Force Tracking($Q_i = 0.1$)

We compare our TOP with this NMPC+WBC baseline and with a Fixed Root reference policy that we fixed the root of robot, and directly execute the upper-body motions, which means the motions can be perfectly executed with the highest control accuracy. We use the motions from dataset $\mathcal{M}$ and unseen dataset $\mathcal{T}$ across methods, and report performance on the following criteria:

- **Time Cost:** Average time cost of per 1500 motion clips (15s of origin data), which shows the time efficiency of executing the target motions.
- **Success Rate:** We define that the robot executes the upper body motion without falling, this motion clip will be labeled as successful motion.
- **Precision:** upper joint position error $\mathbf{E}_{jpe}^{\text{upper}}$, upper end effector position error in world frame $\mathbf{E}_{eepe}^{\text{upper}}$, upper end effort orientation error in world frame $\mathbf{E}_{eeoe}^{\text{upper}}$.
- **Stability:** projected gravity $\mathbf{E}_g$, lower joint acceleration $\mathbf{E}_{acc}^{\text{lower}}$, lower action difference $\mathbf{E}_{action}^{\text{lower}}$.
- **Smoothness:** upper joint acceleration $\mathbf{E}_{acc}^{\text{upper}}$, upper action difference $\mathbf{E}_{action}^{\text{upper}}$.

The results are shown in Table R42. It is worth noting that the output of NMPC+WBC[16] is different from the actions of TOP outputs, so we only use the projected gravity and acceleration to evaluate the NMPC+WBC, and we slow down the target motions of NMPC for better comparison with our TOP method.

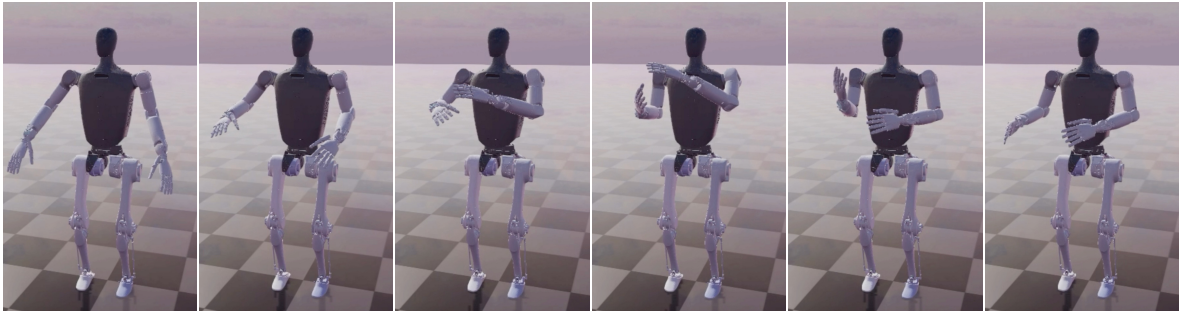
Method	Time Cost ↓	Success Rate ↑	E_{jpe}^{upper} ↓	E_{eepe}^{upper} ↓	E_{eeoe}^{upper} ↓	E_g ↓	E_{acc}^{lower} ↓	E_{action}^{lower} ↓	E_{acc}^{upper} ↓	E_{action}^{upper} ↓
Comparative Results										
Fixed Root (reference)	15.0s	100.0%	0.0130	0.0164	0.0594	1.000	-	-	-	-
NMPC+WBC[16]	35.0s	89.60%	0.0278	0.0437	0.0857	2.938	14.98	-	11.47	-
Ours (TOP)	40.5s	95.30%	0.0269	0.0270	0.0827	2.729	13.45	0.8592	10.41	1.601

Table R42: Comparative results. We execute more than 10000 motion clips from dataset $\mathcal{M}$ and unseen dataset $\mathcal{T}$, and report their time cost, success rate, mean metrics. It is worth noting that the output of NMPC+WBC[16] is different from the actions of TOP outputs, so we only use the projected gravity and acceleration to evaluate the NMPC+WBC, and we slow down the target motions of NMPC for better comparison with our TOP method.

We observe that our proposed TOP consistently achieves lower joint and end-effector position errors compared to the NMPC+WBC baseline, with reductions of approximately 3.2% and 38.2% respectively. This indicates significantly improved pose tracking performance, which is critical for manipulation tasks under dynamic conditions. Moreover, the gravity projection error E_g and lower-body acceleration error E_{acc}^{lower} are also reduced, demonstrating that our method better maintains base stability when interacting with varying external forces or loads. The success rate is increased by 5.7 percentage points, which validates the practical reliability of our planner under diverse scenarios. The only compromise is that our strategy needs to slow down our motions more, so we will lose time efficiency. In short, our TOP achieves better pose tracking and base stability than the NMPC baseline and replay method, while successfully completing the tasks under varying durations and loads.

One of the motions executed by NMPC+WBC are shown in the Figure R30.

Figure R30: **Screenshots of Motions Executed by NMPC+WBC.** We realize the NMPC+WBC in raisim[18] simulator.



While we cannot release our full implementation due to hardware-specific integration of our MPC+WBC baseline, we provide references to open-source bipedal NMPC controllers [32] and quadrupedal MPC+WBC frameworks [31] that follow similar principles.

Additionally, to demonstrate the applicability in **challenging loco-manipulation tasks**, we conducted new real-world tests involving dynamic walking while performing object interactions, such as opening a fridge while stepping forward and pressing a button while turning. These tasks introduce strong coupling between locomotion and upper-body manipulation with significant momentum transfer. The experimental videos have been provided in our supplementary material and open repository (<https://anonymous.4open.science/w/top-258F/>), as shown in Figure R31.



Figure R31: **Real-world loco-manipulation tasks.** Our robot performs door opening while stepping, and pushing a button while turning, requiring simultaneous stability and precise end-effector tracking.

Figure R31 shows the mobile operation scenarios of our TOP policy. In these new test cases, the humanoid robot executes upper-body tasks (e.g., pressing button and open fridge) and omnidirectional walking. The motion prior and time optimization policy work jointly with

the lower-body RL policy to adapt the timestamps, lower-body torques and maintain balance. This demonstrates that the proposed decoupled yet closed-loop framework generalizes well to locomotion plus manipulation tasks.

We sincerely appreciate the reviewer’s suggestion, which helped us strengthen the paper’s experimental evidence and we have incorporated this new experiments and discussion into **Appendix.C**.

Finally, we agree that a more rigorous stability criterion should be provided. We will provide a clear quantitative proof of our decoupling controller in the following response to the **Point 7.2**.

Point 7.2

Data driven method with limited experiment results is not able to prove the stability. Authors are suggested to prove, e.g. using Lyapunov or other analytical theorem, to make the proposed method rigorously. Decoupling upper and lower body often means that some optimized solution would be lost. For example, to pick up a heavy object while keeping balance, usually we would adjust every joints of our body simultaneously to make the CoM above the support polygon (dynamic or quasi-static or static), we do not decouple our upper and lower body, that is not an optimal solution. The stability must be analyzed with the whole body as a complete system.

We appreciate the reviewer’s insightful comment regarding the rigorous proof of stability and the limitations of purely data-driven methods when used alone. We fully agree that any practical humanoid loco-manipulation controller must ensure closed-loop stability for the whole body, especially given the dynamic coupling between the upper and lower-body.

Firstly, to strengthen our analysis beyond pure empirical tests, we have added a quantitative **regression-based analysis** of the relationship between key dynamic variables of precision, stability and time efficiency. Specifically, we have:

- Applied multiple linear regression and polynomial regression fitting to model the quantitative relationship between time efficiency(Δt), stability(projected gravity), and precision(EE position Error).

- Computed the empirical Pareto front[11, 12] based on the experimental data to highlight the set of non-dominated solutions, which illustrates the achievable trade-off boundary between the competing objectives.

Figure R32 shows the updated analysis. In this figure:

- Each scatter point represents an individual real-world manipulation trial with its measured Δt , projected gravity, and end-effector pose error.
- The fitted regression surface captures the overall trend that shorter Δt typically results in slightly increased the projected gravity and pose error due to execute faster upper-body motions.
- The highlighted Pareto front (black) reveals the empirical boundary where no further improvement can be made in one metric without sacrificing at least one of the others.

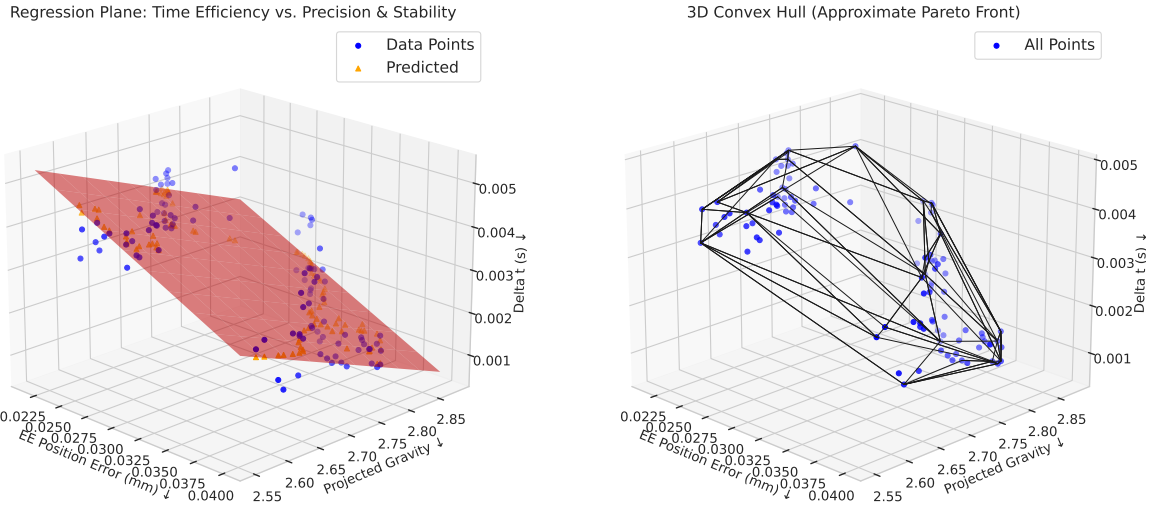


Figure R32: **Regression and Pareto front analysis.** The figure shows the multi-objective trade-off among time, stability, and precision, with the fitted regression surface and the extracted Pareto optimal front (right).

The regression model confirms that the observed improvements in end-effector accuracy are not independent but are directly coupled to the robot’s base stability and Δt of TOP outputs. For instance, motions with longer Δt that means the motions will be slowed down, generally allow the lower-body controller more time to compensate for upper-body

momentum, resulting in lower projected gravity (base drift) and higher end-effector precision. Conversely, pushing the robot to complete motions faster often requires more rapid upper-body motions, which inherently injects higher momentum into the base, slightly degrading stability and precision.

Then, to address the concern about formal stability criteria, we explicitly evaluate the ZMP condition during the experiments. Specifically, we show that our optimized policy keeps the projected ZMP trajectory within the support polygon during task execution, which implies that the system remains dynamically balanced in both static and quasi-static regimes. This complements our empirical results and provides a theoretically grounded indicator of base stability.

We want to explain the definition of ZMP at the beginning. ZMP is the point on the ground where the horizontal component of the moment of the ground reaction forces becomes zero, assuming the ground is a horizontal plane[13]. A necessary condition for the locomotion mechanism to be in dynamic equilibrium is that the candidate ZMP is inside the support polygon. Furthermore, to handle non-planar surfaces, the concept of the Zero Moment Line (ZML) is introduced, which extends the ZMP concept to account for complex contact surfaces[14]. Then we give the calculation process as follows[15]:

$$\boldsymbol{\tau} = \mathbf{p}_{zmp} \times \mathbf{f} + \boldsymbol{\tau}_p \quad (35)$$

$$\dot{\mathbf{P}} = M\mathbf{g} + \mathbf{f} \quad (36)$$

$$\dot{\mathbf{L}} = \mathbf{p}_{com} \times M\mathbf{g} + \boldsymbol{\tau} \quad (37)$$

where $\mathbf{p}_{zmp}$ is the position of ZMP, $\boldsymbol{\tau}_p$ is the moment about the ZMP point, and the $\mathbf{f}$ represents the ground reaction force. P , L , M , $\mathbf{p}_{com}$ represents the linear momentum, the angular momentum, the total mass, the center of mass (CoM) position of the system, respectively, and $\mathbf{g} = [0, 0, -g]^T$. By the definition of ZMP, we have:

$$\tau_{p,x} = \tau_{p,y} = 0 \quad (38)$$

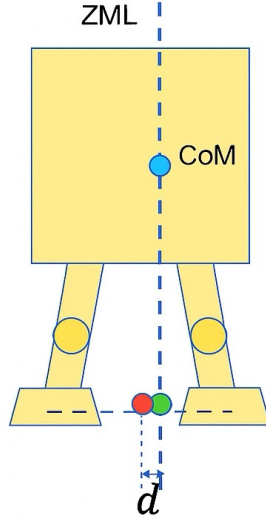
Solving the above equations with respect to $\mathbf{p}_{zmp}$, we have

$$p_{zmp,x} = \frac{Mgp_{com,x} + p_{com,z}\dot{P}_x - \dot{L}_y}{Mg + \dot{P}_z} \quad (39)$$

$$p_{zmp,y} = \frac{Mgp_{com,y} + p_{com,z}\dot{P}_y + \dot{L}_x}{Mg + \dot{P}_z} \quad (40)$$

According to the Eq.39, ZMP at different heights $p_{com,z}$ lies at different locations, whose trajectory forms a line of ZMPs, called ZML [14]. The condition for stable locomotion is that the ZML must cross the support polygon, as shown in Fig R33.

Figure R33: **ZML cross the support polygon.** The red dot represents the approximated center of the support polygon, and the green dot on the ZML is the projected point at the same horizontal height as the red dot. d represents the distance between the center of the support polygon and the projected point of ZML, which can describe the stability of the system. The smaller d is, the more stable the system is.



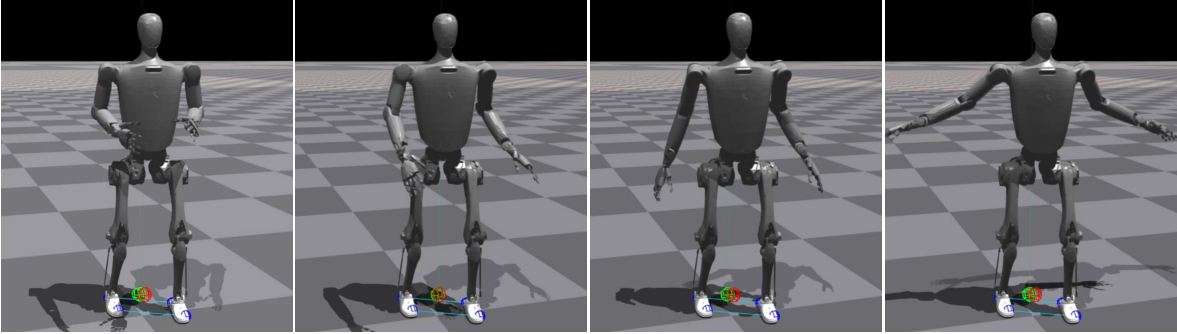
Then, we need to estimate the support polygon of the humanoid robot to determine the relationship between ZML and the support. In general cases, the support polygon is determined by computing the convex hull of all contact points for scenarios with multiple contact points. Therefore, we calculate the ZML lines in simulation, as shown in the Fig R34.

To quantitatively assess the dynamic stability of our policy, we compute the signed distance d between the ZML projection point and the center of the support polygon:

$$d = \min_{p_{csp} \in \mathcal{S}} \|p_{csp} - \text{proj}_{ZML}(p_{csp})\|_2, \quad \text{proj}_{ZML}(p_{csp}) \in \mathbb{R}^2 \quad (41)$$

where $\mathcal{S}$ denotes the support polygon of current contact points, and $\text{proj}_{ZML}(p_{csp})$ is the projected ZML point in the horizontal plane. If $d \in [0, 0.32]$, the ZML point is inside the support polygon; if $d \in (0.32, 0.36]$, it lies near the edge of support polygon; and if $d \in (0.36, +\infty)$, the ZML has exited the support region, indicating potential instability. It is worth noting that the closer d is to 0, the system is more stable and satisfy the ZMP constraint.

Figure R34: **ZML and Support Polygon in Isaacgym.** The red point represents the center of the support polygon, and the green dot on the ZML is the projected point at the same horizontal height as the red dot. The blue points and the light blue lines are the support polygon.



To demonstrate that our time optimization policy (TOP) maintains the Zero-Moment Point (ZMP) condition throughout task execution, we plot the time series of d for representative tasks. As shown in Fig. R35, the distance remains positive during the entire motion, verifying that the ZML stays well within the support polygon at all times, and thus the robot’s base remains dynamically stable despite rapid upper-body motions.

This quantitative evidence demonstrates that our closed-loop policy stabilization effectively ensures that the floating base remains within feasible support constraints, even under varying contact conditions and dynamic upper-body perturbations.

Furthermore, although our design uses a modular decoupling for practical control abstraction, the final controller is **not open-loop**: the RL lower-body policy continuously observes the real-time upper-body motion via the latent motion prior and adapts lower-body actions accordingly. As shown in Table R43, the RL policy is explicitly encouraged to minimize base drift, linear/angular velocity deviations, and physically feasible whole-body motions:

- **Base linear velocity** and **base angular velocity** terms penalize excessive base oscillations.
- **Base orientation** cost stabilizes the base roll/pitch under external moments induced by upper-body motions.
- **Stand still** and **action penalty** terms further ensure smooth, coordinated lower-body

actions.

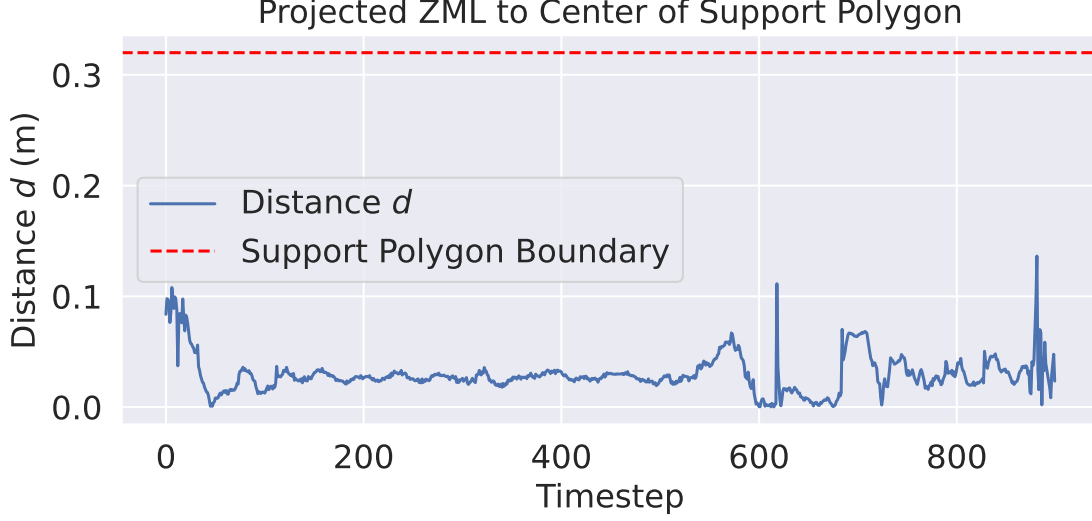


Figure R35: **Stability metric d over time.** The curve shows the distance between the ZML projection and the center of support polygon. The distance remains ≤ 0.32 , confirming that the ZMP condition is always satisfied.

Taken together, these additions the multi-variable regression, the ZMP feasibility check, and the closed-loop reward shaping strengthen the theoretical soundness of our approach.

Thanks to the reviewers for pointing out the problems, which further strengthens the completeness and practicality of our framework, and we have incorporated the regression analysis into **Appendix.B** and the ZMP proof into ZMP proof into **Appendix.H**.

Table R43: Key RL Reward Terms for Stabilization

Term	Expression	Weight
Base linear velocity (xy)	$\exp[-4(\mathbf{v}_{xy}^2)]$	+3.0
Base angular velocity (xy)	$\Sigma \omega_{xy}^2$	-0.1
Base orientation	$\Sigma \mathbf{g}_{xy}^2$	-1.5
Stand still pose	$\exp[-10\ \mathbf{q}_t^{\text{leg}} - \mathbf{q}^{\text{leg,ref}}\ _2]$	+1.0
Action rate penalty	$\ \mathbf{a}_t - \mathbf{a}_{t-1}\ _2^2$	-0.2
Action acceleration penalty	$\ \mathbf{a}_t + \mathbf{a}_{t-2} - 2\mathbf{a}_{t-1}\ _2^2$	-0.2

References

- [1] Kunihiko Fukushima. “Visual feature extraction by a multilayered network of analog threshold elements”. In: *IEEE Transactions on Systems Science and Cybernetics* 5.4 (1969), pp. 322–333.
- [2] Davis Rempe et al. “Humor: 3d human motion model for robust pose estimation”. In: *Proceedings of the IEEE/CVF international conference on computer vision*. 2021, pp. 11488–11499.
- [3] Wenyang Zhou et al. “Emdm: Efficient motion diffusion model for fast and high-quality motion generation”. In: *European Conference on Computer Vision*. Springer. 2024, pp. 18–38.
- [4] Félix G Harvey et al. “Robust motion in-betweening”. In: *ACM Transactions on Graphics (TOG)* 39.4 (2020), pp. 60–1.
- [5] Jungdam Won, Deepak Gopinath, and Jessica Hodgins. “Physics-based character controllers using conditional vaes”. In: *ACM Transactions on Graphics (TOG)* 41.4 (2022), pp. 1–12.
- [6] Agon Serifi et al. “Vmp: Versatile motion priors for robustly tracking motion on physical characters”. In: *Computer Graphics Forum*. Vol. 43. 8. Wiley Online Library. 2024, e15175.
- [7] Guy Tevet et al. “Human motion diffusion model”. In: *arXiv preprint arXiv:2209.14916* (2022).
- [8] Jiefeng Li et al. “GENMO: A GENeralist Model for Human MOTion”. In: *arXiv preprint arXiv:2505.01425* (2025).
- [9] Chenhao Lu et al. “Mobile-television: Predictive motion priors for humanoid whole-body control”. In: *arXiv preprint arXiv:2412.07773* (2024).
- [10] Xuxin Cheng et al. “Expressive whole-body control for humanoid robots”. In: *arXiv preprint arXiv:2402.16796* (2024).
- [11] Kristof Van Moffaert and Ann Nowé. “Multi-objective reinforcement learning using sets of pareto dominating policies”. In: *The Journal of Machine Learning Research* 15.1 (2014), pp. 3483–3512.

- [12] Yusuf Aydin et al. “A computational multicriteria optimization approach to controller design for physical human-robot interaction”. In: *IEEE Transactions on Robotics* 36.6 (2020), pp. 1791–1804.
- [13] Miomir Vukobratović and Branislav Borovac. “Zero-moment point—thirty five years of its life”. In: *International journal of humanoid robotics* 1.01 (2004), pp. 157–173.
- [14] Tilen Brecej and Tadej Petrič. “Zero moment line—universal stability parameter for multi-contact systems in three dimensions”. In: *Sensors* 22.15 (2022), p. 5656.
- [15] Weiji Xie et al. “Humanoid whole-body locomotion on narrow terrain via dynamic balance and reinforcement learning”. In: *arXiv preprint arXiv:2502.17219* (2025).
- [16] Ruben Grandia et al. “Perceptive locomotion through nonlinear model-predictive control”. In: *IEEE Transactions on Robotics* 39.5 (2023), pp. 3402–3421.
- [17] Omid Taheri et al. “GRAB: A dataset of whole-body human grasping of objects”. In: *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part IV 16*. Springer. 2020, pp. 581–600.
- [18] Jemin Hwangbo, Joonho Lee, and Marco Hutter. “Per-contact iteration method for solving contact dynamics”. In: *IEEE Robotics and Automation Letters* 3.2 (2018), pp. 895–902.
- [19] Tairan He et al. “OmniH2O: Universal and Dexterous Human-to-Humanoid Whole-Body Teleoperation and Learning”. In: *arXiv preprint arXiv:2406.08858* (2024).
- [20] Tony Z Zhao et al. “Learning fine-grained bimanual manipulation with low-cost hardware”. In: *arXiv preprint arXiv:2304.13705* (2023).
- [21] Jonas Gehring et al. “Leveraging demonstrations with latent space priors”. In: *arXiv preprint arXiv:2210.14685* (2022).
- [22] Qingwei Ben et al. “Homie: Humanoid loco-manipulation with isomorphic exoskeleton cockpit”. In: *arXiv preprint arXiv:2502.13013* (2025).
- [23] Ilya Loshchilov and Frank Hutter. “Sgdr: Stochastic gradient descent with warm restarts”. In: *arXiv preprint arXiv:1608.03983* (2016).
- [24] Hao Fu et al. “Cyclical annealing schedule: A simple approach to mitigating kl vanishing”. In: *arXiv preprint arXiv:1903.10145* (2019).

- [25] Djork-Arné Clevert, Thomas Unterthiner, and Sepp Hochreiter. “Fast and accurate deep network learning by exponential linear units (elus)”. In: *arXiv preprint arXiv:1511.07289* (2015).
- [26] John Schulman et al. “Proximal policy optimization algorithms”. In: *arXiv preprint arXiv:1707.06347* (2017).
- [27] Xinchun Yan et al. “Mt-vae: Learning motion transformations to generate multimodal human dynamics”. In: *Proceedings of the European conference on computer vision (ECCV)*. 2018, pp. 265–281.
- [28] Mathis Petrovich, Michael J Black, and Gül Varol. “Action-conditioned 3D human motion synthesis with transformer VAE”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2021, pp. 10985–10995.
- [29] Mazeyu Ji et al. “Exbody2: Advanced expressive humanoid whole-body control”. In: *arXiv preprint arXiv:2412.13196* (2024).
- [30] Manuel Y Galliker et al. “Planar bipedal locomotion with nonlinear model predictive control: Online gait generation using whole-body dynamics”. In: *2022 IEEE-RAS 21st International Conference on Humanoid Robots (Humanoids)*. IEEE. 2022, pp. 622–629.
- [31] Farbod Farshidian et al. “OCS2: An open source library for optimal control of switched systems”. In: *Accessed: May 23* (2023).
- [32] pocketxjl. *humanoid-control*. <https://github.com/pocketxjl/humanoid-control.git>. 2024.